

8.04: QUANTUM MECHANICS I

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External Sources

- P. A. M. Dirac. “The Principles of Quantum Mechanics.” 4th Ed. (1958)
- Eiderman, Vladimir. “An Introduction to Complex Analysis and the Laplace Transform.” CRC Press, 2022

1 Math Prerequisites

1.1 Complex Algebra

We recall the two main representations of a complex number.

Property 1.1.1 (Representation of a Complex Number)

Let $z \in \mathbb{C}$.

$$\exists a, b \in \mathbb{R} : z = a + bi \quad (1.1.1)$$

or equivalently

$$\exists r \in \mathbb{R}^+, \varphi \in [0, 2\pi) : z = r \exp(i\varphi) \quad (1.1.2)$$

with

$$a = r \cos \varphi \quad (1.1.3)$$

$$b = r \sin \varphi \quad (1.1.4)$$

We may define the *conjugate* of a complex number:

Definition 1.1.2 (Conjugate of a Complex Number)

Let $z \in \mathbb{C}$ be represented as either $a + bi$ or $r \exp(i\varphi)$. The **conjugate** of z , written as z^* or $\bar{z}$, is given by

$$z^* = \bar{z} = a - bi = r \exp(-i\varphi) \quad (1.1.5)$$

The *magnitude* of a complex number is given by the square root of the product of a complex number and its conjugate.

Definition 1.1.3 (Magnitude of a Complex Number)

Let $z \in \mathbb{C}$. Thus, the **magnitude** of z , written as $|z|$, is given by

$$|z| \equiv \sqrt{zz^*} \quad (1.1.6)$$

We of course observe that $(z^*)^* = z$, and that $z^*z = zz^* \in \mathbb{R}$:

$$zz^* = re^{i\varphi} \cdot re^{-i\varphi} = r^2 \in \mathbb{R} \quad (1.1.7)$$

Property 1.1.4 (Properties of Complex Numbers)

Below are some useful properties of a complex number. Let z_1, z_2 be two complex numbers.

$z = a + bi = r \exp(i\varphi)$:

$$\operatorname{Re}(z) = \frac{z + z^*}{2} \quad (1.1.8) \quad (z_1 z_2)^* = z_1^* z_2^* \quad (1.1.13)$$

$$\operatorname{Im}(z) = \frac{z - z^*}{2i} \quad (1.1.9) \quad |z_1 z_2| = |z_1| |z_2| \quad (1.1.14)$$

$$|z| = \sqrt{(\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2} \quad (1.1.10) \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad (1.1.15)$$

$$r = \sqrt{a^2 + b^2} \quad (1.1.11) \quad \arg(z_1 z_2) = (\arg(z_1) + \arg(z_2)) \mod 2\pi \quad (1.1.16)$$

$$\tan \varphi = \frac{b}{a} \quad (1.1.12) \quad \cos(\theta) \equiv \operatorname{Re}(\exp(i\theta)) \quad (1.1.17)$$

$$\tan \varphi = \frac{b}{a} \quad (1.1.12) \quad \sin(\theta) \equiv \operatorname{Im}(\exp(i\theta)) \quad (1.1.18)$$

Proposition 1.1.5 (Equations with Complex Numbers and Real Variables)

Let $n \in \mathbb{N}$. Let $x_1, \dots, x_{2n} \in \mathbb{R}$. A system of n complex-valued equations with $x_1 \dots x_{2n}$ can be sufficient to find the values of each real variable x_i .

1.2 Delta Function

For more information, see §15 of Dirac.

There are many equivalent definitions of the Delta Function, which is more so a distribution than a proper real-valued function.

Definition 1.2.1 (Delta Function, Limit of a Gaussian)

The standard normal distribution is given by

$$G(x; \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad (1.2.1)$$

We may define the Delta Function $\delta(x)$ according to

$$\delta(x) \equiv \lim_{\sigma \rightarrow 0} G(x; \sigma) \implies \left\{ \delta(x, x \neq 0) \equiv 0 \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(x) dx = 1 \right\} \quad (1.2.2)$$

We may use the Delta Function as a “sifter”:

Property 1.2.2 (Sifting Property)

Let $f(x)$ be a continuous complex-valued function of a real argument. Let $a \in \mathbb{R}$.

$$\int_{-\infty}^{\infty} f(x) \delta(x - a) dx = f(a) \quad (1.2.3)$$

The Heaviside Step Function follows naturally.

Definition 1.2.3 (Heaviside Step Function)

We define $\Theta(x)$ or $u(x)$, the Heaviside Step Function, according to

$$\Theta(x) \equiv u(x) \equiv \int_{-\infty}^x \delta(t) dt = \begin{cases} 0 & x < 0 \\ 1/2 & x = 0 \\ 1 & x > 0 \end{cases} \quad (1.2.4)$$

We may also write the step function as the limit of erf:

$$\text{erf}(x; \sigma) \equiv \int_{-\infty}^x G(t; \sigma) dt \implies \Theta(x) \equiv u(x) \equiv \lim_{\sigma \rightarrow 0} \text{erf}(x; \sigma) \quad (1.2.5)$$

We note several useful properties.

Property 1.2.4 (Delta Function Properties)

Let $a \in \mathbb{R}^+$. Let $b, c \in \mathbb{R}$.

$$\delta(-x) = \delta(x) \quad (1.2.6)$$

$$x\delta(x) = 0 \quad (1.2.7)$$

$$\delta(ax) = \frac{\delta(x)}{a} \quad (1.2.8)$$

$$\delta(x^2 - b^2) = \frac{\delta(x - b) + \delta(x + b)}{2a} \quad (1.2.9)$$

$$\int_{-\infty}^{\infty} \delta(c - x) \delta(x - b) dx = \delta(c - b) \quad (1.2.10)$$

$$f(x) \delta(x - b) = f(b) \delta(x - b) \quad (1.2.11)$$

We may also introduce the Delta function to harvest additional solutions to seemingly complete equations.

Let $x \in \mathbb{R}$. Thus,

$$A = B \implies \frac{A}{x} = \frac{B}{x} + c \cdot \delta(x) \quad (1.2.12)$$

for some unknown c . Additionally,

$$\frac{d}{dx} \log x = \frac{1}{x} - i\pi \delta(x) \quad (1.2.13)$$

Proposition 1.2.5 (Loosened Infinite Integral Convergence Criteria)

Using the delta function, we may write

$$\forall a \in \mathbb{R} : \int_{-\infty}^{\infty} \exp(iax) dx = 2\pi\delta(a) \quad (1.2.14)$$

Proof

Let $a \in \mathbb{R}$, $g \in \mathbb{R}^+$.

$$\int_{-\infty}^{\infty} \exp(iax) dx = \lim_{g \rightarrow \infty} \int_{-g}^g \exp(iax) dx = 2\pi \lim_{g \rightarrow \infty} \left(\frac{\sin(ag)}{\pi a} \right) = 2\pi\delta(a) \quad (1.2.15)$$

where we have used the sine integral kernel $K_g \equiv \sin(ag)/\pi a$. Let f be a smooth function; thus,

$$\int_{-\infty}^{\infty} f(a) \frac{\sin(ag)}{\pi a} da = \int_{-\infty}^{\infty} \frac{f(a) - f(0)}{\pi a} \sin(ag) da + f(0) \int_{-\infty}^{\infty} \frac{\sin(ag)}{\pi a} da \quad (1.2.16)$$

As $g \rightarrow \infty$, the first integral goes to zero since rapid oscillations of $\sin(ag)$ will all cancel¹ and the second one evaluates to 1, giving $f(0)$ exactly as $\delta(a)$ would, and so

$$\int_{-\infty}^{\infty} \exp(iax) dx = 2\pi \int_{-\infty}^{\infty} K_g(a) da = 2\pi\delta(a) \quad (1.2.18)$$

as desired. □

¹Let $h(a) \equiv (f(a) - f(0))/a$; we observe that the numerator of h is smooth and vanishes at $a = 0$; thus, h is bounded at $a = 0$ and also smooth. Applying the Riemann-Lebesgue lemma, which states that

$$\lim_{g \rightarrow \infty} \int_{-\infty}^{\infty} h(a) \sin(ag) da = 0 \quad (1.2.17)$$

we have satisfied the preconditions and so we obtain the desired cancellation of the first integral.

1.3 Fourier, Laplace Transform

1.3.1 Fourier Series

Let $f(t)$ be a periodic function with fundamental period T_0 . It is possible to decompose such a function using a sum of sines and cosines, called the *Fourier Series*:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(2n\pi t/T_0) + \sum_{n=1}^{\infty} b_n \sin(2n\pi t/T_0) \quad (1.3.1)$$

These coefficients are found using the following integrals:

$$a_n = \frac{2}{T_0} \int_0^{T_0/2} f(t) \cos(2n\pi t/T_0) dt \quad ; \quad b_n = \frac{2}{T_0} \int_0^{T_0/2} f(t) \sin(2n\pi t/T_0) dt \quad (1.3.2)$$

We can generalize to complex coefficients such that

$$c_n = \begin{cases} \frac{a_n + ib_n}{2} & n \geq 0 \\ \frac{a_n - ib_n}{2} & n < 0 \end{cases} \quad (1.3.3)$$

There is also an integral form for calculating these coefficients directly:

$$c_n = \frac{1}{T_0} \int_0^{T_0} f(t) e^{-2\pi i t/T_0} dt \quad (1.3.4)$$

From this,

$$a_n = \frac{c_n + c_{-n}}{2} \quad ; \quad b_n = \frac{c_n - c_{-n}}{2j} \quad (1.3.5)$$

It is clear that a function $f(t)$ such that

$$f(t) = \sum_n d_n e^{jn\omega_0 t} \quad (1.3.6)$$

will have coefficients $c_n = d_n$ quite simply. In particular, recalling that

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2} \quad (1.3.7)$$

we see that the Fourier coefficients of $\cos(\omega_0 t)$ will be $c_1 = c_{-1} = 1$ and the rest zero. We can draw this below:

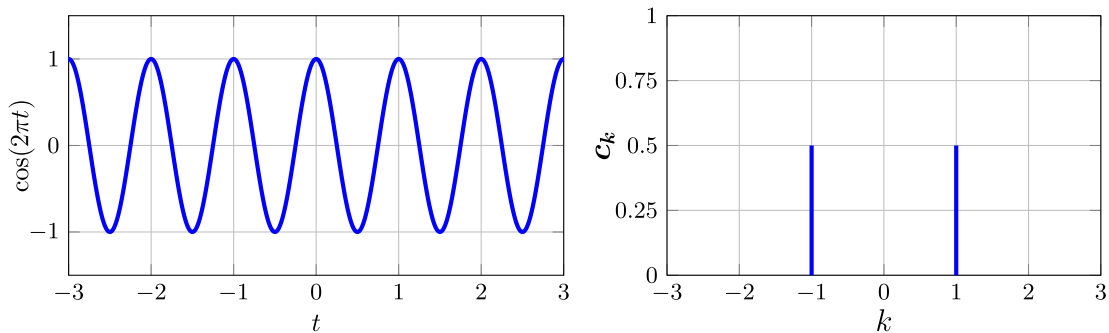


Figure 1. Frequency Spectrum of Cosine.

We introduce an essential theorem:

Theorem 1.3.1 (Parseval's Theorem)

The **Parseval Theorem**^a relates the *energy* of a signal to that of its spectrum:

$$\frac{1}{T_0} \int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2 \quad (1.3.8)$$

^a Marc-Antoine Parseval des Chênes was a French mathematician of the 18th and 19th centuries.

We can also clarify the convergence conditions of the Fourier Series.

Proposition 1.3.2 (Convergence of the Fourier Series)

A sufficient condition for the convergence of the Fourier Series of a periodic signal $x(t)$ is found to be

$$\int_{T_0} |x(t)|^2 dt < \infty \quad (1.3.9)$$

The *Dirichlet conditions* also include a set of sufficient conditions for the convergence of the Fourier Series:

- $x(t)$ is absolutely integrable over a period T_0 :

$$\int_{T_0} |x(t)| dt < \infty \quad (1.3.10)$$

- The number of local maxima and minima of $x(t)$ in a finite interval is also finite.
- The signal $x(t)$ cannot have an infinite number of discontinuities in a finite interval.

When we work with functions having jump discontinuities, such as the square wave, the Fourier Series converges *pointwise* but not *uniformly*, leading to the *Gibbs effect*. Let us study as an example the Fourier series of the periodic signal $x(t)$ defined as

$$x(t \mid t \in [-1/2, 1/2]) = -1 + 2\Theta(t) \quad (1.3.11)$$

and having the graph below:

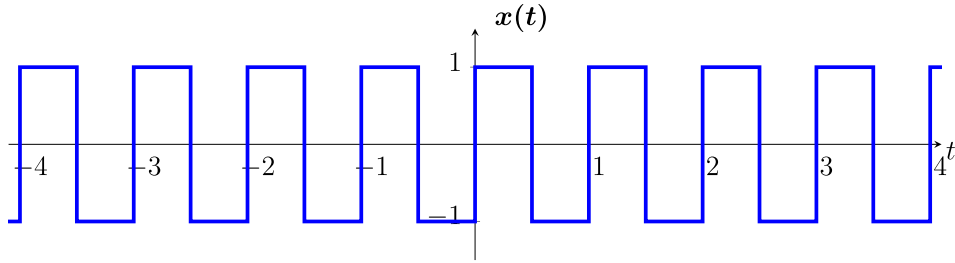


Figure 2. Square wave.

We immediately observe the jumps at each $t = k/2$ for $k \in \mathbb{Z}$. We find the Fourier coefficients:

$$c_n = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-ik2\pi t/T_0} dt = \int_0^{1/2} e^{-ik2\pi t/T_0} dt - \int_{1/2}^1 e^{-ik2\pi t/T_0} dt \quad (1.3.12)$$

$$= \begin{cases} 0 & k = 0 \\ \frac{1 - e^{-i\pi k}}{jk\pi} & k \neq 0 \end{cases} = \begin{cases} 0 & k \text{ is even} \\ \frac{2}{ik\pi} & k \text{ is odd} \end{cases} \quad (1.3.13)$$

The spectrum of $x(t)$ looks like

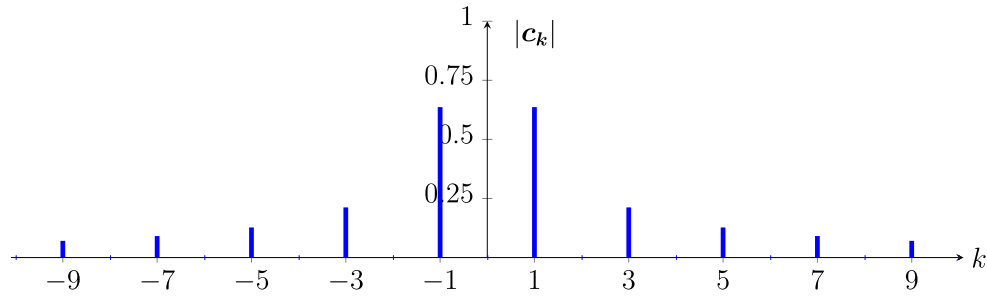


Figure 3. Spectrum of $x(t)$

The successive sums of complex exponentials with the c_k calculated above show the slow convergence around the discontinuities:

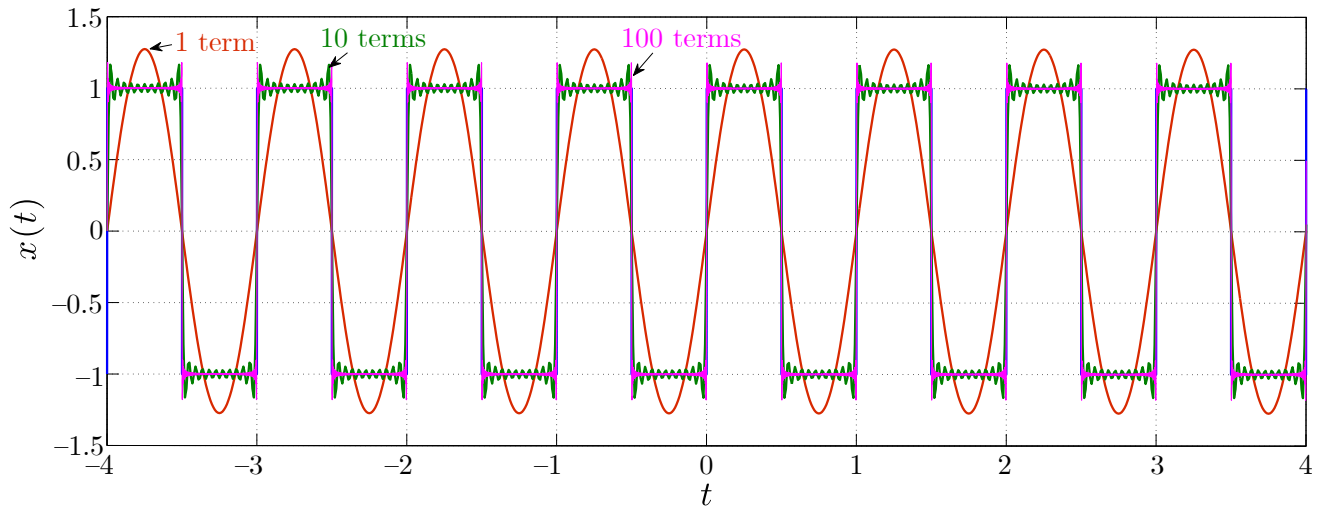


Figure 4. Approximations of $x(t)$ with Fourier Series with 1, 10, and 100 terms.

At each discontinuity, the series approximation maintains an error of approximately 9% compared to the size of the jump, called the *Gibbs* phenomenon. However, the series itself still converges, because the area of the overshooting oscillations tends to zero as the number of terms in the series grows to infinity.

1.3.2 Fourier Transform

First, we introduce some terminology.

Proposition 1.3.3 (Dirichlet Conditions)

The Dirichlet Conditions require the following of a function $f(x) : \mathbb{R} \rightarrow \mathbb{C}$:

- $f(x)$ is absolutely integrable over the real line, satisfying $\int_{-\infty}^{\infty} |f(x)| dx < \infty$.
- Over a finite interval, $f(x)$ has but a finite number of local maxima and minima.
- Over a finite interval, $f(x)$ has but a finite number of discontinuities.

Definition 1.3.4 (Normalized Function)

Let $f(x) : \mathbb{R} \rightarrow \mathbb{C}$ be a complex-valued function of a real-valued argument. We say that $f(x)$ is normalized if

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = 1 \quad (1.3.14)$$

Definition 1.3.5 (Fourier Transform)

Let $f(x)$ be a function satisfying the Dirichlet Conditions. The *Fourier Transform* of $f(x)$, written as $F(\omega)$, is defined by

$$\mathcal{F}\{x\} \equiv F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-j\omega x} dx \quad (1.3.15)$$

We also define the inverse transform:

Definition 1.3.6 (Inverse Fourier Transform)

Let $f(x)$ be a function satisfying the Dirichlet Conditions. Let $F(\omega)$ be its Fourier Transform. It is possible to retrieve the original function $f(x)$ from $F(\omega)$ according to

$$\mathcal{F}^{-1}\{F(\omega)\} \equiv f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\omega) e^{j\omega x} d\omega \quad (1.3.16)$$

Theorem 1.3.7 (Fourier Inversion Theorem)

Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be a function satisfying the Dirichlet Conditions. Thus,

$$f(t) = \iint_{\mathbb{R}^2} e^{2\pi j(t-t')\xi} f(t') dx' d\xi \quad (1.3.17)$$

We may prove that a normalized function $f(x)$ remains normalized under a Fourier Transformation:

Theorem 1.3.8 (Parseval)

Let $f(x) : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying the Dirichlet Conditions. Let $F(\omega) = \mathcal{F}\{f(x)\}$. Thus,

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |F(\omega)|^2 d\omega \quad (1.3.18)$$

Property 1.3.9 (Inverse Fourier Transform Alternate Form)

It is possible to obtain the Inverse Fourier Transform from repeated application of the Fourier Transform:

$$\mathcal{F}^{-1} = \mathcal{F}^3 \quad (1.3.19)$$

1.3.3 Laplace Transform

We can now extend the Fourier Transform to the entire complex plane.

Definition 1.3.10 (Laplace Transform)

Let $f(t)$ be a complex-valued function of a real variable. The *Bilateral Laplace Transform* $\mathcal{L}$ is a functional

returning a complex-valued function $F(s)$ of a complex variable s such that

$$\mathcal{L}\{f(t)\}_s = F(s) = \int_{-\infty}^{\infty} f(t)e^{-st} dt \quad (1.3.20)$$

The *Unilateral Laplace Transform* is a similar functional $\mathcal{L}_{\text{oneside}}$ returning a complex-valued function $G(s)$ of a complex variable s such that

$$\mathcal{L}_{\text{oneside}}\{f(t)\}_s = G(s) = \int_0^{\infty} f(t)e^{-st} dt \quad (1.3.21)$$

It is possible to characterize constraints on $f(t)$ such that $F(s)$ or $G(s)$ exist.

Theorem 1.3.11 (Convergence of the Laplace Transform – Eiderman 269)

If $f(t)$ is a piecewise continuous function with growth index σ_0 , then the integral below converges absolutely for all p in the half-plane $\text{Re}(s) > \sigma_0$, and it converges uniformly on every half-plane $\text{Re}(s) > \sigma_1$ where $\sigma_1 > \sigma_0$. The function $F(s)$ defined by the expression below is analytic in the half-plane $\text{Re}(s) > \sigma_0$.

$$F(s) = \mathcal{L}_{\text{oneside}}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt \quad (1.3.22)$$

where the *growth index* is defined as follows:

Definition 1.3.12 (Growth Index – Eiderman 266)

The growth index σ_0 of a piecewise-differentiable function $f(t)$ satisfying $(t < 0) \Rightarrow f(t) = 0$ is the real number separating the sets $\sigma \in A \equiv (-\infty, \sigma_0)$ and $\sigma \in B \equiv (\sigma_0, \infty)$ such that

$$\forall \sigma \in A \equiv (-\infty, \sigma_0) : \exists M \in \mathbb{R} : \forall t \in \mathbb{R} : |f(t)| \leq Me^{\sigma t} \quad (1.3.23)$$

(exclusive) or

$$\forall \sigma \in B \equiv (\sigma_0, \infty) : \exists M \in \mathbb{R} : \forall t \in \mathbb{R} : |f(t)| \leq Me^{\sigma t} \quad (1.3.24)$$

We observe that even though the theorem applies to the one-sided Laplace transform,

$$\mathcal{L}_{\text{oneside}}\{f(t)\} = \mathcal{L}_{\text{twoside}}\{f(t)u(t)\} \quad (1.3.25)$$

since for any integrable $g(t)$ (for example $g(t) = f(t)e^{-st}$),

$$\mathcal{L}_{\text{twoside}}\{g(t)u(t)\} = \int_{-\infty}^{\infty} g(t)u(t) dt = \int_0^{\infty} g(t) dt = \mathcal{L}_{\text{oneside}}\{g(t)\} \quad (1.3.26)$$

Proposition 1.3.13 (Conditions on the Growth Index)

Let $f(t)$ be a piecewise differentiable function such that $\exists g(t)$ piecewise differentiable such that $f(t) = g(t)u(t)$. Thus,

$$\lim_{t \rightarrow \infty} |f(t)|e^{-\sigma t} = 0 \implies \exists M \in \mathbb{R} : \forall t \in \mathbb{R} : |f(t)| \leq Me^{\sigma t} \quad (1.3.27)$$

We now take the inverse Laplace Transform of a function $F(s)$, using the method of residues.

Theorem 1.3.14 (Residue of Complex-valued Function on the Complex Plane: Eiderman 7.18)

1. Suppose that z_0 is a pole of the function f of order $n \geq 1$. Then,

$$\operatorname{res}_{z_0} f = \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} ((z - z_0)^n f(z))^{(n-1)} \quad (1.3.28)$$

where $^{(n-1)}$ represents the $n-1^{\text{th}}$ derivative.

2. Suppose that on a punctured neighborhood of z_0 a function $f(z)$ takes the form

$$f(z) = \frac{h(z)}{(z - z_0)^n} \quad (1.3.29)$$

where h is analytic on a neighborhood of z_0 , and $h(z_0) \neq 0$ (so that f has a pole of order n at z_0). Then,

$$\operatorname{res}_{z_0} f = \frac{h^{(n-1)}(z_0)}{(n-1)!} \quad (1.3.30)$$

We can use the residue to find the inverse Laplace Transform:

Theorem 1.3.15 (Inverse Laplace Transform using Residues: Eiderman 9.29)

Suppose that $F(s) \equiv \mathcal{L}_{\text{oneside}}\{f(t)\}$, and that $F(s)$ is a rational function with poles at $p_1, \dots, p_n$. Then,

$$f(t) = \sum_k \operatorname{res}_{p_k} (F(s)e^{st}) \quad (1.3.31)$$

1.3.4 Properties of the Fourier Transform

It is convenient to note the following properties of the Fourier Transform.

Property 1.3.16 (Properties of the Fourier Transform)

Let $f, g : \mathbb{R} \rightarrow \mathbb{C}$ two functions satisfying the Dirichlet conditions. Let $a, b \in \mathbb{C}$.

$$\mathcal{F}\{af + bg\} = a\mathcal{F}\{f\} + b\mathcal{F}\{g\} \quad (1.3.32)$$

$$\mathcal{F}\{f(x - \alpha)\} = e^{-i\omega\alpha} \mathcal{F}\{f\} \quad (1.3.33)$$

$$\mathcal{F}\{f(\alpha x)\} = \frac{1}{|\alpha|} \mathcal{F}\{f\}(\omega/\alpha) \quad (1.3.34)$$

$$\mathcal{F}\{f(-x)\} = \mathcal{F}\{f\}(-\omega) \quad (1.3.35)$$

$$\mathcal{F}\{f(\alpha x - \beta)\} = \frac{e^{-i\omega\beta/\alpha}}{|\alpha|} \mathcal{F}\{f\}(\omega/\alpha) \quad (1.3.36)$$

$$\mathcal{F}\{e^{2\pi i \alpha t} f(x)\} = \mathcal{F}\{f\}(\omega - 2\pi\alpha) \quad (1.3.37)$$

1.3.5 Properties of the Laplace Transform

Proposition 1.3.17 (Properties of the Laplace Transform)

Let f, g be two functions satisfying the Dirichlet Conditions. Let F, G be their Laplace Transformations. Let $a, b \in \mathbb{C}$.

Linearity:	$af(t) + bg(t) \longrightarrow aF(s) + bG(s)$
Uncertainty:	$f(t/a) \longrightarrow aF(as)$
Time Delay:	$f(t-a)u(t-a) \longrightarrow e^{-as}F(s)$
Temporal Phase:	$e^{-at}f(t) \longrightarrow F(s+a)$
1st Derivative:	$f'(t) \longrightarrow sF(s) - f(0)$
n^{th} Derivative:	$f^{(n)}(t) \longrightarrow s^n F(s) - \sum_{k=0}^{n-1} p^{n-k-1} f^{(k)}(0)$
Integral:	$\int_0^t f(\tau) d\tau \longrightarrow \frac{F(s)}{s}$
Multiplication by t:	$tf(t) \longrightarrow -F'(s)$
Division by t:	$\frac{f(t)}{t} \longrightarrow \int_s^\infty F(p) dp$
Convolution:	$f \star g \longrightarrow FG$
Multiplication:	$fg \longrightarrow \frac{F \star G}{2\pi j}$
Duality:	$F(t) \longrightarrow f(-s)$
Conjuguate:	$\overline{f(t)} \longrightarrow \overline{F(\overline{s})}$

For convolution in Laplace space, the convolution integral is evaluated along the vertical line $c - j\infty \rightarrow c + j\infty$ for c in the region of convergence. See 16.002.review for further details, as well as Eiderman.

1.4 Linear Algebra

We first define two useful tensors, i.e. matrices whose representation depends on a coordinate system choice.

Definition 1.4.1 (Unit Tensor $\Leftrightarrow$ Kronecker Delta)

The *unit tensor* or *Kronecker Delta* is a two-index number, given by

$$i, j \in \mathbb{N} : \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (1.4.1)$$

Definition 1.4.2 (Permutation Tensor $\Leftrightarrow$ Levi-Civita Symbol)

The *permutation tensor* or *Levi-Civita Symbol* is given by

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } ijk = \{(1, 2, 3), (2, 3, 1), (3, 1, 2)\} \\ -1 & \text{if } ijk = \{(3, 2, 1), (2, 1, 3), (1, 3, 2)\} \\ 0 & \text{otherwise} \end{cases} \quad (1.4.2)$$

We recall the notions of vectors and matrices, in particular that matrices do not generally commute, i.e.

$$AB \neq BA \quad (1.4.3)$$

for $A, B \in \mathbb{M}_{n \times n}$.

We may characterize a set of vectors as linearly (in)dependent.

Definition 1.4.3 (Linear Independence)

A set of vectors $\{\mathbf{v}_k\} \subset \mathbb{C}^n$ is said to be **linearly independent** if

$$\forall i : \nexists \{c_k\} : \mathbf{v}_i = \sum_{k \neq i} c_k \mathbf{v}_k \quad (1.4.4)$$

Otherwise, the set is called **linearly dependent**. The set of all linear combinations of the vectors $\{\mathbf{v}_k\}$ is called the *span* of $\{\mathbf{v}_k\}$.

We may then characterize the span:

Definition 1.4.4 (Basis)

A set of vectors $\{\mathbf{v}_k\} \subset \mathbb{C}^n$ is said to be a **basis** for a vector space $\mathcal{A}$ if every element $\mathbf{a} \in \mathcal{A}$ can be expressed as a linear combination of the vectors $\{\mathbf{v}_k\}$.

For example, the simplest basis for $\mathbb{R}^2$ is $\{[0, 1], [1, 0]\}$.

Definition 1.4.5 (Inner Product)

The *inner product* of two vectors $\mathbf{a}, \mathbf{b}$, expressed as a sum of coefficients and basis vectors in some basis $\{\hat{\mathbf{e}}_i\}$ as $\mathbf{a} = \sum a_i \hat{\mathbf{e}}_i \in \mathbb{C}^n$ and $\mathbf{b} = \sum b_i \hat{\mathbf{e}}_i \in \mathbb{C}^n$ is given by

$$\mathbf{a} \cdot \mathbf{b} \equiv \sum a_i^* b_i \in \mathbb{C} \quad (1.4.5)$$

We may write an analogous definition for functions:

Definition 1.4.6 (Inner Product for Functions)

Let $f(x), g(x)$ be functions over some simply-connected domain D with values in $\mathbb{C}$. The *inner product* of these two functions is given by

$$\langle f | g \rangle = \int_D f^*(x) g(x) dx \quad (1.4.6)$$

We may now characterize bases in terms of inner products between components:

Definition 1.4.7 (Orthonormal Basis)

A vector is *normalized* if its inner product with itself is 1. Two vectors are *orthogonal* if the inner product between them is 0. A basis is thus *orthonormal* if

- all basis vectors $\hat{\mathbf{e}}_i$ are normalized, i.e. $||\hat{\mathbf{e}}_i|| \equiv \sqrt{\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_i} = 1$, and
- the inner product between any pair of distinct basis vectors is 0, i.e. $\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \delta_{ij}$.

Definition 1.4.8 (Eigenvalue, Eigenvector)

Let $A \in \mathbb{M}_{n \times n}$. We call $\{\lambda, \mathbf{v}\}$ an eigenvalue and eigenvector pair if

$$A\mathbf{v} = \lambda\mathbf{v} \quad (1.4.7)$$

We may find eigenvalues according to the following procedure.

Proposition 1.4.9 (Eigenvalues of a Matrix)

Let $A \in \mathbb{M}_{n \times n}$. Let I be the identity matrix. Thus, any eigenvalues λ of A must satisfy

$$|\lambda I - A| = 0 \quad (1.4.8)$$

The eigenvectors with an eigenvalue λ belong to the kernel $\text{Nul}(\lambda I - A)$.

We define the determinant.

Definition 1.4.10 (Determinant)

The *determinant* $|\mathbf{A}|$ of a matrix $\mathbf{A}$ is a complex scalar that satisfies the following properties:

1. For two $n \times n$ matrices $\mathbf{A}, \mathbf{B}$: $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$.
2. If the column vectors of an $n \times n$ matrix $\mathbf{A}$ do not form a basis (i.e. $\mathbf{A}$ is not full rank), then $|\mathbf{A}| = 0$.
3. For a 2×2 matrix $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is given by

$$|\mathbf{A}| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \quad (1.4.9)$$

Proposition 1.4.11 (Determinant of a 3x3 Matrix)

For a 3×3 matrix, the determinant can be quickly found to be

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - (ceg + afh + bdi) \quad (1.4.10)$$

If we write each element of the above matrix as a_{ij} , then

$$|a_{ij}| = \sum_{i,j,k=1}^3 \epsilon_{ijk} a_{1i} a_{2j} a_{3k} \quad (1.4.11)$$

Proposition 1.4.12 (Determinant of a Diagonal Matrix)

A matrix is said to be diagonal if all nonzero elements lie on the diagonal. The determinant of such a matrix is the product of the diagonal elements.

Proposition 1.4.13 (Eigenvalues of a Diagonal Matrix)

Let $\mathbf{\Lambda}$ be a diagonal matrix. The eigenvalues of $\mathbf{\Lambda}$ are simply the diagonal entries, and the corresponding eigenvectors are the standard orthonormal basis vectors $\langle 1, 0 \cdots 0 \rangle \cdots \langle 0, \cdots 0, 1 \rangle$.

Property 1.4.14 (Determinant From Eigenvalues)

Consider a matrix $\mathbf{A}$ with eigenvalues $\{\lambda_j\}$. The determinant of $\mathbf{A}$ is the product of these eigenvalues.

It is of use to define the matrix exponential:

Definition 1.4.15 (Matrix Exponential)

Let $\mathbf{A} \in \mathbb{M}_{n \times n}$. Let $t \in \mathbb{R}$. We define

$$\exp(\mathbf{A}t) \equiv \sum_{n=0}^{\infty} \frac{\mathbf{A}^n t^n}{n!} \quad (1.4.12)$$

Proposition 1.4.16 (Diagonalization of a Matrix)

Let $\mathbf{A}$ be an $n \times n$ matrix with n linearly independent eigenvectors. It is possible to factor $\mathbf{A}$ into a $\mathbf{PDP}^{-1}$ factorization where $\mathbf{P}$ is a matrix with the eigenvectors of $\mathbf{A}$ as columns and $\mathbf{D}$ is a matrix with the eigenvalues of $\mathbf{A}$ in the diagonal. Note that if the eigenvectors are orthogonal, then it is possible to normalize them and obtain a $\mathbf{PDP}^T$ factorization.

Proposition 1.4.17 (Matrix Exponential From Diagonalization)

Suppose the matrix $\mathbf{A}$ can be diagonalized into $\mathbf{PDP}^{-1}$. Thus,

$$\exp(\mathbf{A} \cdot t) = \exp(\mathbf{PDP}^{-1} \cdot t) = \exp(\mathbf{P}(\mathbf{D} \cdot t)\mathbf{P}^{-1}) = \mathbf{P} \exp(\mathbf{D} \cdot t) \mathbf{P}^{-1} \quad (1.4.13)$$

$$= \mathbf{P} \begin{bmatrix} \exp(D_{11}t) & & \\ & \ddots & \\ & & \exp(D_{nn}t) \end{bmatrix} \mathbf{P}^{-1} \quad (1.4.14)$$

Proposition 1.4.18 (Matrix Exponential Using the Laplace Transform)

Let $A \in \mathbb{M}_{n \times n}$. Let $t \in \mathbb{R}$. Thus,

$$\exp(At) = \mathcal{L}^{-1} \{ (sI - A)^{-1} \} \quad (1.4.15)$$

This last method has the benefit of not depending on the convergence of A^n , and is wholly analytical.

1.5 Statistics

We require the notion of expectation and uncertainty, as well as Bayes' Law.

Definition 1.5.1 (Expected Value)

Let $\{(x_k, p_k)\}$ be a discrete probability distribution where the probability of the random variable X having the value value x_k is p_k . The **expectation** or **expected value** of x is thus

$$\langle x \rangle = \sum x_k p_k \quad (1.5.1)$$

For a continuous distribution, let $f(x)$ be the probability distribution such that $P(X \in [a, b]) = \int_a^b f(x) dx$.

Thus,

$$\langle x \rangle = \int_{-\infty}^{\infty} x f(x) dx \quad (1.5.2)$$

Definition 1.5.2 (Uncertainty)

The uncertainty Δx of a random variable x is given by

$$\Delta x \equiv \sqrt{\langle x^2 \rangle - \langle x \rangle^2} \quad (1.5.3)$$

Definition 1.5.3 (Conditional Probability)

The conditional probability of A given B is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (1.5.4)$$

Law 1.5.4 (Bayes' Law)

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} \quad (1.5.5)$$

Property 1.5.5 (Gaussian Integrals)

The Gaussian Distribution is defined by

$$G(x; \sigma) \equiv \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad (1.5.6)$$

Several convenient integrals arise:

$$\int_{-\infty}^{\infty} G(x; \sigma) dx = 1 \quad \mathcal{F}\{G(x; \sigma)\} = G\left(k; \frac{1}{\sigma}\right) \quad (1.5.7)$$

$$\forall \alpha, \beta \in \mathbb{C} : \operatorname{Re}\{\alpha\} > 0 : \begin{cases} \int_{-\infty}^{\infty} \exp(-\alpha(x - \beta)^2) dx = \sqrt{\frac{\pi}{\alpha}} \\ \int_{-\infty}^{\infty} \exp(-\alpha x^2 + \beta x) dx = \sqrt{\frac{\pi}{\alpha}} \exp\left(\frac{\beta^2}{4\alpha}\right) \\ \int_{-\infty}^{\infty} x^2 \exp(-\alpha x^2) dx = \frac{\sqrt{\pi}}{2\alpha^{3/2}} \end{cases} \quad (1.5.8)$$

Proof

The first of the Gaussian integrals is obtained through the polar trick, by which we introduce a double integral and then convert to polar coordinates. The second can be obtained from the first by completing the square. The third is obtained by differentiating under the integral sign. $\square$

2 Operators

2.1 Bra-Ket Vectors

See Dirac §5, 6 for further reading.

2.1.1 From Experiment to Mathematical Representation

One may observe from experiments that certain differences arise between large-scale “classical” phenomena and small-scale “quantum” phenomena, most notably in that quantum systems must be modeled as a superposition of discrete states.

We wish to model these states; the most obvious choice is a vector, since these are easily added to produce more vectors. We must however have the ability to work with an unspecified (possibly infinite) number of dimensions.

Definition 2.1.1 (Ket Vector)

A *ket vector* or *ket* is a kind of vector used to describe a quantum state. Most generally, it has an infinite number of dimensions. A ket describing the state A is written as $|A\rangle$.

We may treat ket vectors like any other vector, forming linear combinations, most generally with complex coefficients. For example, $|R\rangle$ below would be a linear combination of $|A\rangle$ and $|B\rangle$ with coefficients c_1 and c_2 respectively:

$$|R\rangle = c_1|A\rangle + c_2|B\rangle \quad (2.1.1)$$

We may also integrate such kets should they be parametrized by a continuous variable, say, position x :

$$\int |x\rangle dx = |Q\rangle \quad (2.1.2)$$

Linear (in)dependence carries over exactly as in linear algebra.

Axiom 2.1.2 (Quantum State)

Each state of a dynamical system at a particular time corresponds to a ket vector, the correspondence being such that if a state results from the superposition of certain other states, its corresponding ket vector is expressible linearly in terms of the corresponding ket vectors of the other states, and conversely.

We require thus that our superposition process be independent of order, perfectly satisfied by the usage of vectors. We may also now take the notion of (in)dependence and apply it to states as well as ket vectors (these are now equivalent).

Axiom 2.1.3 (Superposition of a State with Itself)

A state superposed with itself cannot form a new state, but only the original state over again.

Should we superpose the same state twice, we simply obtain a scaling:

$$c_1|A\rangle + c_2|A\rangle = (c_1 + c_2)|A\rangle \quad (2.1.3)$$

Should $c_1 + c_2 = 0$, we obtain a cancelling-out i.e. destructive interference; otherwise, the state $(c_1 + c_2)|A\rangle$ cannot be distinguished from either $c_1|A\rangle$ or $c_2|A\rangle$ according to the last assumption above, and so we obtain the following proposition.

Proposition 2.1.4 (State Invariance With Respect to Scaling)

Suppose a quantum state A is represented by the ket $|A\rangle$. The ket $c|A\rangle$ for any $c \in \mathbb{C} \setminus \{0\}$ corresponds to the same quantum state A .

We note that the zero ket does not correspond to a variant of some state with zero amplitudes, but rather no state at all.

Proposition 2.1.5 (Ratios of Coefficients in a Superposition)

Consider a set of ket vectors $\{|A_i\rangle\}$, and let $|R\rangle$ be their superposition with some coefficients $c_i \in \mathbb{C}$. Only the ratios c_i/c_j are needed along with the ket vector set $\{|A_i\rangle\}$ to fully specify the state $|R\rangle$.

2.1.2 Bra Vectors

For any vector space, we may define a “dual” vector space that serve to complement the previous vector space. We shall call these “bra” vectors.

Consider a linear function ϕ of a ket vector, returning a scalar (most generally complex), so that

$$\phi(|A\rangle + |A'\rangle) = \phi(|A\rangle) + \phi(|A'\rangle) \quad (2.1.4)$$

$$\text{and } \phi(c|A\rangle) = c\phi(|A\rangle) \quad (2.1.5)$$

We may thus describe ϕ as a scalar product of $|A\rangle$ with a new vector: such a new vector would correspond to each linear function ϕ satisfying the above properties.

Definition 2.1.6 (Bra Vector)

Suppose ϕ is a linear function mapping a ket vector to a scalar. We may define the *bra vector* $\langle B|$ such that

$$\langle B|A\rangle \equiv \phi(|A\rangle) \quad (2.1.6)$$

for each ket vector $|A\rangle$. A bra vector is considered to be completely defined when its scalar product with every ket vector is given.

Proposition 2.1.7 (Completeness of a Bracket Expression)

A complete bracket expression is one with a bra at the far left and a ket at the far right. Any complete bracket denotes a number. Any incomplete bracket expression denotes a vector, of the bra or ket kind according to whether it contains the first or second part of the brackets. A third kind of expression, namely

$$|A\rangle\langle B| \quad (2.1.7)$$

denotes a linear operator, a function that takes a ket on the right or a bra on the left, and returns the respective kind of vector.

Thus, we may obtain some properties that arise directly from the definition:

Property 2.1.8 (Linearity of Brackets)

Let $\langle B|$ be a bra vector and $|A\rangle$ be a ket vector. Thus, according to the definition of bra vector “multiplication” with ket vectors,

$$\langle B|\{|A\rangle + |A'\rangle\} = \langle B|A\rangle + \langle B|A'\rangle \quad (2.1.8)$$

$$\langle B|\{c|A\rangle\} = c\langle B|A\rangle \quad (2.1.9)$$

Proposition 2.1.9 (Zero Bra)

A bra $\langle P|$ is said to be “zero” (i.e. $\langle P| = 0$) if

$$\forall |A\rangle : \langle P|A\rangle = 0 \quad (2.1.10)$$

The same linear properties that hold for kets also hold for bras.

Axiom 2.1.10 (Bra as the Conjugate Transpose of a Ket)

We constrain the degree of freedom on the nature of bra vectors according to the following assumption. We assume a one-to-one correspondence between bras and kets; let $|A\rangle$ and $|A'\rangle$ be ket vectors, and let $\langle A|$ and $\langle A'|$ be the corresponding bra vectors. We require

$$\forall c \in \mathbb{C} : c^* \langle A| \longleftrightarrow c|A\rangle \quad (2.1.11)$$

$$\langle A + A'| \longleftrightarrow |A + A'\rangle \equiv |A\rangle + |A'\rangle \quad (2.1.12)$$

We can thus characterize the bra $\langle A|$ corresponding to a ket $|A\rangle$ as the “conjugate imaginary” or “complex transpose” of the ket vector.

We may use this one-to-one correspondence:

Proposition 2.1.11 (Characterization of a State with a Bra Vector)

Since states may be characterized by a ket vector alone, and there exists a one-to-one correspondence between ket vectors and bra vectors, we may thus characterize a state fully with a bra vector.

We may now combine bras and kets and deduce certain essential properties.

Proposition 2.1.12 (Inner Product of a Bra and a Ket)

We may describe the space of ket vectors as an “inner product space” according to the linear algebra definition by which we define the inner product of two kets as the product of the corresponding bra vector on the left with the original ket vector on the right, such that

$$|A\rangle \cdot |B\rangle \equiv \langle A|B\rangle \in \mathbb{C} \quad (2.1.13)$$

This is equivalent to

$$\mathbf{a} \cdot \mathbf{b} = \left(\sum a_i \hat{\mathbf{e}}_i \right) \cdot \left(\sum b_i \hat{\mathbf{e}}_i \right) = \sum a_i^* b_i \in \mathbb{C} \quad (2.1.14)$$

From these we deduce

Property 2.1.13 (Bracket Inversion)

Let $\langle B|$ be a bra vector and $|A\rangle$ be a ket vector. Thus,

$$\langle B|A\rangle = \overline{\langle A|B\rangle} \equiv (\langle A|B\rangle)^* \quad (2.1.15)$$

Thus, the inner product of a ket vector with itself is a real number, since it is equal to its complex conjugate.

Property 2.1.14 (Length of a Ket)

The inner product of a ket with itself is given by

$$\langle A|A\rangle > 0 \quad (2.1.16)$$

except when $\langle A| = 0$.

Definition 2.1.15 (Normalization, Orthogonality of Kets)

We say a ket is normalized when

$$\langle A|A\rangle = 1 \quad (2.1.17)$$

We say two kets $|A\rangle, |B\rangle$ are orthogonal when

$$\langle A|B\rangle = 0 \quad (2.1.18)$$

We note that normalization is not enough to specify a ket entirely; an arbitrary *phase factor* $e^{i\gamma}$ for $\gamma \in \mathbb{R}$ can be multiplied.

2.2 Linear Operators

See Dirac §7, 8 for further reading.

2.2.1 Operators, Commutation

We now consider functions that take a ket and return a ket, called “operators”.

Definition 2.2.1 (Linear Operator)

A (linear) operator is a linear function which takes a ket on the right and evaluates to a ket. An operator $\hat{a}$ mapping $|A\rangle$ to $|B\rangle$ will be written as $\hat{a}|A\rangle = |B\rangle$. An operator $\hat{a}$ is completely defined when the result of its application to every ket vector is given.

For a linear operator, we deduce the following properties:

$$\hat{\alpha}(|A\rangle + |A'\rangle) = \hat{\alpha}|A\rangle + \hat{\alpha}|A'\rangle \quad (2.2.1)$$

$$\hat{\alpha}(c|A\rangle) = c\hat{\alpha}|A\rangle \quad (2.2.2)$$

We may add linear operators one to another in the following way:

$$(\hat{\alpha} + \hat{\beta})|A\rangle = \hat{\alpha}|A\rangle + \hat{\beta}|A\rangle \quad (2.2.3)$$

The distributive axiom of multiplication is thus satisfied. The product of two linear operators is associative, but not generally commutative:

$$\hat{\alpha}\hat{\beta}|A\rangle = \hat{\alpha}(\hat{\beta}|A\rangle) = (\hat{\alpha}\hat{\beta})|A\rangle \quad (2.2.4)$$

and

$$\hat{\alpha}\hat{\beta} \neq \hat{\beta}\hat{\alpha} \quad (\text{in general}) \quad (2.2.5)$$

Definition 2.2.2 (Commutation for Operators)

Two operators $\hat{\eta}$ and $\hat{\xi}$ are said to *commute* if

$$\forall |A\rangle : \hat{\eta}\hat{\xi}|A\rangle = \hat{\xi}\hat{\eta}|A\rangle \quad (2.2.6)$$

$$\iff \hat{\eta}\hat{\xi} = \hat{\xi}\hat{\eta} \quad (2.2.7)$$

Multiplication by a scalar is a special case of a linear operator, most notably in that it commutes with all linear operators.

We may also consider linear operators' behavior when multiplied with a bra on the left. From the definition of the bras, we obtain that

$$(\langle B|\hat{\alpha})|A\rangle = \langle B|(\hat{\alpha}|A\rangle) \rightarrow \langle B|\hat{\alpha}|A\rangle \quad (2.2.8)$$

We may return to the product of a ket on the left and bra on the right, namely $|A\rangle\langle B|$; we observe that it is a linear operator in that it operates linearly with kets on the right and bras on the left:

$$|A\rangle\langle B|P\rangle = |A\rangle \cdot \phi_B(|P\rangle) = \phi_{AB}(|P\rangle) \quad (2.2.9)$$

$$\langle Q|A\rangle\langle B| = \phi_A(\langle Q|) \cdot \langle B| = \phi_{BA}(\langle Q|) \quad (2.2.10)$$

where we use ϕ to represent a linear function.

Axiom 2.2.3 (Operators as Dynamical Variables)

We assume that linear operators correspond to dynamical variables^a at some particular time. We note the most important consequence: these are subject to an algebra in which the commutative axiom of multiplication does **not** hold.

^aExamples of dynamical variables are coordinates, velocity, momentum, and angular momentum, energy, etc.

As a consequence, it is convenient to write dynamical variables and operators interchangeably without notational differences.

2.2.2 Conjugate Relations

Let $\langle P|$ be a bra vector and $\hat{\alpha}$ an operator. Consider the ket $|A\rangle$ that corresponds to $\langle P|\hat{\alpha}$. We know that $|A\rangle$ will depend antilinearly on $\langle P|$ and so it will depend linearly on $|P\rangle$. We can thus consider it the result of a linear operation, denoting that operation as the result of the “adjoint” of $\hat{\alpha}$, denoted by $\hat{\alpha}^\dagger \Leftrightarrow \bar{\alpha}$.

Definition 2.2.4 (Adjoint of a Linear Operator)

Let $\hat{\alpha}$ be a linear operator. We define $\hat{\alpha}^\dagger \Leftrightarrow \bar{\alpha}$, its *Hermitian Adjoint*, according to

$$\langle B|\hat{\alpha}^\dagger|A\rangle = (\langle A|\hat{\alpha}|B\rangle)^* \quad (2.2.11)$$

Proposition 2.2.5 (Adjoint of an Adjoint)

For any linear operator $\hat{\alpha}$, the adjoint of the adjoint is the original operator.

$$(\hat{\alpha}^\dagger)^\dagger = \hat{\alpha} \quad (2.2.12)$$

Proof

Apply the definition of the adjoint to $\hat{\alpha}^\dagger$:

$$\langle B|(\hat{\alpha}^\dagger)^\dagger|A\rangle = (\langle A|\hat{\alpha}^\dagger|B\rangle)^* = ((\langle A|\hat{\alpha}|B\rangle)^*)^* = \langle B|\hat{\alpha}|A\rangle \quad (2.2.13)$$

Since this is true for any bra $\langle B|$ and ket $|A\rangle$,

$$\langle B|(\hat{\alpha}^\dagger)^\dagger = \langle B|\hat{\alpha} \implies (\hat{\alpha}^\dagger)^\dagger = \hat{\alpha} \quad (2.2.14)$$

as desired. $\square$

We thus can call the adjoint the “conjugate transpose linear operator.”

Definition 2.2.6 (Hermitian Operator)

We say that an operator $\hat{\alpha}$ is *Hermitian* or *Self-Adjoint* if

$$\hat{\alpha} = \hat{\alpha}^\dagger \quad (2.2.15)$$

Such operators are said to be “real.”

We write additional properties of linear operators and their adjoints.

Proposition 2.2.7 (Adjoint of a Product of Operators)

Consider a bra $\langle Q|$, ket $|P\rangle$, and two linear operators β, α . We omit the carets for simplicity. Thus,

$$\beta^\dagger \alpha^\dagger = (\alpha\beta)^\dagger \quad (2.2.16)$$

Proof

We write

$$\langle Q|\beta^\dagger \alpha^\dagger|P\rangle = \overline{\langle P|\alpha\beta|Q\rangle} = \langle Q|(\alpha\beta)^\dagger|P\rangle \quad (2.2.17)$$

Since this holds for any ket $|P\rangle$ and any bra $\langle Q|$, we have

$$\beta^\dagger \alpha^\dagger = (\alpha\beta)^\dagger \quad (2.2.18)$$

as desired. $\square$

We note that for any two linear operators ξ, η , even if those operators are real, the composition $\xi\eta$ is not real. However,

$$\overline{\xi\eta + \eta\xi} = \xi\eta + \eta\xi \quad (2.2.19)$$

$$\overline{i(\xi\eta - \eta\xi)} = i(\xi\eta - \eta\xi) \quad (2.2.20)$$

Proposition 2.2.8 (Commuting Operators are Hermitian)

Suppose ξ and η are two linear operators such that $\xi\eta = \eta\xi$ i.e. they commute. Then,

$$\overline{\xi\eta} = \xi\eta \quad (2.2.21)$$

Furthermore, if ξ is real, then

$$\forall n \in \mathbb{N} : \overline{\xi^n} = \xi^n \quad (2.2.22)$$

i.e. ξ^n is real.

We may generalize the adjoint of a product to n terms:

Proposition 2.2.9 (Adjoint of a Product of Linear Operators)

Let $\{\hat{\alpha}_i\}$ be a set of linear operators. Thus,

$$\left(\prod_{i=0}^n \alpha_i \right)^\dagger = \prod_{i=0}^n \alpha_{n-i}^\dagger \quad (2.2.23)$$

Proposition 2.2.10 (Adjoint of the Ket-Bra Linear Operator)

Consider a bra $\langle B|$ and a ket $|A\rangle$. Thus,

$$\overline{|A\rangle\langle B|} = |B\rangle\langle A| \quad (2.2.24)$$

Proof

Let $\langle P|$ be a general bra. Thus, the conjugate imaginary (corresponding ket) of

$$\langle P|(|A\rangle\langle B|) = \langle P|A\rangle\langle B| \quad (2.2.25)$$

is

$$\overline{\langle P|A\rangle}\langle B| = \langle A|P\rangle\langle B| = |B\rangle\langle A|P\rangle = (|B\rangle\langle A|)|P\rangle \quad (2.2.26)$$

The conjugate imaginary (here corresponding bra) of the last expression is

$$\langle P|(\overline{|B\rangle\langle A|}) \quad (2.2.27)$$

and so

$$\langle P|(|A\rangle\langle B|) = \langle P|(\overline{|B\rangle\langle A|}) \Rightarrow |A\rangle\langle B| = \overline{|B\rangle\langle A|} \quad (2.2.28)$$

as desired. $\square$

We may begin to put the pieces together.

Theorem 2.2.11 (Power Annihilation)

Suppose ξ is a real linear operator and

$$\exists |P\rangle : \xi^m |P\rangle = 0 \quad (2.2.29)$$

and $m \in \mathbb{N}$. Then,

$$\xi |P\rangle = 0 \quad (2.2.30)$$

2.3 Eigenvalues, Eigenstates

See Dirac §9 for further reading.

2.3.1 Premise

As for traditional linear algebra, we may study solutions $|P\rangle$ and a to the equation

$$\hat{a}|P\rangle = a|P\rangle \quad (2.3.1)$$

for some operator $\hat{a}$, where we ignore the trivial solution $|P\rangle = 0$. Equations of this kind are called **eigenvalue equations**².

We first notice that for any eigenvector $|P\rangle$ with an eigenvalue a with respect to some operator $\hat{a}$, any scalar multiple of $|P\rangle$ is also an eigenvector with that same eigenvalue.

We may develop theory for real operators.

Proposition 2.3.1 (Eigenvalues of a Hermitian Operator are Real)

Suppose ξ is a real operator. All eigenvalues of this operator are real.

Proof

We multiply by the eigenvector on the left:

$$\xi|P\rangle = a|P\rangle \Rightarrow \langle P|\xi|P\rangle = a\langle P|P\rangle \quad (2.3.3)$$

We take the adjoint:

$$\langle P|\xi^\dagger|P\rangle = \overline{\langle P|\xi|P\rangle} = \overline{a\langle P|P\rangle} = \bar{a}\langle P|P\rangle \quad (2.3.4)$$

We know that $\xi^\dagger = \xi$, and so $\langle P|\xi^\dagger|P\rangle = \langle P|\xi|P\rangle$, which implies $\bar{a} = a$. $\square$

Thus, $\langle P|\xi = a\langle P|$ if $\xi|P\rangle = a|P\rangle$.

Property 2.3.2 (Notation of Eigenvalues)

Let ξ be a real linear operator. The eigenvalues of ξ are denoted ξ' , ξ'' , ξ^r , etc. We write eigenkets $|\xi'\rangle$; should two eigenkets have the same eigenvalues, then we write them as $|\xi'1\rangle$, $|\xi'2\rangle$, etc.

Since the linear operators are in fact dynamical variables, we can call the eigenvectors or eigenkets as *eigenstates*.

Theorem 2.3.3 (Orthogonality of Eigenkets of a Hermitian Operator)

Two eigenvectors of a real dynamical variable belonging to different eigenvalues are orthogonal.

Proof

Let $|\xi'\rangle$ and $|\xi''\rangle$ be two eigenkets of the real dynamical variable ξ , belonging to the eigenvalues ξ' and ξ'' respectively. In equation form,

$$\xi|\xi'\rangle = \xi'|\xi'\rangle \quad (2.3.5)$$

$$\xi|\xi''\rangle = \xi''|\xi''\rangle \quad (2.3.6)$$

We take the adjoint of the first equation to obtain

$$\langle \xi'|\xi = \xi'\langle \xi'| \quad (2.3.7)$$

We multiply by $|\xi''\rangle$ on the right to obtain

$$\langle \xi'|\xi|\xi''\rangle = \xi'\langle \xi'|\xi''\rangle \quad (2.3.8)$$

²The most well known eigenvalue equation is perhaps

$$\frac{d}{dx}f(x) = f(x) \quad (2.3.2)$$

for some function f , the solution of which is of course the exponential function. This equation seeks the eigenfunction (equivalent to eigenvector) of the differentiation operator, with the eigenvalue 1. The solution to the problem with the eigenvalue a is of course $\exp(ax)$.

Multiplying the second eigenvalue equation on the left by $\langle \xi' |$ gives

$$\langle \xi' | \xi | \xi'' \rangle = \xi'' \langle \xi' | \xi'' \rangle \quad (2.3.9)$$

We may subtract the last two equations to obtain

$$(\xi' - \xi'') \langle \xi' | \xi'' \rangle = \langle \xi' | \xi | \xi'' \rangle - \langle \xi' | \xi | \xi'' \rangle = 0 \quad (2.3.10)$$

which requires either $\langle \xi' | \xi'' \rangle = 0$ or $\xi' = \xi''$, as desired. $\square$

We have not established the existence of eigenvector/eigenvalue pairs. In general this is intractable, but a special case in which the real linear operator ξ satisfies a “characteristic polynomial” can allow for certain conclusions.

Proposition 2.3.4 (Eigenvalues from a Characteristic Polynomial)

Suppose ξ is a real linear operator that satisfies the following polynomial,

$$\phi(\xi) = \xi^n + a_1 \xi^{n-1} + a_2 \xi^{n-2} + \cdots + a_n = 0 \quad (2.3.11)$$

where a_i are all scalars (possibly complex). Suppose that ϕ is the simplest equation that ξ can satisfy. It can thus be shown that

1. The number of eigenvalues of ξ is n .
2. The eigenkets of ξ form a complete basis for the space of all eigenkets.

Proof

We may begin by factorizing the characteristic polynomial into the form

$$\phi(\xi) = (\xi - c_1)(\xi - c_2) \cdots (\xi - c_n) \quad (2.3.12)$$

Let $\chi_r(\xi)$ be the remainder of division of $\phi(\xi)$ by $(\xi - c_r)$, so that

$$\forall r \in \{1, 2, \dots, n\} : \phi(\xi) = (\xi - c_r)\chi_r(\xi) \quad (2.3.13)$$

Thus, for any ket $|P\rangle$,

$$(\xi - c_r)\chi_r(\xi)|P\rangle = \phi(\xi)|P\rangle = 0 \quad (2.3.14)$$

Since $\chi_r(\xi)|P\rangle$ cannot vanish for every $|P\rangle$ (since this would imply that ξ satisfies a polynomial of degree $n-1$ rather than n). Consider a $|P\rangle$ for which $\chi_r(\xi)|P\rangle$ does not vanish; this would imply that $\chi_r(\xi)|P\rangle$ is a eigenket of ξ with the eigenvalue c_r . Thus, we have proven that there are at least n eigenvalues of ξ .

To prove that none other than the c_r are possible eigenvalues, suppose that ξ' is an eigenvalue with a corresponding eigenket of $|\xi'\rangle$, and so

$$\xi|\xi'\rangle = \xi'|\xi'\rangle \quad (2.3.15)$$

$$\Rightarrow \phi(\xi)|\xi'\rangle = \phi(\xi')|\xi'\rangle \quad (2.3.16)$$

Since the left hand side of the second equation is zero, we must have $\phi(\xi') = 0$ and so ξ' is one of the c_r .

We prove that all the roots are distinct. Suppose that a root c_s has a multiplicity m ; thus, the polynomial $\phi'(\xi)$ defined by

$$\phi'(\xi) = \frac{\phi(\xi)}{(\xi - c_s)^{m-1}} \quad (2.3.17)$$

will have the same number of distinct eigenvalues, and is a simpler polynomial than ϕ , but ξ still satisfies it. We have thus proved the first point.

The proof of the second point is as follows. Let $\chi_r(c_r)$ be the number obtained when c_r is substituted for ξ in the algebraic expression $\chi_r(\xi)$. Since each c_i is different, $\chi_r(c_r)$ cannot vanish. Consider the following expression:

$$\sum_r \frac{\chi_r(\xi)}{\chi_r(c_r)} - 1 \quad (2.3.18)$$

If we replace ξ by c_s , each term with $r \neq s$ will have a zero numerator, and the term with $r = s$ will have value 1, since the numerator will be equal to the denominator. Thus, the expression evaluates to 0 for $\xi = c_s$ for each $s \in \{1 \cdots n\}$.

We can thus apply an arbitrary ket:

$$|P\rangle = \sum_r \frac{1}{\chi_r(c_r)} \chi_r(\xi) |P\rangle \quad (2.3.19)$$

Thus, we have expressed an arbitrary ket $|P\rangle$ in terms of eigenkets of ξ : each $\chi_r(\xi)|P\rangle$ is an eigenket with eigenvalue c_r , as shown previously. Thus, the theorem is proved. $\square$

2.4 Observables

See Dirac §10, 11, and 12 for further reading.

2.4.1 From Operators Back to Dynamical Variables

We must naturally restrict the value of the measurement of a dynamical variable to a real one, since a complex one would require two measurements, between which the state might change as a result of the observation.

Axiom 2.4.1 (Eigenstates of Dynamical Variables)

If a dynamical system is in an eigenstate of a real dynamical variable ξ , belonging to an eigenvalue ξ' , then a measurement of ξ will certainly give as result the number ξ' . Conversely, if the system is in a state such that the measurement of a real dynamical variable ξ is certain to give one particular result, then the state is an eigenstate of ξ and the result of the measurement is the eigenvalue of ξ to which this eigenstate belongs.

When a measurement of a real dynamical variable ξ is made, experiment has shown that this collapses a superposition of states into the one associated with the measured value.

Proposition 2.4.2 (Possible Measurement Values)

Any result of a measurement of a real dynamical variable is one of its eigenvalues. Every eigenvalue of a dynamical variable is a possible result of a measurement of that variable for some state of the system.

Proposition 2.4.3 (Measureability of a Dynamical Variable)

In order to be measurable, the set of eigenvectors of a dynamical variable must form a basis, also called a “complete set”.

Definition 2.4.4 (Observable)

A real dynamical variable is said to be an *observable* if and only if its eigenstates form a complete set.

We may mathematically characterize this definition. The eigenvalues of a real dynamical variable ξ may consist of a finite or infinite discrete set of numbers, or all of the numbers in some range. In the discrete case, the dependence of a state on the eigenstates of ξ is one of a linear combination, i.e.

$$\exists \{c_r\} : |P\rangle = \sum_r c_r |\xi^r\rangle \quad (2.4.1)$$

In the continuous case, we use integration:

$$|P\rangle = \int f(\xi') |\xi' c\rangle d\xi' + \sum_r c_r |\xi^r d\rangle \quad (2.4.2)$$

where $f(\xi')$ is a complex-valued function defined over the continuous eigenvalues ξ' . We note that generally it is quite difficult to prove a quantity is an observable, and so “observability” is assumed for some quantities such as energy.

Proposition 2.4.5 (Observability of the Ket-Bra Product)

Let $|A\rangle$ be a normalized ket. The operator $|A\rangle\langle A|$ is thus an observable with eigenvalues 0 and 1 with $|A\rangle$ corresponding to 1 and all other kets orthogonal to $|A\rangle$ corresponding to 0.

Proof

We observe that the square of the described operator is the original operator:

$$(|A\rangle\langle A|)^2 = |A\rangle\langle A|A\rangle\langle A| = |A\rangle\langle A| \quad (2.4.3)$$

Thus, the algebraic equation

$$\xi^2 - \xi = 0 \quad (2.4.4)$$

is satisfied, and so the eigenvalues are 1 and 0. One quickly observes that $|A\rangle$ is an eigenvector with eigenvalue 1, and that any ket $|P\rangle$ such that $\langle A|P\rangle = 0$ is an eigenvector with eigenvalue 0, as desired. $\square$

Suppose we now have an infinite number of eigenvalues which lie in some domain D . Suppose that for $|\xi'x\rangle \neq 0$, Thus,

$$\int \langle \xi'x | \xi''x \rangle d\xi'' > 0 \quad (2.4.5)$$

2.4.2 Functions of Observables

Let ξ be an observable. Should we create a new observable $k\xi$ where k is a real number, we would measure $k\xi'$ with the new observable if we would have measured ξ' with the original one. We may of course generalize to any function $f(\xi)$ where we would measure $f(\xi')$ in the case of measurement of ξ' from ξ .

Let us define $f(\xi)$ in general as that linear operator satisfying

$$f(\xi)|\xi'\rangle = f(\xi')|\xi'\rangle \quad (2.4.6)$$

for every eigenket $|\xi'\rangle$ of ξ and $f(\xi')$ is a number (not necessarily real). This function will also satisfy the equivalent of Eq. (2.4.2):

$$f(\xi)|P\rangle = \int f(\xi')|\xi'c\rangle d\xi' + \sum_r f(\xi^r)|\xi^r d\rangle \quad (2.4.7)$$

We may define the conjugate transpose of $f(\xi)$, namely $\overline{f(\xi)}$, according to

$$\langle \xi' | \overline{f(\xi)} = \bar{f}(\xi') \langle \xi' | \quad (2.4.8)$$

Proposition 2.4.6 (Functions of an Observable)

Any function of $f(\xi)$ of an observable ξ is also an observable.

Proof

Thus, for an arbitrary $|P\rangle$:

$$\langle \xi'' | \overline{f(\xi)} | P \rangle = \bar{f}(\xi'') \langle \xi'' | P \rangle \quad (2.4.9)$$

$$= \int \bar{f}(\xi'') \langle \xi'' | \xi' c \rangle d\xi' + \sum_r \bar{f}(\xi'') \langle \xi'' | \xi^r d \rangle \quad (2.4.10)$$

$$= \int \bar{f}(\xi'') \langle \xi'' | \xi' c \rangle d\xi' + \bar{f}(\xi'') \langle \xi'' | \xi'' d \rangle \quad (2.4.11)$$

Similarly,

$$\langle \xi'' | \bar{f}(\xi) | P \rangle = \int \bar{f}(\xi') \langle \xi'' | \xi' c \rangle d\xi' + \bar{f}(\xi'') \langle \xi'' | \xi'' d \rangle \quad (2.4.12)$$

Since the integrands vanish for $\xi' \neq \xi''$, we have that the integrands of the last two equations are the same and so

$$\langle \xi'' | \overline{f(\xi)} | P \rangle = \langle \xi'' | \bar{f}(\xi) | P \rangle \quad (2.4.13)$$

which implies that

$$\overline{f(\xi)} = \bar{f}(\xi) \quad (2.4.14)$$

which then implies that f is real; since each eigenstate of ξ is also an eigenstate of $f(\xi)$. Thus, $f(\xi)$ is an observable, as desired. $\square$

Definition 2.4.7 (Inverse of an Observable)

Suppose that α is an observable where 0 is not an eigenvalue of α . Thus, the inverse α^{-1} exists and satisfies

$$\alpha^{-1}|\alpha'\rangle = (\alpha')^{-1}|\alpha'\rangle \quad (2.4.15)$$

We also have

$$\alpha\alpha^{-1} = \alpha^{-1}\alpha = 1 \quad (2.4.16)$$

Proposition 2.4.8 (Inverse of a Composition of Operators)

The inverse of a product $\alpha\beta$ of operators is

$$(\alpha\beta)^{-1} = \beta^{-1}\alpha^{-1} \quad (2.4.17)$$

Definition 2.4.9 (Square Root of an Observable)

The square root of an observable α always exists; it is real if α has no negative eigenvalues. It satisfies

$$\sqrt{\alpha}|\alpha'\rangle = \pm\sqrt{\alpha'}|\alpha'\rangle \quad (2.4.18)$$

Property 2.4.10 (Composition of Square Roots)

Let α be an observable. Thus, the composition of the square root of α with itself gives α back: $\sqrt{\alpha}\sqrt{\alpha} = \alpha$.

2.4.3 Expectation of an Observable

Let ξ be an observable. In classical mechanics, the observable is simply a scalar that one measures; in quantum mechanics, we can loosely say that for a state $|x\rangle$, the observable has a value given by

$$\langle \xi \rangle \equiv \langle x | \xi | x \rangle \quad (2.4.19)$$

which is uniquely determined if $|x\rangle$ is normalized (any phase factor will cancel). We clarify below.

Definition 2.4.11 (Expectation of an Observable)

Let ξ be an observable. We have previously established that measurements of ξ must yield eigenvalues of ξ only. We can find the **average value** of such a measurement should it be conducted an infinite number of times on a state $|x\rangle \equiv \sum c_j |\xi^r\rangle$ according to

$$\langle \xi \rangle \equiv \langle x | \xi | x \rangle = \sum |c_j|^2 \xi^r |\xi^r\rangle \quad (2.4.20)$$

We note that it is not guaranteed that one may ever measure $\langle \xi \rangle$: if it is not an eigenvalue, it cannot be measured.

Proposition 2.4.12 (Probability of Measuring an Particular Eigenvalue)

Consider an observable ξ and a state $|x\rangle \equiv \sum c_j |\xi^r\rangle$. The probability of measuring the eigenvalue ξ^r is given by $|c_r|^2$.

Proof

Let a be some real number. Let $f(\xi)$ be $\delta_{\xi r}$ (the unit tensor indexed by ξ and r) as defined in the mathematical prerequisites section. Thus, the average value of this function is just the probability of ξ having the value r :

$$\langle \delta_{\xi r} \rangle = \langle x | \delta_{\xi r} | x \rangle = |c_r|^2 \quad (2.4.21)$$

as desired. $\square$

Definition 2.4.13 (Probability Density)

Consider an observable ξ with an infinite number of eigenvalues in some range. The probability density $\psi(a)$ is given by

$$\psi(a)da \equiv \langle x | \chi(\xi) | x \rangle \quad (2.4.22)$$

where $\chi(\xi)$ is the function having the value 1 if $\xi \in [a, a + da]$ and 0 otherwise.

2.4.4 Representations

See Dirac §20 for further reading.

Definition 2.4.14 (Representative of a Linear Operator)

A representative of a linear operator is found by expliciting its effects on an arbitrary ket in terms of projections onto its eigenkets.

Definition 2.4.15 (Projection Operator)

For a basis of kets $|n'\rangle$, one finds the coefficient $c_{n'}$ in the linear combination of such kets describing some arbitrary state ψ according to

$$c_{n'} |n'\rangle = P_n |\psi\rangle \quad (2.4.23)$$

where $P_n \equiv |n'\rangle\langle n'|$ is the *projection operator*.

Concretely, let ξ be an observable. Let $f(\xi)$ be any function of this observable. Let $|P\rangle$ be an arbitrary ket. We can formalize the wavefunction representation of a ket according to the probability density function definition, recalling that inner products are effectively projections, and so we find the probability of a state being in some eigenstate ξ' of an observable ξ by calculating that inner product:

$$\psi(\xi') \equiv \langle \xi' | P \rangle \quad (2.4.24)$$

and so we can write

$$|P\rangle = |\psi(\xi)\rangle \quad (2.4.25)$$

Thus,

$$\langle \xi' | f(\xi) | P \rangle = f(\xi') \psi(\xi') \quad (2.4.26)$$

Any ket thus can be represented as a function of ξ s multiplied into a *standard ket* $|\rangle$, which satisfies

$$\langle \xi' | \rangle = 1 \quad \forall \xi' \quad (2.4.27)$$

In this vein, if $|\xi''\rangle$ is an eigenket of some observable ξ , we write it according to

$$|\xi''\rangle = \delta_{\xi_1 \xi_1''} \dots \delta_{\xi_v \xi_v''} \delta(\xi_{v+1} - \xi_{v+1}'') \dots \delta(\xi_u - \xi_u'') \quad (2.4.28)$$

where $\xi_1 \dots \xi_v$ are the discrete eigenvalues of ξ and $\xi_{v+1} \dots \xi_u$ are the continuous eigenvalues and the standard ket is written $|\rangle$ rather than $|\rangle$.

Property 2.4.16 (Expected Value Shortcut)

Let ξ be an observable. Thus, $\langle \xi^2 \rangle \equiv \langle \psi | \xi^2 | \psi \rangle = |\xi | \psi \rangle|^2$.

2.5 Commutators

See Dirac §13, 21 for further reading.

2.5.1 Commutability, Compatibility

We recall that two operators ξ, η commute if

$$\xi\eta = \eta\xi \quad (2.5.1)$$

Definition 2.5.1 (Simultaneous Eigenstate)

Consider two observables ξ, η . A ket $|A\rangle$ is an *simultaneous eigenstate* of ξ and η if

$$\exists \xi' : \xi|A\rangle = \xi'|A\rangle \iff \exists \eta' : \eta|A\rangle = \eta'|A\rangle \quad (2.5.2)$$

Proposition 2.5.2 (Simultaneous Eigenstates of Commuting Observables)

Consider two observables ξ and η . If ξ and η commute, then there exist simultaneous eigenstates, and they form a basis.

Proof

Let ξ and η be two commuting observables. Thus, the eigenkets of either form a basis. We write an arbitrary eigenket of η , $|\eta'\rangle$, in terms of the eigenkets of ξ :

$$|\eta'\rangle = \int |\xi'\eta'c\rangle d\xi' + \sum_r |\xi^r\eta'd\rangle \quad (2.5.3)$$

We write

$$0 = (\eta - \eta')|\eta'\rangle \quad (2.5.4)$$

$$= \int (\eta - \eta')|\xi'\eta'c\rangle d\xi' + \sum_r (\eta - \eta')|\xi^r\eta'd\rangle \quad (2.5.5)$$

We multiply the summand by ξ on the left:

$$\xi(\eta - \eta')|\xi^r\eta'd\rangle = (\eta - \eta')\xi|\xi^r\eta'd\rangle = (\eta - \eta')\xi^r|\xi^r\eta'd\rangle = \xi^r(\eta - \eta')|\xi^r\eta'd\rangle \quad (2.5.6)$$

Thus, it is an eigenket of ξ with the eigenvalue ξ^r . Thus, the integrand is similarly an eigenket of ξ with the eigenvalue ξ' . The expression in Eq. (2.5.5) must equal zero, and so the only way this is possible is if every term is zero, and so

$$(\eta - \eta')|\xi'\eta'c\rangle = 0 \Rightarrow \eta|\xi'\eta'c\rangle = \eta'|\xi'\eta'c\rangle \quad (2.5.7)$$

and so the eigenkets of ξ are also eigenkets of η , as desired. $\square$

Proposition 2.5.3 (Commutation from a Basis of Simultaneous Eigenstates)

If ξ and η are two observables such that their simultaneous eigenstates form a complete set (basis), then ξ and η commute. This is the converse of the previous proposition.

Proof

Let $|\xi'\eta'\rangle$ be a simultaneous eigenket; thus

$$(\xi\eta - \eta\xi)|\xi'\eta'\rangle = (\xi'\eta' - \eta'\xi')|\xi'\eta'\rangle = 0 \quad (2.5.8)$$

We can expand an arbitrary ket $|P\rangle$ in terms of the simultaneous eigenstates; thus,

$$(\xi\eta - \eta\xi)|P\rangle = 0 \quad (2.5.9)$$

Since $|P\rangle$ was arbitrary, we have

$$\xi\eta = \eta\xi \quad (2.5.10)$$

and so they commute, as desired. $\square$

Property 2.5.4 (Observation of Commuting Dynamical Variables)

In the special case when the two observables commute, the observations are to be considered as non-interfering or compatible, in such a way that one can give a meaning to the two observations being made simultaneously and can discuss the probability of any particular results being obtained.

From the point of view of general theory, any two or more commuting observables may be counted as a single observable, the result of a measurement of which consists of two or more numbers.

2.5.2 Comparison to Poisson Brackets

We briefly review the language of classical mechanics (Lagrangian, Hamiltonian): we work with generalized coordinates q_r and generalized momenta p_r , also called canonical coordinates and momenta.

Definition 2.5.5 (Poisson Bracket)

In classical mechanics, the *Poisson Bracket* of any two dynamical variables u and v is given by

$$[u, v] \equiv \sum_r \left(\frac{\partial u}{\partial q_r} \frac{\partial v}{\partial p_r} - \frac{\partial u}{\partial p_r} \frac{\partial v}{\partial q_r} \right) \quad (2.5.11)$$

Poisson brackets of any two classical dynamical variables exist.

Property 2.5.6 (Properties of Poisson Brackets)

Let u and v be two classical dynamical variables. We can write

$$[u, v] = -[v, u] \quad (2.5.12)$$

Let c be a constant. Thus,

$$[u, c] = 0 \quad (2.5.13)$$

The Poisson Brackets are bilinear:

$$[u_1 + u_2, v] = [u_1, v] + [u_2, v] \quad (2.5.14)$$

$$[u, v_1 + v_2] = [u, v_1] + [u, v_2] \quad (2.5.15)$$

$$[u_1 u_2, v] = [u_1, v] u_2 + u_1 [u_2, v] \quad (2.5.16)$$

$$[u, v_1 v_2] = [u, v_1] v_2 + v_1 [u, v_2] \quad (2.5.17)$$

Property 2.5.7 (Jacobi Identity)

Let u, v, w be three classical dynamical variables. Thus,

$$[u, [v, w]] + [v, [w, u]] + [w, [u, v]] = 0 \quad (2.5.18)$$

We wish to introduce a quantum version of the Poisson Brackets. We may express $[u_1 u_2, v_1 v_2]$ in two different ways:

$$[u_1 u_2, v_1 v_2] = [u_1, v_1 v_2] u_2 + u_1 [u_2, v_1 v_2] \quad (2.5.19)$$

$$= ([u_1, v_1] v_2 + v_1 [u_1, v_2]) u_2 + u_1 ([u_2, v_1] v_2 + v_1 [u_2, v_2]) \quad (2.5.20)$$

$$= [u_1, v_1] v_2 u_2 + v_1 [u_1, v_2] u_2 + u_1 [u_2, v_1] v_2 + u_1 v_1 [u_2, v_2] \quad (2.5.21)$$

and

$$[u_1 u_2, v_1 v_2] = [u_1 u_2, v_1] v_2 + v_1 [u_1 u_2, v_2] \quad (2.5.22)$$

$$= [u_1, v_1] u_2 v_2 + u_1 [u_2, v_1] v_2 + v_1 [u_1, v_2] u_2 + v_1 u_1 [u_2, v_2] \quad (2.5.23)$$

These two are equal and so we can write

$$[u_1, v_1] (u_2 v_2 - v_2 u_2) = (u_1 v_1 - v_1 u_1) [u_2, v_2] \quad (2.5.24)$$

We have chosen the dynamical variables arbitrarily. Thus,

$$u_1 v_1 - v_1 u_1 = i\hbar[u_1, v_1] \quad \text{and} \quad u_2 v_2 - v_2 u_2 = i\hbar[u_2, v_2] \quad (2.5.25)$$

where $\hbar$ is a real number independent of any operator/dynamical variable.

Definition 2.5.8 (Commutator)

The *commutator* of any two operators in quantum mechanics is given by

$$[u, v] = uv - vu \quad (2.5.26)$$

The quantum Poisson Bracket $[u, v]$ is defined such that

$$[u, v] = uv - vu = i\hbar[u, v] \quad (2.5.27)$$

Proposition 2.5.9 (Quantum PB Properties)

The Quantum Poisson Bracket, $[u, v]$, satisfies all of the properties enumerated for classical Poisson Brackets.

We now build the bridge from classical mechanics to quantum mechanics.

Proposition 2.5.10 (Canonical Coordinates/Momenta PBs)

Let $\{q_r\}$ be a set of canonical coordinates, and $\{p_r\}$ the corresponding canonical momenta. Thus,

$$[q_r, q_s] = 0 \quad [p_r, p_s] = 0 \quad [q_r, p_s] = \delta_{rs} \quad (2.5.28)$$

We write the equivalent for quantum mechanics:

Law 2.5.11 (Fundamental Quantum Conditions)

Let $\{q_r\}$ be the set of dynamical variables corresponding to coordinates let $\{p_r\}$ be the set of dynamical variables corresponding to momenta. Thus,

$$[q_r, q_s] = 0 \quad [p_r, p_s] = 0 \quad [q_r, p_s] = i\hbar\delta_{rs} \quad (2.5.29)$$

Proposition 2.5.12 (CM as a Limiting Case of QM)

Classical Mechanics can be regarded as the limiting case of Quantum Mechanics as $\hbar \rightarrow 0$.

Axiom 2.5.13 (Independence of Degrees of Freedom)

While we have shown that for quantum systems with classical analogues, dynamical variables referring to different degrees of freedom commute, we assume this to hold generally.

Theorem 2.5.14 (Uncertainty Product from the Commutator)

Let $|\psi\rangle$ be a quantum state. Let $\hat{A}, \hat{B}$ be two observables. Thus,

$$(\Delta A)_\psi^2 (\Delta B)_\psi^2 \geq \left\langle \frac{1}{4} \left| [\hat{A}, \hat{B}] \right|^2 \right\rangle_\psi \quad (2.5.30)$$

2.6 Unitary Transformations

Let U be a linear operator with a reciprocal U^{-1} . For α an arbitrary linear operator, let us define

$$\tilde{\alpha} \equiv U\alpha U^{-1} \quad (2.6.1)$$

We wish to characterize those U that transform any real α into a real $\tilde{\alpha}$. From the similarity transform, we have

$$\tilde{\alpha}U = U\alpha \Rightarrow \overline{U}\tilde{\alpha} = \alpha\overline{U} \Rightarrow \overline{U}\tilde{\alpha}U = \alpha\overline{U}U \quad (2.6.2)$$

From the first expression above, we have

$$\overline{U}(\tilde{\alpha}U) = \overline{U}U\alpha \quad (2.6.3)$$

Thus,

$$\overline{U}U\alpha = \alpha\overline{U}U \quad (2.6.4)$$

which implies that $\overline{U}U$ commutes with any linear operator, since any linear operator can be expressed as a linear combination of two real ones (first added to i times the second). Thus, $\overline{U}U$ must be a number. It must be a real number, since

$$\overline{\overline{U}U} = \overline{U}\overline{\overline{U}} = \overline{U}U \quad (2.6.5)$$

We can suppose this number is 1 without any loss of generality.

Definition 2.6.1 (Unitary Transformation)

An operator U is said to be *unitary* if it satisfies the following four properties:

$$\overline{U}U = 1 \quad U = \overline{U}^{-1} \quad \overline{U} = U^{-1} \quad U^{-1}\overline{U}^{-1} = 1 \quad (2.6.6)$$

These are equivalent to

$$\overline{\alpha} = \alpha \implies \tilde{\tilde{\alpha}} = \alpha \quad (2.6.7)$$

where

$$\tilde{\alpha} \equiv U\alpha U^{-1} \quad (2.6.8)$$

Let us explore some properties.

Property 2.6.2 (Properties of Similarity Transformations)

Let U be a linear operator. Let operators with a tilde be those transformed according to

$$\tilde{\alpha} = U\alpha U^{-1} \quad (2.6.9)$$

Thus,

- The eigenvalues of α are the same as those of $\tilde{\alpha}$, though the eigenkets are transformed according to

$$|\tilde{\alpha}'\rangle = U|\alpha'\rangle \quad (2.6.10)$$

- $\widetilde{\alpha_1 + \alpha_2} = \tilde{\alpha}_1 + \tilde{\alpha}_2$

- $\widetilde{\alpha_1\alpha_2} = \tilde{\alpha}_1\tilde{\alpha}_2$

Proof

From the premise,

$$\tilde{\alpha} = U\alpha U^{-1} \quad (2.6.11)$$

Thus,

$$\alpha|\alpha'\rangle = \alpha'|\alpha'\rangle \Rightarrow \tilde{\alpha}U|\alpha'\rangle = U\alpha U^{-1}U|\alpha'\rangle = U\alpha|\alpha'\rangle = \alpha'U|\alpha'\rangle \quad (2.6.12)$$

and so $U|\alpha'\rangle$ is an eigenket of $\tilde{\alpha}$ with the same eigenvalue α' . The distribution into addition and multiplication is quickly shown:

$$\widetilde{\alpha_1 + \alpha_2} = U(\alpha_1 + \alpha_2)U^{-1} = U\alpha_1 U^{-1} + U\alpha_2 U^{-1} = \tilde{\alpha}_1 + \tilde{\alpha}_2 \quad (2.6.13)$$

$$\widetilde{\alpha_1 \alpha_2} = U\alpha_1 \alpha_2 U^{-1} = U\alpha_1 U^{-1} U\alpha_2 U^{-1} = \tilde{\alpha}_1 \tilde{\alpha}_2 \quad (2.6.14)$$

as desired. $\square$

Proposition 2.6.3 (Classical Analogue of Unitary Transformations)

Unitary transformations in quantum mechanics are the analogue of contact transformations in the classical theory.

Property 2.6.4 (Properties of Unitary Transformations)

Let U be a unitary transformation, let α be any operator.

- U^{-1} is also unitary.
- The successive application of two unitary transformations is also unitary.

We are now fully mathematically equipped to describe elementary QM.

3 Classical to Quantum

3.1 Classical Bohr Model

We begin with a thought experiment: consider the Bohr model of a Hydrogen atom, but classically. First, classical mechanics provides us the following tool:

Theorem 3.1.1 (Virial)

Let there be a system of N particles with kinetic energy K and potential energy U , where U is homogeneous of degree k . Then,

$$2\overline{K} = k\overline{U} = \overline{\sum_a^N \mathbf{r}_a \cdot \frac{\partial U}{\partial \mathbf{r}_a}} \quad (3.1.1)$$

where the bar denotes averaging over an infinite time period.

In the Bohr model, we consider the electrostatic potential energy field $V = -e^2/r$. The relation between the kinetic energy and potential energy is thus given by the Virial Theorem according to $k = -1$:

$$\overline{K} = -E \quad \text{and} \quad \overline{V} = 2E \quad (3.1.2)$$

where E is the total energy $\overline{K} + \overline{V}$.

Let us suppose that the magnitude ℓ of the electron's angular momentum is quantized, equal to $n\hbar$ for some integer n .

Proposition 3.1.2 (Bohr Radii, Energy Levels)

The quantized values for the total energy and associated orbit radii r_n are given by

$$r_n = \frac{n^2 \lambda}{\alpha} \quad (3.1.3)$$

and

$$E_n = -\frac{E_e}{2n^2 \alpha^2} \quad (3.1.4)$$

where $\alpha \equiv e^2/\hbar c$ is the fine structure constant, $E_e \equiv m_e c^2$ is the electron rest energy, $\lambda \equiv \hbar/(m_e c)$.

Proof

Let $m \equiv m_e$ for legibility. We begin with the definition of the kinetic energy:

$$K = \frac{mv^2}{2} \quad (3.1.5)$$

In terms of angular momentum ℓ , this is

$$K = \frac{\ell^2}{2mr_n^2} \quad (3.1.6)$$

This is equal to $-\overline{V}/2$ according to the Virial earlier stated:

$$\frac{\ell^2}{2mr_n^2} = -\frac{1}{2} \left(-\frac{e^2}{r_n} \right) = \frac{e^2}{2r_n} \quad (3.1.7)$$

We rearrange and open $\ell \equiv n\hbar$:

$$r_n = \frac{n^2 \hbar^2}{me^2} \quad (3.1.8)$$

We can now express this in terms of the desired quantities:

$$\boxed{r_n} = \frac{n^2 \hbar^2}{me^2} = n^2 \frac{\hbar}{mc} \cdot \frac{mc}{1} \cdot \frac{\hbar}{me^2} = n^2 \lambda \cdot \frac{\hbar c}{e^2} = \boxed{\frac{n^2 \lambda}{\alpha}} \quad (3.1.9)$$

The energy E_n is given simply by

$$\boxed{E_n} = \frac{V}{2} = -\frac{e^2}{2r_n} = -\frac{e^2}{2} \cdot \frac{me^2}{n^2\hbar^2} = -\frac{mc^2e^4}{2n^2c^2\hbar^2} = \boxed{-\frac{E_e}{2n^2\alpha^2}} \quad (3.1.10)$$

as desired. $\square$

We try to find why the classical description is not enough by finding the decay time of an electron orbiting a proton in the Bohr model.

Proposition 3.1.3 (Bohr Atom: Lost Energy per Orbit)

Assuming a circular orbit, for a non-relativistic electron, the energy $|\Delta E|$ lost per revolution is small compared to the electron's kinetic energy K ; the ratio $|\Delta E|/K$ is given by

$$\frac{|\Delta E|}{K} = \frac{8\pi}{3} \frac{v^3}{c^3} \quad (3.1.11)$$

Proof

A non-relativistic, accelerating electron radiates energy at a rate given by the Larmor formula:

$$\dot{E} = -\frac{2}{3} \frac{e^2 a^2}{c^3} \quad (3.1.12)$$

A circular orbit gives

$$a = \frac{v^2}{r} = \frac{F}{m} = -\frac{e^2}{mr^2} \quad (3.1.13)$$

Thus,

$$\dot{E} = -\frac{2}{3} \frac{e^2}{c^3} \left(-\frac{e^2}{mr^2} \right)^2 = -\frac{2}{3} \frac{e^2}{c^3} \frac{e^4}{m^2 r^4} = -\frac{2}{3} \frac{e^6}{m^2 c^3 r^4} \quad (3.1.14)$$

The period is given by

$$T = \frac{2\pi r}{v} \quad (3.1.15)$$

where

$$v = \sqrt{\frac{e^2}{mr}} \quad (3.1.16)$$

which gives

$$T = 2\pi r \sqrt{\frac{mr}{e^2}} \quad (3.1.17)$$

Thus,

$$\Delta E = \dot{E}T = -\frac{2}{3} \frac{e^6}{m^2 c^3 r^4} \cdot 2\pi r \sqrt{\frac{mr}{e^2}} = \frac{4\pi e^5 m^{1/2} r^{3/2}}{3m^2 c^3 r^4} = \frac{4\pi}{3} \frac{e^5}{m^{3/2} c^3 r^{5/2}} \quad (3.1.18)$$

The kinetic energy K is given by

$$K = \frac{mv^2}{2} = \frac{m}{2} \frac{e^2}{mr} = \frac{e^2}{2r} \quad (3.1.19)$$

and so the desired ratio is

$$\frac{\Delta E}{K} = \frac{4\pi}{3} \frac{e^5}{m^{3/2} c^3 r^{5/2}} \cdot \frac{2r}{e^2} = \frac{8\pi}{3} \cdot \frac{e^3}{m^{3/2} c^3 r^{3/2}} = \boxed{\frac{8\pi}{3} \frac{v^3}{c^3}} \quad (3.1.20)$$

Proposition 3.1.4 (Bohr Atom Classical Decay Time)

The classical decay time for a Bohr atom from a radius of 50 pm to 1 pm is roughly 1.3×10^{-11} s, decaying at a non-relativistic speed.

Proof

We find the time elapsed:

$$\Delta t = \int dt = \int \left(\frac{dr}{dt} \right)^{-1} dr \quad (3.1.21)$$

The differential $\dot{r}$ is obtained from

$$\dot{E} = \frac{dE}{dr} \dot{r} \Rightarrow \dot{r} = -\frac{4}{3} \frac{e^4}{m^2 c^3} \frac{1}{r^2} \quad (3.1.22)$$

Thus,

$$\boxed{\Delta t} = \int -\frac{3}{4} \frac{m^2 c^3}{e^4} r^2 dr = -\frac{3}{4} \frac{m^2 c^3}{e^4} \left(\frac{r^3}{3} \right) \Big|_{r_0}^{r_f} = \boxed{\frac{m_e^2 c^3}{4e^4} (r_0^3 - r_f^3)} \quad (3.1.23)$$

For 50 pm $\rightarrow$ 1 pm, we have

$$\boxed{\Delta t = 1.3 \times 10^{-11} \text{ s}} \quad (3.1.24)$$

This gives

$$\dot{r} \simeq \frac{49 \times 10^{-12}}{1.3 \times 10^{-11}} \approx 3.77 \frac{\text{m}}{\text{s}} \ll c \quad (3.1.25)$$

which is **non-relativistic.**

3.1.1 Mach-Zehnder Interferometer

A simple device with which state superposition may be observed is the **Mach-Zehnder Interferometer**, developed in the 1890s.

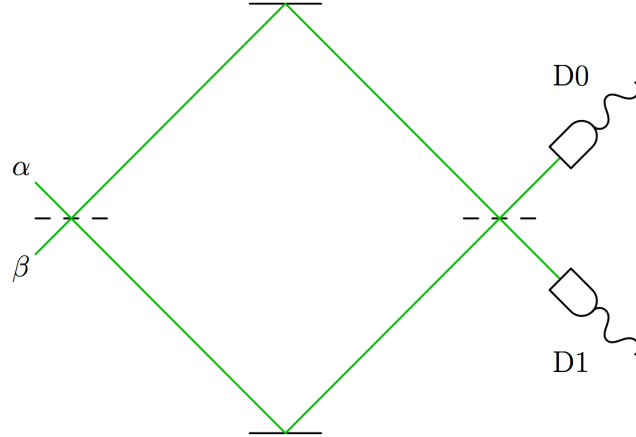


Figure 5. Mach-Zehnder Interferometer.

An incident beam of light arrives at α . We can model the beam splitters (dashed lines) using the following matrices:

$$\text{BS1} : \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{BS2} : \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (3.1.26)$$

These act on a vector $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ describing the state of the two paths. We require the conservation of the light energy, and so

$$|\alpha|^2 + |\beta|^2 = 1 \quad (3.1.27)$$

before and after any beam splitter action. We may model a dielectric that induces a phase change φ in the upper path as

$$\begin{bmatrix} e^{i\varphi} & 0 \\ 0 & 1 \end{bmatrix} \quad (3.1.28)$$

Should the lower path be stopped, for example, the operator would be

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad (3.1.29)$$

The probability of detecting a photon at D_0 or D_1 is given by the composition of the beam splitters and any other items in the correct order, for example if there were no other features and the probability of a photon in the upper path before the first beam splitter was α and the probability for the lower path was β then

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ -\alpha \end{bmatrix} \quad (3.1.30)$$

Property 3.1.5 (MZI Probability Distribution Without Features)

A state $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ introduced in a Mach-Zehnder Interferometer without dielectrics, blockages, or other features is given by $\begin{pmatrix} \beta \\ -\alpha \end{pmatrix}$.

Suppose we send one photon at β . We can now block the lower path with the matrix described above between the two beam splitters, giving the following.

Property 3.1.6 (MZI Probability Distribution with Lower Path Blocked)

Suppose one photon is sent at β . The probability distributions are given by

Outcome	P
Received at D_0	1
Received at D_1	0

Table 1. MZI Dist. for Free Lower Path.

Outcome	P
Received at D_0	1/4
Received at D_1	1/4
Blocked	1/2

Table 2. MZI Dist. for Blocked Lower Path.

We may also introduce a phase shift δ_l in the lower branch and a phase shift δ_u in the upper branch. The matrices would be

$$\begin{bmatrix} \exp(i\delta_u) & \\ & \exp(i\delta_l) \end{bmatrix} \quad (3.1.31)$$

and so

$$\begin{pmatrix} \alpha_f \\ \beta_f \end{pmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} e^{i\delta_u} & \\ & e^{i\delta_l} \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \alpha(e^{i\delta_l} - e^{i\delta_u}) + \beta(e^{i\delta_u} + e^{i\delta_l}) \\ -\alpha(e^{i\delta_u} + e^{i\delta_l}) + \beta(e^{i\delta_u} - e^{i\delta_l}) \end{bmatrix} \quad (3.1.32)$$

Thus,

$$P_0 = |\alpha_f|^2 \quad (3.1.33)$$

$$= \alpha_f \alpha_f^* \quad (3.1.34)$$

$$= \frac{1}{4} (\alpha (e^{i\delta_l} - e^{i\delta_u}) + \beta (e^{i\delta_u} + e^{i\delta_l})) (\alpha^* (e^{-i\delta_l} - e^{-i\delta_u}) + \beta^* (e^{-i\delta_u} + e^{-i\delta_l})) \quad (3.1.35)$$

$$\begin{aligned} &= \frac{\alpha\alpha^*}{4} (e^{i\delta_l} e^{-i\delta_l} - e^{i\delta_u} e^{-i\delta_l} - e^{-i\delta_u} e^{i\delta_l} + e^{i\delta_u} e^{-i\delta_u}) \\ &+ \frac{\beta\beta^*}{4} (e^{i\delta_u} e^{-i\delta_u} + e^{i\delta_u} e^{-i\delta_l} + e^{-i\delta_u} e^{i\delta_l} + e^{i\delta_l} e^{-i\delta_l}) \\ &+ \frac{\alpha^*\beta}{4} (e^{i\delta_u} e^{-i\delta_l} - e^{i\delta_u} e^{-i\delta_u} + e^{i\delta_l} e^{-i\delta_l} - e^{-i\delta_u} e^{i\delta_l}) \\ &+ \frac{\alpha\beta^*}{4} (e^{i\delta_l} e^{-i\delta_u} - e^{i\delta_l} e^{-i\delta_l} + e^{i\delta_u} e^{-i\delta_u} - e^{-i\delta_l} e^{i\delta_u}) \end{aligned} \quad (3.1.36)$$

$$= |\alpha|^2 \sin^2 \frac{\delta_u - \delta_l}{2} + |\beta|^2 \cos^2 \frac{\delta_u - \delta_l}{2} + \text{Im} \{ \alpha\beta^* \} \sin(\delta_u - \delta_l) \quad (3.1.37)$$

Similarly,

$$P_1 = |\alpha|^2 \cos^2 \frac{\delta_u - \delta_l}{2} + |\beta|^2 \sin^2 \frac{\delta_u - \delta_l}{2} - \text{Im} \{ \alpha\beta^* \} \sin(\delta_u - \delta_l) \quad (3.1.38)$$

Property 3.1.7 (MZI Probability Distribution for Phase Shifts)

The probability distribution of receiving a photon at either D_0 or D_1 of a Mach-Zehnder Interferometer given an initial state α, β is given by

Outcome	P
Received at D_0	$ \alpha ^2 \sin^2 \frac{\delta_u - \delta_l}{2} + \beta ^2 \cos^2 \frac{\delta_u - \delta_l}{2} + \text{Im} \{ \alpha\beta^* \} \sin(\delta_u - \delta_l)$
Received at D_1	$ \alpha ^2 \cos^2 \frac{\delta_u - \delta_l}{2} + \beta ^2 \sin^2 \frac{\delta_u - \delta_l}{2} - \text{Im} \{ \alpha\beta^* \} \sin(\delta_u - \delta_l)$

3.1.2 Elitzur-Vaidman Bombs

We may apply these to “Elitzur-Vaidman Bombs” which detonate when a photon passes, but our detector has a change of testing whether the bomb is “broken” i.e. acts like a blockage without actually detonating it. Suppose that 80% of the bombs in some stock are defective; if one is chosen at random and the photon is detected at D_0 , then we can apply Bayes’ Law knowing

Faulty	p	Working	p
explosion	0	explosion	1/2
D_0	1	D_0	1/4
D_1	0	D_1	1/4

Table 3. Faulty EV Bombs are equivalent to no bombs. **Table 4.** Working EV Bombs are equivalent to a block.

and

$$P(\text{defective}) = 0.8 \quad P(D_0 \mid \text{defective}) = 1 \quad P(D_0) = 1 \cdot 0.8 + \frac{1}{4} \cdot 0.2 = 0.85 \quad (3.1.39)$$

to find

$$P(\text{defective} \mid D_0) = \frac{1 \cdot 0.8}{0.85} = \frac{80}{85} = \boxed{\frac{16}{17}} \quad (3.1.40)$$

3.2 Experimental Evidence

3.2.1 Photoelectric Effect

The *photoelectric effect* is an excellent macroscopic quantum effect, clearly highlighting limitations of classical theory.

Definition 3.2.1 (Photoelectric Effect)

Light of sufficiently short wavelength impinging on a negatively-charged metal plate releases electrons; the electron current is proportional to the intensity of the light. Light below a certain cutoff frequency ν_0 does not release any electrons, regardless of intensity.

Classically, photons of any frequency should release electrons, and the current would be proportional to the intensity and not the frequency, since kinetic energy would depend on the intensity alone. The quantum explanation relies instead on the square potential in which electrons in metal lie, shown below.

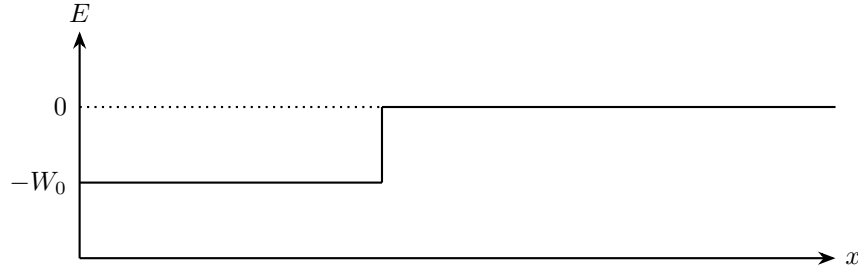


Figure 6. Potential well of an electron in a metal.

Each particular metal has a characteristic workfunction W_0 . From energy conservation, we can argue

$$h\nu = W_0 + W_{\text{kin}} \quad (3.2.1)$$

where $h\nu$ is the energy of the incoming photon, W_0 is the work function of the metal, and W_{kin} is the leftover kinetic energy. Since kinetic energy cannot be negative, the smallest frequency for which electrons may be released is $\nu_0 = W_0/h$. The kinetic energy of a photon being determined by ν implies that intensity is simply an electron count per unit time and area, and so the electron current will be proportional to intensity and not ν .

The reverse process, in which an electron is rapidly decelerated by impact with some substrate, can produce X-rays of only certain frequencies: there exists a minimum wavelength for which light can be produced. The initial energy of the electron comes from acceleration through a potential difference:

$$q_e V_0 = h\nu_{\text{max}} = \frac{hc}{\lambda_{\text{min}}} \quad (3.2.2)$$

3.2.2 Compton Scattering

Compton scattering highlights the momentum of light waves.

Definition 3.2.2 (Compton Scattering)

Light may elastically scatter off of an electron, deviating some angle θ from its initial trajectory. This is known as *Compton Scattering*. Momentum and energy are conserved.

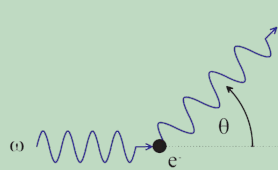


Figure 7. Compton Scattering.

Experimentally, for photons with $\theta = \pi$ (backscattering), it has been observed that the wavelength shift $\Delta\lambda$ is constant and invariant of the initial frequency, having a value of 4.85×10^{-12} m.

Proposition 3.2.3 (Compton Scattering Wavelength Shift)

Consider a photon of wavelength λ_0 scattering off an electron. Let the final wavelength be λ_1 , the scattering angle θ . Thus,

$$\Delta\lambda = \lambda_1 - \lambda_0 = \lambda_c(1 - \cos\theta) \quad (3.2.3)$$

for

$$\lambda_c = \frac{h}{m_e c} \quad (3.2.4)$$

the Compton Wavelength of the electron.

Proof

We begin from the conservation of momentum and the conservation of energy. Let unprimed quantities represent pre-collision values, and primed quantities post-collision values.

$$\begin{cases} \hbar\mathbf{k} = \mathbf{p}' + \hbar\mathbf{k}' \\ E + \hbar\omega = E' + \hbar\omega' \end{cases} \quad (3.2.5)$$

where $\omega = c|\mathbf{k}|$ and $\omega' = c|\mathbf{k}'|$. We also know that the energy E of the electron is given by

$$E = m_e c^2 \quad ; \quad E' = \sqrt{m_e^2 c^4 + (p')^2 c^2} \quad (3.2.6)$$

Let the $\mathbf{x}$ axis be the coincident with the initial direction of the photon, and presume the electron is at rest. Let θ be the angle between the final trajectory of the photon and the original one, also called the scattering angle. Thus,

$$\mathbf{p}' = \hbar(\mathbf{k} - \mathbf{k}') = \hbar \left(k \begin{bmatrix} 1 \\ 0 \end{bmatrix} - k' \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \right) = \hbar \begin{bmatrix} k - k' \cos\theta \\ -k' \sin\theta \end{bmatrix} \quad (3.2.7)$$

We can calculate the value of $p^2 = |\mathbf{p}|^2$:

$$p^2 = \hbar^2 (k^2 - 2kk' \cos\theta + (k')^2 \cos^2\theta + (k')^2 \sin^2\theta) \quad (3.2.8)$$

$$= \hbar^2 (k^2 + (k')^2 - 2kk' \cos\theta) \quad (3.2.9)$$

We can plug this into the expression for the conservation of energy:

$$m_e c^2 + \hbar c k = \hbar c k' + \sqrt{\hbar^2 c^2 (k^2 + (k')^2 - 2kk' \cos\theta) + m_e^2 c^4} \quad (3.2.10)$$

We take the square of both sides:

$$m_e^2 c^4 + \hbar^2 c^2 (k - k')^2 + 2m_e c^2 \hbar c (k - k') = \hbar^2 c^2 (k^2 + (k')^2 - 2kk' \cos\theta) + m_e^2 c^4 \quad (3.2.11)$$

$$\Rightarrow \hbar^2 c^2 (k^2 + (k')^2 - 2kk' - k^2 - (k')^2 + 2kk' \cos\theta) = -2m_e c^2 \hbar c (k - k') \quad (3.2.12)$$

$$\Rightarrow \hbar(kk')(1 - \cos\theta) = m_e c (k - k') \quad (3.2.13)$$

$$\Rightarrow \frac{k - k'}{kk'} = \frac{\hbar}{m_e c} (1 - \cos\theta) \quad (3.2.14)$$

We express the left hand side in terms of λ and λ' :

$$\frac{k - k'}{kk'} = \frac{\frac{2\pi}{\lambda} - \frac{2\pi}{\lambda'}}{\frac{4\pi^2}{\lambda\lambda'}} \quad (3.2.15)$$

$$= \frac{2\pi}{4\pi^2} \cdot (\lambda\lambda') \cdot \frac{\lambda' - \lambda}{\lambda\lambda'} \quad (3.2.16)$$

$$= \frac{\Delta\lambda}{2\pi} \quad (3.2.17)$$

Thus,

$$\boxed{\Delta\lambda} = \frac{2\pi\hbar}{m_e c} (1 - \cos\theta) = \frac{h}{m_e c} (1 - \cos\theta) = \boxed{\lambda_c(1 - \cos\theta)} \quad (3.2.18)$$

as desired. □

Proposition 3.2.4 (Compton Scattering Energy Shift)

Consider a photon of energy E_0 and wavelength λ_0 scattering off an electron. Let the final energy be E_1 , final wavelength λ_1 , and the scattering angle θ . Thus,

$$\Delta E = -\frac{E_0^2}{E_e(1 - \cos \theta)^{-1} + E_0} \quad (3.2.19)$$

for

$$E_e = m_e c^2 \quad (3.2.20)$$

the rest energy of the electron.

Proof

We start in terms of frequency ν :

$$\Delta E = h\Delta\nu = hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_0} \right) = \frac{hc}{\lambda_0 \lambda_1} (\lambda_0 - \lambda_1) = -\frac{hc\Delta\lambda}{\lambda_0(\lambda_0 + \Delta\lambda)} = -\frac{hc}{\lambda_0 \left(\frac{\lambda_0}{\Delta\lambda} + 1 \right)} = -\frac{E_0}{\frac{\lambda_0}{\Delta\lambda} + 1} \quad (3.2.21)$$

We continue by expressing λ_0 in terms of E_0 :

$$\Delta E = -\frac{E_0}{\frac{\lambda_0}{\Delta\lambda} + 1} = -\frac{E_0}{\frac{hc}{E_0 \Delta\lambda} + 1} = -\frac{E_0^2}{\frac{hc}{(1 - \cos \theta)h/(m_e c^2)} + E_0} = -\frac{E_0^2}{\frac{m_e c^2}{1 - \cos \theta} + E_0} \quad (3.2.22)$$

Let E_e be the rest energy of the electron, $E_e \equiv m_e c^2$. Thus,

$$\Delta E = -\frac{E_0^2}{E_e(1 - \cos \theta)^{-1} + E_0} \quad (3.2.23)$$

as desired. □

3.2.3 Back to Theory: De Broglie, Fermat

In the spirit of the wave-particle duality evidenced in light, we extend this to all particles in quantum mechanics.

Definition 3.2.5 (De Broglie Waves)

Consider a particle with momentum p . The *De Broglie Wave* associated with this particle has a wavelength of

$$\lambda_{dB} = \frac{h}{p} \quad (3.2.24)$$

where h is the Planck Constant ($6.626 \times 10^{-34} \text{ m}^2 \cdot \text{kg/s}$).

Evidence for electrons and other massive particles acting as waves appeared at the start of the 20th century as experimentalists compared electron diffraction and X-ray diffraction, as well as the double slit experiment conducted with electrons (yielding the same pattern as for light). In the late 17th century, Fermat formulated the following concept for the choice of path for light.

Axiom 3.2.6 (Fermat's Principle)

Light takes the path where there is no first-order change in the traveling time for nearby paths.

We explore the effects. Consider light refracting through a glass. According to Snell's Law, the path will deviate in some $n > 1$ glass. Consider an observer on the other side of this glass; the straight-line path between the observer and the source will not be the light path at all; thus, how does the light know which path to take to arrive to the observer?

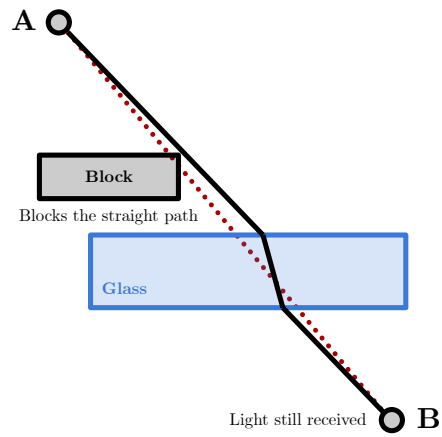


Figure 8. Light passing through glass to an observer, even with a block in the straight path.

We may approach this by modeling every portion of a light path as an isotropic point source; interference will eliminate all but the final path, which is usually straight. In quantum mechanics, this same principle can be used to not only identify massive particle paths but also constrain the Bohr model orbits.

4 Wavefunctions

4.1 Definitions, Expectations

We have established that even point particles are waves; thus, it becomes impossible to exactly locate the particle. We introduce a *probability density function* (PDF) $|\psi|^2$ for particles.

Definition 4.1.1 (Wavefunction)

A *wavefunction* is a representation of a quantum state in the form of a function

$$\Psi(\mathbf{r}, t) : \mathbb{R}^3 \times \mathbb{R}^+ \rightarrow \mathbb{C} \quad (4.1.1)$$

whose modulus squared is the probability density of finding a particle at a location $\mathbf{r}$ at a time t . In one dimension, we write

$$\Psi(r, t) : \mathbb{R} \times \mathbb{R}^+ \rightarrow \mathbb{C} \quad (4.1.2)$$

It is customary to separate, if possible, the time dependence:

$$\Psi(r, t) = \psi(r) \cdot f(t) \quad (4.1.3)$$

where $f(t)$ is some function of time.

Since $|\psi(x)|^2$ is a PDF, we may apply the machinery of statistics in the case of ψ being square-integrable.

Proposition 4.1.2 (Expected Position in 1D)

The *expectation* of position for $\psi(x)$ a 1D time-independent wavefunction is given by

$$\langle x \rangle = \langle \psi | x | \psi \rangle = \int_{-\infty}^{\infty} x |\psi(x)|^2 dx \quad (4.1.4)$$

Proof

We can describe the state $|\psi\rangle$ using the function $\psi(x)$, and so the integral above is simply the inner product of the operator result $\hat{x}|\psi\rangle$ with $\langle\psi|$. $\square$

Proposition 4.1.3 (Expected Value in 1D)

The expectation of a general function $f(x)$ for $\psi(x)$ a 1D time-independent wavefunction is given by

$$\langle f(x) \rangle = \langle \psi | f | \psi \rangle = \int_{-\infty}^{\infty} f(x) |\psi(x)|^2 dx \quad (4.1.5)$$

We thus introduce the normalization requirement of a wavefunction, since it is a probability distribution:

Property 4.1.4 (Normalization Condition)

A wavefunction that is square-integrable must be normalized, satisfying

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1 \quad (4.1.6)$$

This implies that normalized wavefunctions must decay at infinity, satisfying $\lim_{x \rightarrow \pm\infty} \psi(x) = 0$.

4.2 Momentum, Uncertainty

Consider a particle in a 1D box of size L . We thus require that $\psi(x) = 0$ at the boundary, namely $x = 0$ and $x = L$. Thus,

$$\psi(x) = \sum_{n=1}^{\infty} c_n \sin(k_n x) \quad (4.2.1)$$

for

$$k_n = \frac{n\pi}{L} \quad (4.2.2)$$

We generalize to complex coefficients and complex exponentials as a basis:

$$\psi(x) = \sum_{n=-\infty}^{\infty} c_n \exp(ik_n x) \quad (4.2.3)$$

for $k_n = 2n\pi/L$ and $n \in \mathbb{Z}$. We start to see the wavelength of the harmonics appear in the form of wavevector k_n .

We may write a position-based wavefunction in terms of momenta, noting that momentum and position are Fourier-conjugate variables according to the De Broglie wave theory, as hinted by the particle-in-a-box case.

Proposition 4.2.1 (Wavefunction in Momentum Basis)

Let $\psi(x)$ be a 1D time-independent wavefunction in position basis. We write the momentum representation as a function of either wavevector k , $\tilde{\phi}(k)$, or momentum p , $\phi(p)$. The wavefunction in position and wavevector space are Fourier conjugates:

$$\psi(x) = \mathcal{F}^{-1}\{\tilde{\phi}(k)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{\phi}(k) e^{ikx} dk \quad (4.2.4)$$

Conversely,

$$\tilde{\phi}(k) = \mathcal{F}\{\psi(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x) e^{-ikx} dx \quad (4.2.5)$$

The momentum representation is similarly found:

$$\phi(p) = \frac{\mathcal{F}\{\psi(x)\}|_{p \equiv \hbar k}}{\sqrt{\hbar}} = \frac{1}{\sqrt{2\hbar\pi}} \int_{-\infty}^{\infty} \psi(x) e^{-ipx/\hbar} dx = \frac{\tilde{\phi}(k)}{\sqrt{\hbar}} \quad (4.2.6)$$

$$\psi(x) = \frac{\mathcal{F}^{-1}\{\phi(p)\}}{\sqrt{\hbar}} = \frac{1}{\sqrt{2\hbar\pi}} \int_{-\infty}^{\infty} \phi(p) e^{ipx/\hbar} dp \quad (4.2.7)$$

According to the construction of the momentum representation, we may apply Fourier uncertainty relations between conjugate variables to obtain

$$\Delta x \Delta k \geq \frac{1}{2} \quad (4.2.8)$$

We also recall that angular frequency and time are Fourier conjugates.

Proposition 4.2.2 (Heisenberg Uncertainty Principle)

Consider a quantum state $|\psi\rangle$. The uncertainty, defined in Def. 1.5.2, of position and momentum are related according to

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (4.2.9)$$

Energy and time are similarly related:

$$\Delta t \Delta E \geq \frac{\hbar}{2} \quad (4.2.10)$$

The momentum distribution $\phi(p)$ gives a probability density function as well when its magnitude is squared.

Proposition 4.2.3 (Gaussian as the Optimal Uncertainty Solution)

Let $G(x; \mu, \sigma)$ be the standard normalized Gaussian distribution, namely

$$G(x; \sigma) \equiv \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad (4.2.11)$$

Let $w_0 > 0$, let $k_0 = 1/(2w_0)$. The position and momentum wavefunctions of the form

$$\psi(x) = \sqrt{G(x; w_0)} \quad ; \quad \phi(k) = \sqrt{G(p; k_0)} \quad (4.2.12)$$

optimize the Heisenberg Uncertainty Principle, that is,

$$\Delta p \Delta x = \frac{\hbar}{2} \geq \frac{\hbar}{2} \quad (4.2.13)$$

Proof

According to the Gaussian distribution properties, the Fourier transform of a Gaussian is also a gaussian with the reciprocal width. In this case, we take the Fourier transform of the reciprocal of a Gaussian, so we expect a similar but not quite exact result. We will obtain an extra factor of two:

$$\phi(k) \equiv \mathcal{F}\{\psi(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{(2\pi)^{1/4} w_0^{1/2}} \exp\left(-\frac{x^2}{4w_0^2}\right) e^{-ikx} dx \quad (4.2.14)$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{(2\pi)^{1/4} w_0^{1/2}} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{4w_0^2} - ikx\right) dx \quad (4.2.15)$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{(2\pi)^{1/4} w_0^{1/2}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{4w_0^2} (x^2 + ikx4w_0^2)\right) dx \quad (4.2.16)$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{(2\pi)^{1/4} w_0^{1/2}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{4w_0^2} (x + 2kw_0^2 i)^2 - k^2 w_0^2\right) dx \quad (4.2.17)$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{(2\pi)^{1/4} w_0^{1/2}} e^{-k^2 w_0^2} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{4w_0^2} (x + 2kw_0^2 i)^2\right) dx \quad (4.2.18)$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{(2\pi)^{1/4} w_0^{1/2}} \exp(-k^2 w_0^2) \cdot \sqrt{4w_0^2 \pi} \quad (4.2.19)$$

$$= \frac{2\sqrt{\pi} w_0}{\sqrt{2\pi}} \cdot \frac{1}{(2\pi)^{1/4} w_0^{1/2}} \exp(-k^2 w_0^2) \quad (4.2.20)$$

$$= \sqrt{\frac{2w_0}{\sqrt{2\pi}}} \cdot \exp(-k^2 w_0^2) \quad (4.2.21)$$

We can match the expression to find k_0 :

$$\frac{k^2}{4k_0^2} = k^2 w_0^2 \Rightarrow \boxed{k_0 = \frac{1}{2w_0}} \quad (4.2.22)$$

Thus,

$$\phi(k) = \sqrt{\frac{2w_0}{\sqrt{2\pi}}} \cdot \exp(-k^2 w_0^2) = \boxed{\frac{1}{\sqrt[4]{2\pi}\sqrt{k_0}} \cdot \exp\left(-\frac{k^2}{4k_0^2}\right)} \quad (4.2.23)$$

Thus, the uncertainty relation is

$$k_0 w_0 = \frac{1}{2} \Rightarrow \Delta k \Delta x = \frac{1}{2} \Rightarrow \boxed{\Delta p \Delta x = \frac{\hbar}{2}} \quad (4.2.24)$$

as desired. $\square$

Theorem 4.2.4 (Parseval for QM)

Parseval's general theorem gives

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = \int_{-\infty}^{\infty} |\tilde{\phi}(k)|^2 dk = \int_{-\infty}^{\infty} |\phi(p)|^2 dp \quad (4.2.25)$$

Thus, if one of the representations is normalized, all are normalized.

4.2.1 Many-slit Interference

Let n slits have size b and spacing d . We can apply the same small angle approximation, such that the phase difference between light from two adjacent slits is

$$\Delta\varphi_{j,j-1} = kd \sin \theta = \frac{2\pi d \sin \theta}{\lambda} \quad (4.2.26)$$

Thus, we add the results from the n slits, each with its phase:

$$\psi_{\text{net}} = \sum \psi_j = \psi_0 \sum \left(e^{2\pi i d \sin \theta / \lambda} \right)^j = \psi_0 \frac{1 - \exp\left(i \frac{2\pi n d \sin \theta}{\lambda}\right)}{1 - \exp\left(i \frac{2\pi d \sin \theta}{\lambda}\right)} \quad (4.2.27)$$

Thus, the intensity will be proportional to

$$I \propto \left| \frac{1 - \exp\left(i \frac{2\pi n d \sin \theta}{\lambda}\right)}{1 - \exp\left(i \frac{2\pi d \sin \theta}{\lambda}\right)} \right|^2 \quad (4.2.28)$$

If the slit size (b) is appreciable, one must include a *Fraunhofer* term, or far-field diffraction. For a slit, this is obtained by considering the slit as a continuous distribution of point sources, such that

$$d\psi = \frac{A}{r_1} e^{i(kr_1 - \omega t)} dx \quad (4.2.29)$$

for $r_1 = r - x \sin \theta$ integrated from $-b/2 \rightarrow b/2$:

$$\psi \simeq \frac{A}{r} e^{i(kr - \omega t)} \int_{-b/2}^{b/2} e^{-ikx \sin \theta} dx = \frac{2A}{r} e^{i(kr - \omega t)} \frac{1}{k \sin \theta} \sin\left(\frac{kb}{2} \sin \theta\right) \quad (4.2.30)$$

Let $\psi' = kb \sin \theta$;

$$\psi = \frac{A}{r} e^{i(kr - \omega t)} b \operatorname{sinc}(\psi'/2\pi) \quad (4.2.31)$$

where

$$\operatorname{sinc}(x) \equiv \frac{\sin \pi x}{\pi x} \quad (4.2.32)$$

Thus, the intensity becomes

$$I_{\text{single}} \propto \operatorname{sinc}^2 \left(\frac{b \sin \theta}{\lambda} \right) \quad (4.2.33)$$

Thus, for n slits, for $\sin \theta \simeq \tan \theta$, the intensity on a screen a distance Z from the slits as a function of the vertical axis (parallel to that of the slits) is given by

$$I_n(X) = I_0 \left| \frac{1 - \exp \left(i \frac{2\pi n d X}{Z \lambda} \right)}{1 - \exp \left(i \frac{2\pi d X}{Z \lambda} \right)} \right|^2 \operatorname{sinc}^2 \left(\frac{b X}{Z \lambda} \right) \quad (4.2.34)$$

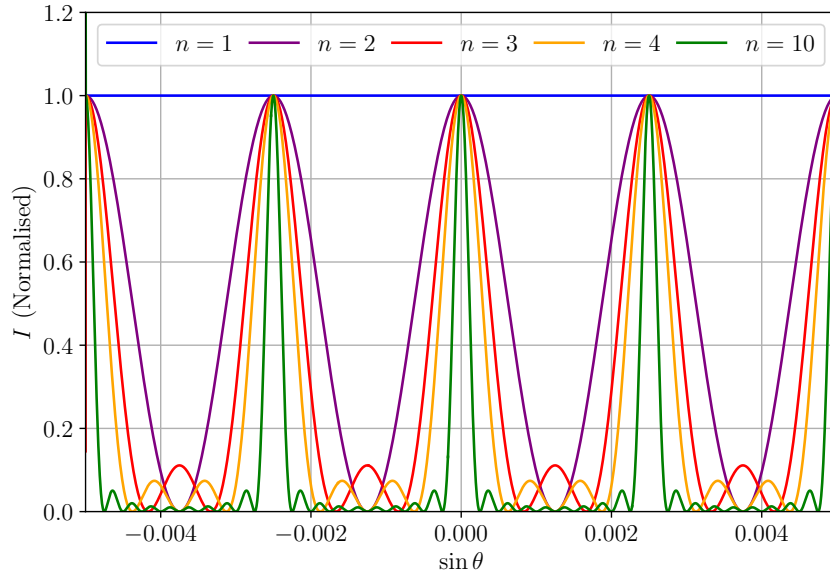


Figure 9. Interference from n slits.

Consider a two-slit setup with an electron beam impinging on the slits, with a screen far away. We note that should we try to ascertain with a photon beam (light) through which slit an electron passed, the introduced momentum from the measurement will entirely wash out the interference pattern since the uncertainty is much larger than the amplitude of the interference.

In particular, scattering an electron with a photon causes a phase shift, which when averaged leads to a washout of the interference pattern. For A some normalization constant, k the electron wavevector, $k_{\text{in}}, k_{\text{out}}$ the incident and scattered photon wavevectors, L_1, L_2 the distances from the slits to the point of interest on the screen, and $\mathbf{x}_1, \mathbf{x}_2$ the slit position vectors, the intensity in the detector plane is given by

$$|\psi'_D|^2 = 2|A|^2 \left(1 + \cos \left(k(L_1 - L_2) + (\mathbf{k}_{\text{out}} - \mathbf{k}_{\text{in}})(\mathbf{x}_1 - \mathbf{x}_2) \right) \right) \quad (4.2.35)$$

4.3 Bohr Model

4.3.1 Premise

We begin with experimental results.

Definition 4.3.1 (Balmer Series)

In 1885, Balmer found an empirical description for several emission lines of Hydrogen:

$$\nu_k = \frac{c}{\lambda_k} = R_\infty \left(\frac{1}{2^2} - \frac{1}{k^2} \right) : k = 3, 4, 5 \dots \quad (4.3.1)$$

where R_∞ is the Rydberg constant, 3.29×10^{15} Hz. The lines are labeled α , β , etc in order of increasing k .

We now know these to correspond to electron shell jumps to and from the $n = 2$ shell.

Definition 4.3.2 (Lyman Series)

The Lyman Series lies in the UV spectrum, corresponding to electron shell jumps to and from the $n = 1$ shell:

$$\nu_k = \frac{c}{\lambda_k} = R_\infty \left(1 - \frac{1}{k^2} \right) : k = 3, 4, 5 \dots \quad (4.3.2)$$

Definition 4.3.3 (Paschen Series)

The Paschen Series lies in the infrared, corresponding to electron shell jumps to and from the $n = 3$ shell:

$$\nu_k = \frac{c}{\lambda_k} = R_\infty \left(\frac{1}{3^2} - \frac{1}{k^2} \right) : k = 3, 4, 5 \dots \quad (4.3.3)$$

We can thus extrapolate the n -th series:

$$\nu_{nk} = \frac{c}{\lambda_{nk}} = R_\infty \left(\frac{1}{n^2} - \frac{1}{k^2} \right) : k = n + 1, n + 2, \dots \quad (4.3.4)$$

The Bohr Model can explain these series according to three postulates.

Axiom 4.3.4 (Bohr Postulates)

The Bohr Model of Hydrogen postulates the following characterizations of the mechanisms producing the observed series:

1. Atoms have certain stable states (stationary states); light is emitted or absorbed when the atom makes a transition between stationary states.
2. If two stationary states have energies E_j , E_k for $E_j > E_k$, then the light emitted on the transition $E_j \rightarrow E_k$ has the frequency

$$\nu_{jk} = \frac{E_j - E_k}{h} \quad (4.3.5)$$

3. For circular orbits, the stationary states can be determined from a quantization condition for the angular momentum, $mvr = n\hbar$ for $n \in \mathbb{N}$, where r is the classical orbit, and v is the classical velocity for that orbit; the integer n is called the quantum number for that orbit.

We may describe the Bohr energy levels classically, without relativistic effects and without further quantum mechanics than those principles enumerated above. The circular orbit of an electron can be characterized according to the Coulomb Force that it is bound by:

$$\frac{mv^2}{r} = \frac{Zq^2}{4\pi\epsilon_0 r^2} \Rightarrow m^2 v^2 r^2 = \frac{Zq^2}{4\pi\epsilon_0} mr \quad (4.3.6)$$

The Bohr quantization of angular momentum requires

$$n^2 \hbar^2 = (mvr)^2 = \frac{Zq^2}{4\pi\epsilon_0} mr \quad (4.3.7)$$

For the n th Bohr orbit, the radius r_n is thus described by

$$r_n = \frac{4\pi\epsilon_0}{mZq^2} n^2 \hbar^2 \quad (4.3.8)$$

and so the n th orbital velocity is

$$v_n = \frac{n\hbar}{mr_n} = \frac{Zq^2}{4\pi\epsilon_0} \frac{1}{n\hbar} \quad (4.3.9)$$

The energy is

$$E_n = \frac{mv_n^2}{2} - \frac{Zq^2}{4\pi\epsilon_0 r_n} = -\frac{m}{2} \left(\frac{Zq^2}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2 \hbar^2} \quad (4.3.10)$$

We can introduce the dimensionless *fine-structure constant* α , given by

$$\alpha = \frac{q^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137} \quad (4.3.11)$$

Proposition 4.3.5 (Bohr Atom)

The Bohr atom model states that the n th energy level has the following properties:

$$r_n = \frac{\hbar}{mcZ\alpha} n^2 \quad v_n = \frac{Z\alpha c}{n} \quad E_n = -\frac{mc^2}{2} \frac{(Z\alpha)^2}{n^2} \quad (4.3.12)$$

These hold only for

$$v \ll c \quad \text{and} \quad Z \ll 100 \quad (4.3.13)$$

Definition 4.3.6 (Bohr Radius)

The *Bohr Radius* is the smallest allowed orbit for a Hydrogen atom, given by

$$a_0 = \frac{\hbar}{mc\alpha} = 0.53 \text{ \AA} \quad (4.3.14)$$

For an atom of atomic number Z , the radius and binding energy are

$$r_1 = \frac{a_0}{Z} \quad E_1 = -\frac{mc^2}{2} (Z\alpha)^2 = -Z^2 \cdot 13.6 \text{ eV} \quad (4.3.15)$$

We note that in the above calculation one should use the reduced mass μ , given by

$$\mu = (m^{-1} + M^{-1})^{-1} \quad (4.3.16)$$

instead of only the mass of the nucleus; however, this is only relevant for small atoms such as hydrogen and deuterium, which can be distinguished by their spectral lines.

From these expressions we can retrieve the Rydberg constant.

Proposition 4.3.7 (Rydberg Constant)

The experimentally measured Rydberg Constant can be expressed in terms of fundamental quantities as follows:

$$R_\infty \equiv \frac{mc^2 \alpha^2}{2h} \quad (4.3.17)$$

4.3.2 Correspondence Principle

The transition frequency for $m = n + 1 \rightarrow n$ as $n \rightarrow \infty$ can be found, to leading order in n , to be

$$\nu_{n+1 \rightarrow n} = R_\infty \left(\frac{1}{n^2} - \frac{1}{(n+1)^2} \right) = R_\infty \frac{(n+1)^2 - n^2}{n^2(n+1)^2} = R_\infty \frac{2n+1}{n^2(n+1)^2} \approx \frac{2}{n^3} R_\infty \quad (4.3.18)$$

The electron orbit frequency is given by

$$f_{n+1} \approx f_n = \frac{v_n}{2\pi r_n} = \frac{\alpha c}{n} \frac{1}{2\pi} \frac{m c \alpha}{\hbar} \frac{1}{n^2} = \frac{m c^2 \alpha^2}{\hbar n^3} = \frac{2}{n^3} R_\infty \quad (4.3.19)$$

These expressions match !

Theorem 4.3.8 (Bohr Correspondence Principle)

For large n (large principal quantum number), the frequency emitted by the Bohr atom on a transition $n \rightarrow n+1$ equals the classical emission frequency by the oscillating electron.

We again see a classical mechanics as a limiting case of quantum mechanics.

We can motivate the Bohr quantization condition from considering standing waves on a closed string. We require an integer number n of wavelengths on the string, which is a circumference of a circle of radius r :

$$2\pi r = n\lambda = \frac{2\pi}{k} n = \frac{2\pi\hbar}{p} n \Rightarrow rp = n\hbar \quad (4.3.20)$$

The momentum p is the product mv :

$$mvr = n\hbar \quad (4.3.21)$$

exactly as desired.

4.3.3 Wilson-Sommerfeld Quantization, QHO First Glance

Proposition 4.3.9 (Wilson-Sommerfeld Quantization Condition)

$$\oint p \, dq = nh : n \in \mathbb{N} \quad (4.3.22)$$

We can generalize the Bohr Quantization Condition to elliptical orbits: consider $L = mvr$ as a canonical momentum with corresponding canonical coordinate angle ϕ . Thus,

$$\oint p \, dq = \oint L \, d\phi = L \cdot 2\pi \Rightarrow mvr = n\hbar \quad (4.3.23)$$

and we obtain the same constraint. We observe that non-integer numbers of wavelengths per loop will result in destructive interference, in the same vein as Fermat's principle.

We may apply this to the quantum harmonic oscillator (to be discussed in detail later); consider a particle with a total energy E fixed by

$$E = \frac{p^2}{2m} + \frac{m\omega^2 x^2}{2} \quad (4.3.24)$$

The magnitude of the momentum is given by

$$|p(x)| = \sqrt{2m} \sqrt{E - \frac{m\omega^2 x^2}{2}} \quad (4.3.25)$$

The Wilson-Sommerfeld quantization applied to this system gives

$$\oint p \, dq = \oint p(x) \, dx \quad (4.3.26)$$

$$= 2 \int_{-x_0}^{x_0} \sqrt{2m} \cdot \sqrt{E - \frac{m\omega^2 x^2}{2}} \, dx \quad (4.3.27)$$

$$= 2\sqrt{2mE} \int_{-x_0}^{x_0} \sqrt{1 - \frac{m\omega^2}{2E} x^2} \, dx \quad (4.3.28)$$

$$= 2\sqrt{2mE} \int_{-x_0}^{x_0} \sqrt{1 - \left(\frac{x}{x_0}\right)^2} \, dx \quad (4.3.29)$$

$$= 2\sqrt{2mE}x_0 \int_{-1}^1 \sqrt{1 - y^2} \, dy \quad (4.3.30)$$

$$= \sqrt{2mE}\pi x_0 \quad (4.3.31)$$

$$= \frac{2\pi E}{\omega} \quad (4.3.32)$$

$$= nh \quad (4.3.33)$$

Thus, the quantized energy levels of the QHO are given by

$$E_n = n\hbar\omega \quad (4.3.34)$$

These are not totally accurate (we shall see later that $E_n = (n + 1/2)\hbar\omega$) but the step size is correct. The mistake arises from the lack of a true classical analogue (which this method attempts to harness) to the QHO.

5 1D Schrödinger Equation

5.1 The Momentum Operator

See Dirac §22 for further reading. Consider a dynamical system with 1 dimension. Returning to the language of operators, commutators, and quantum Poisson brackets, we recall that

$$[q, p] = 1 \quad (5.1.1)$$

since the canonical coordinate q and canonical momentum p correspond to the same degree of freedom. Thus,

$$[q, p] \equiv i\hbar[q, p] \Rightarrow qp - pq = i\hbar \quad (5.1.2)$$

Let us describe the state of such a 1D quantum system with a wavefunction $\psi(q)$, writing the ket as $\psi(q)\rangle$. Since ψ is a function, we may take its derivative³, denoting it as

$$\frac{d\psi}{dq}\rangle \quad (5.1.3)$$

Since the derivative is a linear operator, we can represent the above ket as the result of the linear operator d/dq applied to $\psi\rangle$:

$$\frac{d}{dq}\psi\rangle = \frac{d\psi}{dq}\rangle \quad (5.1.4)$$

We have thus defined this derivative operator by characterizing the full range over the domain of all kets. We can then apply it to a bra:

$$\left(\langle\psi_b(q)\frac{d}{dq}\right)\psi_k\rangle = \langle\psi_b\left(\frac{d}{dq}\psi_k\right)\rangle \quad (5.1.5)$$

for all (ket) functions ψ_k . Thus, evaluating the inner product according to linear algebra rules gives

$$\int_{-\infty}^{\infty} \langle\psi_b\frac{d}{dq}|q'\rangle dq' \psi_k(q') = \int_{-\infty}^{\infty} \psi_b(q') dq' \frac{d\psi_k(q')}{dq'} \quad (5.1.6)$$

Integration by parts gives

$$\int_{-\infty}^{\infty} \langle\psi_b\frac{d}{dq}|q'\rangle dq' \psi_k(q') = \cancel{(\psi_k(q)\psi_b(q))}\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{d\psi_b(q')}{dq'} dq' \psi(q') \quad (5.1.7)$$

$$= - \int_{-\infty}^{\infty} \langle\frac{d\psi_b(q)}{dq}|q'\rangle dq' \psi(q') \quad (5.1.8)$$

where we have assumed that the wavefunction decays to zero at infinity⁴. We thus have achieved a symmetry in the forms, which allows us to conclude that

$$\langle\psi_b\frac{d}{dq} = -\langle\frac{d\psi_b}{dq} \quad (5.1.9)$$

for any bra $\langle\psi_b$.

Proposition 5.1.1 (Conjugate Transpose of the Derivative)

The conjugate transpose of the differentiation operator, that is, the d/dq operator on the right of a bra vector, has the meaning of minus differentiation with respect to q . The derivative is thus a **pure imaginary linear operator**.

³It is assumed that it is at least piecewise differentiable, with steps described with delta functions as needed.

⁴We have not lost generality since wavefunctions that do not decay to zero can be modeled as an infinite superposition of square-integrable wavefunctions, and so this holds for each individually.

As such, we seek to find a representation of d/dq . From the decomposition of any eigenket into a product of delta functions and the standard ket,

$$|q''\rangle = \delta(q - q'') \quad (5.1.10)$$

$$\Rightarrow \frac{d}{dq}|q''\rangle = \frac{d}{dq}\delta(q - q'') \quad (5.1.11)$$

$$\Rightarrow \langle q' | \frac{d}{dq} | q'' \rangle = \frac{d}{dq'} \delta(q' - q'') \quad (5.1.12)$$

Property 5.1.2 (Representative of the Derivative)

The operator d/dq in position space has the representation

$$\langle q' | \frac{d}{dq} | q'' \rangle = \frac{d}{dq'} \delta(q' - q'') \quad (5.1.13)$$

Proposition 5.1.3 (Commutator of the Derivative with its Variable)

The commutator of the derivative operator with its independent variable is unity:

$$\left[\frac{d}{dq}, q \right] = 1 \quad (5.1.14)$$

Proof

We find the commutator using the product rule:

$$\left[\frac{d}{dq}, q \right] \psi \equiv \frac{d}{dq} q \psi - q \frac{d}{dq} \psi \quad (5.1.15)$$

$$= \psi + q \cancel{\frac{d\psi}{dq}} - q \cancel{\frac{d\psi}{dq}} \quad (5.1.16)$$

$$= \psi \quad (5.1.17)$$

Thus,

$$\left[\frac{d}{dq}, q \right] \psi = \psi \Rightarrow \left[\frac{d}{dq}, q \right] = 1 \quad (5.1.18)$$

as desired. $\square$

We are free to extend this work to n dimensions, by which we have

$$\frac{\partial}{\partial q_r} \psi = \frac{\partial \psi}{\partial q_r} \quad \text{and} \quad \frac{\partial}{\partial q_r} = 0 \quad (5.1.19)$$

As for 1 dimension,

$$\langle \psi_b | \frac{\partial}{\partial q_r} = - \langle \frac{\partial \psi_b}{\partial q_r} \quad (5.1.20)$$

The commutator thus takes the form

$$\left[\frac{\partial}{\partial q_r}, q_s \right] = \delta_{rs} \quad (5.1.21)$$

to keep separate degrees of freedom independent. By rules of multivariable calculus, Clairaut's Theorem allows for

$$\left[\frac{\partial}{\partial q_r}, \frac{\partial}{\partial q_s} \right] = 0 \quad (5.1.22)$$

We are ready to put the pieces together.

Theorem 5.1.4 (Momentum Operator as a Derivative)

The momentum operator $\hat{p}_r$ can be represented as

$$\hat{p}_r = -i\hbar \frac{\partial}{\partial q_r} \quad (5.1.23)$$

Proof

We have “defined” the momentum operator in terms of the Quantum Conditions (Law 2.5.11):

$$[q_r, p_s] = i\hbar \delta_{rs} \quad (5.1.24)$$

We have previously shown that

$$\left[\frac{\partial}{\partial q_r}, q_s \right] = \delta_{rs} \quad (5.1.25)$$

which implies

$$\left[-i\hbar \frac{\partial}{\partial q_r}, q_s \right] = -i\hbar \delta_{rs} \quad (5.1.26)$$

as desired. We have also shown that

$$\left[\frac{\partial}{\partial q_r}, \frac{\partial}{\partial q_s} \right] = 0 \quad (5.1.27)$$

which implies

$$\left[-i\hbar \frac{\partial}{\partial q_r}, -i\hbar \frac{\partial}{\partial q_s} \right] = 0 \quad (5.1.28)$$

and so

$$[p_r, p_s] = 0 \quad (5.1.29)$$

as desired. Thus, the operators $-i\hbar \frac{\partial}{\partial q_r}$ and $\hat{p}_r$ are indistinguishable, as desired. $\square$

Property 5.1.5 (Heisenberg from the Commutator of Position and Momentum)

Applying Th. 2.5.14 to the position and momentum operators, we have

$$(\Delta x)_\psi^2 (\Delta p)_\psi^2 \geq \frac{1}{4} \langle \hbar \rangle_\psi^2 = \frac{\hbar^2}{4} \quad (5.1.30)$$

which is exactly the Heisenberg Uncertainty Principle.

Proposition 5.1.6 (Schrödinger’s Representation)

There exists a representation of the position operators q_r so that they are diagonal and

$$\hat{p}_r = -i\hbar \frac{\partial}{\partial q_r} \quad (5.1.31)$$

Proof

In general, the momentum operator is in fact written as

$$p_r = i\hbar \frac{\partial}{\partial q_r} + \frac{\partial F(q)}{\partial q_r} \quad (5.1.32)$$

A suitable choice of phase factors $\gamma(q)$ can eliminate this function F by causing it to become constant. We observe that one could write

$$p_r = i\hbar \frac{\partial}{\partial q_r} + f_r(q) \quad (5.1.33)$$

for some real function $f_r(q)$ of position which must in turn satisfy

$$\partial_r f_s = \partial_s f_r \quad (5.1.34)$$

and so

$$f_r = \frac{\partial F}{\partial q_r} \quad (5.1.35)$$

for some F independent of r .

A suitable choice of phase factors in the diagonal operator representation can be found to make F vanish: an $\exp(i\gamma)$ factor where γ is a real function of position gives this alternative representation (labeled with tilde),

$$\widetilde{\rangle} \equiv \exp(-i\gamma)\rangle \Rightarrow \widetilde{\left(\frac{\partial}{\partial q_r}\right)} = \frac{\partial}{\partial q_r} + i\frac{\partial \gamma}{\partial q_r} \quad (5.1.36)$$

an F of the form

$$F = \hbar\gamma + \text{const} \quad (5.1.37)$$

so that

$$\hat{p}_r = -i\hbar \widetilde{\left(\frac{\partial}{\partial q_r}\right)} \quad (5.1.38)$$

as originally desired, up to a constant shared phase factor. $\square$

As a consequence of this framework, we have that any state, classically expressed with canonical coordinates and momenta can now be represented in terms of coordinates alone:

$$f(q_1 \dots q_n, p_1 \dots p_n) = f\left(q_1 \dots q_n, -i\hbar \frac{\partial}{\partial q_1} \dots -i\hbar \frac{\partial}{\partial q_n}\right) \quad (5.1.39)$$

hence justifying the wavefunction as a function of only coordinates being sufficient to fully describe a quantum state.

Property 5.1.7 (Momentum Applied to the Standard Ket)

$$\hat{p}_r \rangle = 0 \quad (5.1.40)$$

Proof

We trivially have that

$$\langle \frac{\partial}{\partial q_r} | \rangle = 0 \quad (5.1.41)$$

since $\rangle$ is equivalent to $\psi = 1$ everywhere. Since

$$\hat{p}_r = -i\hbar \frac{\partial}{\partial q_r} \quad (5.1.42)$$

we have

$$\langle p_r | \rangle = -i\hbar \langle \frac{\partial}{\partial q_r} | \rangle = 0 \quad (5.1.43)$$

as desired. $\square$

Property 5.1.8 (Commutator of Momentum with a Function of Position)

$$[f(\hat{x}, \hat{y}, \hat{z}), \hat{p}_i] = i\hbar \partial_{x_i} f \quad (5.1.44)$$

Property 5.1.9 (Momentum applied to Position Eigenstates)

$$\hat{p}_r |q'_1 \dots q'_n\rangle = i\hbar \frac{\partial}{\partial q'_r} |q'_1 \dots q'_n\rangle \quad (5.1.45)$$

Proof

We start with an arbitrary state ψ :

$$\langle q'_1 \dots q'_n | \frac{\partial}{\partial q'_r} \psi \rangle = \langle q'_1 \dots q'_n | \frac{\partial \psi}{\partial q'_r} \rangle = \frac{\partial}{\partial q'_r} \langle q'_1 \dots q'_n | \psi \rangle \quad (5.1.46)$$

Thus,

$$\langle q'_1 \dots q'_n | \hat{p}_r \rangle = -i\hbar \frac{\partial}{\partial q'_r} \langle q'_1 \dots q'_n | \quad (5.1.47)$$

and so

$$\hat{p}_r |q'_1 \dots q'_n\rangle = i\hbar \frac{\partial}{\partial q'_r} |q'_1 \dots q'_n\rangle \quad (5.1.48)$$

as desired. $\square$

From here we can recover the Fourier integrals of the wave paradigm, should we so desire.

Proposition 5.1.10 (Observability of Momentum)

Momentum is an observable.

Proof

We first write the inner product of a position eigenstate with a momentum eigenstate:

$$\langle q' | p' \rangle = c' \exp\left(\frac{ip'q'}{\hbar}\right) \quad (5.1.49)$$

for $c' = c(p')$ some function independent of q' but possibly dependent on p' . Unfortunately these do not satisfy the Dirichlet conditions (i.e. in particular do not decay to 0 at infinity) which in fact prevents orthogonality in the strictest sense:

$$\langle p' | p'' \rangle = \int_{-\infty}^{\infty} \langle p' | q' \rangle dq' \langle q' | p'' \rangle = \overline{c'} c'' \int_{-\infty}^{\infty} \exp\left(\frac{-i(p' - p'')q'}{\hbar}\right) dq' \quad (5.1.50)$$

This integral does not converge in the usual sense; however, we can apply the loosened convergence criteria in Prop.

1.2.5, recalling $\forall a \in \mathbb{R} : \int_{-\infty}^{\infty} \exp(iax) dx = 2\pi\delta(a)$ to obtain

$$\langle p' | p'' \rangle = \overline{c'} c'' \cdot 2\pi\delta\left(\frac{p' - p''}{\hbar}\right) = \overline{c'} c'' \hbar \delta(p' - p'') = |c'| \hbar \delta(p' - p'') \quad (5.1.51)$$

using delta function properties. Since we require

$$\langle p' | p'' \rangle = \delta(p' - p'') \quad (5.1.52)$$

we have that $|c| = 1/\sqrt{\hbar}$; since this specifies c up to a phase factor, we then have

$$\langle q' | p' \rangle = \frac{1}{\sqrt{\hbar}} \exp\left(\frac{ip'q'}{\hbar}\right) \quad (5.1.53)$$

Thus, it is true that

$$p' \neq p'' \implies \langle p' | p'' \rangle = 0 \quad (5.1.54)$$

and so eigenstates of the momentum operator are orthogonal (but unnormalizable). Thus, momentum is an observable, as desired. $\square$

Theorem 5.1.11 (Position Operator in Terms of Momentum)

The r th position operator can be expressed as a linear operator applied to momentum as follows:

$$q_r = i\hbar \frac{\partial}{\partial p_r} \quad (5.1.55)$$

Proposition 5.1.12 (Momentum and Position as Fourier Conjugates)

Either of the representatives of the position or momentum operators is given, apart from numerical coefficients, by the amplitudes of the Fourier components of the other.

Let $|X\rangle$ be an arbitrary state. We can apply the above work to write

$$\langle p'|X\rangle = \frac{1}{\sqrt{h}} \int_{-\infty}^{\infty} \exp\left(-\frac{iq'p'}{h}\right) dq \langle q'|X\rangle \quad (5.1.56)$$

$$\langle q'|X\rangle = \frac{1}{\sqrt{h}} \int_{-\infty}^{\infty} \exp\left(\frac{iq'p'}{h}\right) dp' \langle p'|X\rangle \quad (5.1.57)$$

which are clearly Fourier integrals, as were derived in the case of the square well in Section 4. For a square-integrable wavefunction localized in some area in both position and momentum space, called a “wave packet”, the relation between the widths $\Delta q'$ and $\Delta p'$ of the wave packet is that of $\Delta p' \Delta q' \geq h$, the Heisenberg Uncertainty Principle⁵.

Proposition 5.1.13 (Conservation of Momentum)

In free space, the Momentum and Hamiltonian operators commute.

Proof

In free space, the Hamiltonian is but $\hat{H} = \hat{p}^2/2m$. Thus, we write $[\hat{p}, \hat{H}] = [\hat{p}, \hat{p}^2/2m] = \frac{1}{2m} (\hat{p}^3 - \hat{p}^3) = 0$. $\square$

5.1.1 Energy Measurements

In line with the assumptions based on experiment that have led to the derivation of the entire theory, we discuss what happens when one measures energy.

- A **strong** measurement, with sufficient resolution to resolve exactly which eigenstate a system would be in, collapses the superposition of states into the one measured.
- A **weak** measurement, without sufficient resolution to resolve a particular eigenstate, results in a state given by the Bayesian probability associated the initial state as a prior given the measurement.

Property 5.1.14 (Expected Value of Energy)

Consider a quantum state $|\psi\rangle$ expressed as a superposition of energy eigenstates $|u_n\rangle$. The expectation of the energy is given by

$$\langle E \rangle \equiv \langle \hat{H} \rangle = \langle \psi | \hat{H} | \psi \rangle = \sum |c_n|^2 \langle u_n | \hat{H} | u_n \rangle = \sum |c_n|^2 E_n \quad (5.1.58)$$

where

$$c_n \equiv \langle u_n | \psi \rangle \quad (5.1.59)$$

In wavefunction notation, we may write

$$\langle E \rangle = \int_{-\infty}^{\infty} \psi^*(x) \hat{H} \psi(x) dx \quad (5.1.60)$$

⁵The fact that $\hbar/2$ is replaced here by h arises from a different definition of the widths.

We may apply a similar logic to find an expected value for momentum.

Property 5.1.15 (Expected Value of Momentum)

Let $\phi(k)$ be the wavenumber (momentum divided by $\hbar$) probability distribution function for some quantum state $|\psi\rangle$. Thus,

$$\langle p \rangle = \langle \psi | \hat{p} | \psi \rangle = \hbar \langle \psi | \hat{k} | \psi \rangle = \hbar \int_{-\infty}^{\infty} \phi^*(k) k \phi(k) dk \quad (5.1.61)$$

We may also write the kinetic energy using this expectation, according to $T \equiv p^2/2m$:

$$\langle T \rangle = \frac{\langle \hat{p}^2 \rangle}{2m} = \frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \phi^*(k) k^2 \phi(k) dk = -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \psi^*(x) \partial_x^2 \psi(x) dx \quad (5.1.62)$$

according to the position-basis representation of the momentum operator, presented in Thm. 5.1.4.

We may also perform the opposite, i.e. measure the expected position based on a momentum distribution.

Property 5.1.16 (Expected Value of Position)

Let $\phi(p)$ be the momentum probability distribution function for some quantum state $|\psi\rangle$, and let $\psi(x)$ be the wavefunction (position probability distribution function). The expected value of position is given equivalently by

$$\langle x \rangle = \langle \psi | x | \psi \rangle = \int_{-\infty}^{\infty} \psi^*(x) x \psi(x) dx = i\hbar \int_{-\infty}^{\infty} \phi^*(p) \partial_p \phi(p) dp \quad (5.1.63)$$

where we have used Thm. 5.1.11 to represent the position operator $\hat{x}$ as $i\hbar\partial_p$.

5.2 1D S.E, T.I.S.E

We wish to convert the quantum conditions into a differential equation that allows us to predict the behavior in space and time of a wavefunction.

Axiom 5.2.1 (Assumptions Regarding Time-Dependent Behavior)

In an undisturbed system, we assume that

- For a state $|At_0\rangle$ at some time t_0 , states $|At\rangle$ are “determined by” the state $|At_0\rangle$ by some numerical factor.
- Each state $|Pt\rangle$ has the same length as $|Pt_0\rangle$.

Let $|At\rangle$ and $|Bt\rangle$ be two states depending on the time t . The principle of superposition gives

$$|Rt\rangle = c_1|At\rangle + c_2|Bt\rangle \quad (5.2.1)$$

Thus, in general, the state of the system $|Pt\rangle$ is a linear function of the constituent states at that time t , which are themselves “determined by” the state at some t_0 up to a numerical factor, which implies the existence of a linear operator T that advances time:

$$|Pt\rangle = T|Pt_0\rangle \quad (5.2.2)$$

T must depend on t and t_0 only, and not on the state $|P\rangle$ in question. The assumption regarding length causes T to be a phase factor only, since we must have

$$\langle Qt|Pt\rangle = \langle Qt_0|Pt_0\rangle \quad (5.2.3)$$

Proposition 5.2.2 (Time Progression is Unitary)

Let T be the time advancement operator. Thus, $\overline{T}T = 1$.

We may thus write the time derivative as an operator:

$$\frac{d}{dt_0}|Pt_0\rangle = \left(\lim_{t \rightarrow t_0} \frac{T - 1}{t - t_0}\right)|Pt_0\rangle \quad (5.2.4)$$

Thus,

$$i\hbar \frac{d}{dt}|Pt\rangle = H(t)|Pt\rangle \quad (5.2.5)$$

This is the most general form of the **Schrödinger Equation**. We assume that $H(t)$ is the *energy* of the system.

Axiom 5.2.3 (Schrödinger Equation)

We have shown that the energy operator allows for the time progression. Decomposing the energy into kinetic energy, given classically by $\hat{p}^2/2m$, and potential energy, some $V(x)$, we have

$$i\hbar \partial_t \Psi(x, t) = \left(\frac{\hat{p}^2}{2m} + V(x)\right) \Psi(x, t) = \left(-\frac{\hbar^2}{2m} \partial_x^2 + V(x)\right) \Psi(x, t) \quad (5.2.6)$$

This PDE is called the **Schrödinger Equation** (SE). The eigenvalue differential equation without time dependence is called the Time-Independent Schrödinger Equation (TISE):

$$(E - V(x))\psi(x) = -\frac{\hbar^2}{2m} \partial_x^2 \psi(x) \quad (5.2.7)$$

We note that the Schrödinger Equation is *non-relativistic*.

Definition 5.2.4 (Hamiltonian)

The *Hamiltonian* operator is the same as in classical mechanics; it satisfies

$$\hat{H}|\psi\rangle = E|\psi\rangle \quad (5.2.8)$$

where E is the energy associated with the state $|\psi\rangle$.

Property 5.2.5 (Boundary Conditions for the SE)

Since the SE is first-order in time and second-order in position, an initial-time state as well as two position boundary conditions are requisite. Also, the SE is linear, and so any superpositions of solutions are also solutions.

Proposition 5.2.6 (Time Dependence of an Energy Eigenstate)

Consider a wavefunction $\psi(x)$ satisfying the TISE for a given potential $V(x)$, i.e.

$$\hat{H}\psi(x) = E\psi(x) \quad (5.2.9)$$

Thus, the time dependence is

$$\Psi(x, t) = \psi(x) \exp\left(-\frac{iEt}{\hbar}\right) \quad (5.2.10)$$

Proof

We write the Schrödinger Equation:

$$i\hbar \partial_t \Psi = \hat{H}\Psi \quad (5.2.11)$$

We separate the time and spatial dependences⁶ so that

$$\Psi(x, t) = T(t)\psi(x) \quad (5.2.12)$$

Thus,

$$i\hbar\partial_t\Psi = i\hbar T'(t)\psi(x) \quad (5.2.13)$$

and

$$\hat{H}\Psi = T(t)\hat{H}\psi(x) = T(t)E\psi(x) = ET(t)\psi(x) \quad (5.2.14)$$

Thus,

$$i\hbar T'(t)\cancel{\psi(x)} = ET(t)\cancel{\psi(x)} \Rightarrow T'(t) = \frac{E}{i\hbar}T(t) \Rightarrow T(t) = C_0 \exp\left(-\frac{iEt}{\hbar}\right) \quad (5.2.15)$$

The constant is specified implicitly to be 1 according to $\Psi(x, t_0) = \psi(x)$, which gives the desired solution. $\square$

Proposition 5.2.7 (Expectation Value of Potential)

Suppose one can expand a potential $\hat{V}(\hat{x})$ as a power series:

$$V(x) = \sum_{n=0}^{\infty} a_n x^n \quad (5.2.16)$$

The potential expectation can thus be expressed not only in position basis as

$$\langle V \rangle = \int_{-\infty}^{\infty} \psi^*(x) V(x) \psi(x) dx \quad (5.2.17)$$

but also in momentum basis as

$$\langle V \rangle = \int_{-\infty}^{\infty} \phi^*(p) \left(\sum_{n=0}^{\infty} a_n (i\hbar\partial_p)^n \right) \phi(p) dp \quad (5.2.18)$$

5.3 Infinite Square Well

We consider a potential that forces a particle to reside in a finite interval.

Definition 5.3.1 (Infinite Square Well)

The potential $V(x)$ given by

$$V(x) = \begin{cases} 0 & x \in [0, a] \\ \infty & \text{elsewhere} \end{cases} \quad (5.3.1)$$

is known as the *infinite square well*.

We also introduce some nomenclature:

Definition 5.3.2 (Bound State)

A *bound state* is a state in which the energy E is less than the local potential $V(x)$ at some x in the domain; such states are not classically allowed but are solutions in quantum mechanics.

We assume some physical aspects of solutions.

Axiom 5.3.3 (Continuity at a Potential Barrier)

We assume that wavefunctions ψ and their position derivatives ψ' are continuous at finite potential steps.

⁶While this is perhaps not the most general solution, we can construct the most general solution from a superposition of an infinite number of such separated solutions, in the spirit of Taylor, Fourier, Laurent, etc series.

Proposition 5.3.4 (Energy Eigenstates of the Infinite Square Well)

The stationary states of the infinite square well are given by

$$u_n = \sqrt{\frac{2}{a}} \sin\left(n\pi \frac{x}{a}\right) \cdot (\Theta(x) - \Theta(x-a)) \quad (5.3.2)$$

Proof

We write the time-independent Schrödinger Equation:

$$(E - V(x))\psi = -\frac{\hbar^2}{2m} \partial_x^2 \psi \quad (5.3.3)$$

We observe the wavefunction must be zero outside the well since the energy cannot be infinite. Thus, we consider only $x \in [0, a]$; in this range, $V(x) = 0$:

$$\partial_x^2 \psi = -\frac{2mE}{\hbar^2} \psi \quad (5.3.4)$$

This is a second order linear differential equation with the boundary conditions $\psi(0) = \psi(a) = 0$. Thus, the basis for solutions is the sine function, of the form

$$\psi(x) = k_n \sin\left(\frac{n\pi x}{a}\right) \quad (5.3.5)$$

We choose k_n so that

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1 \quad (5.3.6)$$

In this case, we have

$$1 = \int_{-\infty}^{\infty} |\psi|^2 dx = \int_0^a |k_n|^2 \sin^2\left(\frac{n\pi x}{a}\right) dx = |k_n|^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = |k_n|^2 \frac{a}{2} \quad (5.3.7)$$

Thus,

$$k_n = \sqrt{\frac{2}{a}} \quad (5.3.8)$$

for all n , so we find that the solutions to the TISE are

$$u_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) & x \in [0, a] \\ 0 & \text{elsewhere} \end{cases} = \boxed{\sqrt{\frac{2}{a}} \sin\left(n\pi \frac{x}{a}\right) \cdot (\Theta(x) - \Theta(x-a))} \quad (5.3.9)$$

as desired. We observe that these are orthonormal, since sine functions of different frequencies are orthogonal. $\square$

Proposition 5.3.5 (Quantized Energies of the Infinite Square Well)

The energies in the quantized levels of the infinite square well are given by

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2ma^2} \quad (5.3.10)$$

Proof

We quite simply evaluate the TISE:

$$E = -\frac{\hbar^2}{2m} \frac{\partial_x^2 \psi}{\psi} = -\frac{\hbar^2}{2m} \cdot \frac{-n^2 \pi^2 / a^2 \cdot \sin(n\pi x/a)}{\sin(n\pi x/a)} = \frac{\hbar^2 n^2 \pi^2}{2ma^2} \quad (5.3.11)$$

as desired. $\square$

Property 5.3.6 (ISW: Decomposition of an Arbitrary State)

Consider an arbitrary state $|\psi\rangle$ with wavefunction $\psi(x)$ residing in an infinite square well of size a . This state can be decomposed into energy eigenstates according to

$$|\psi\rangle = \sum c_n |u_n\rangle = \sum |u_n\rangle \langle u_n | \psi \rangle : c_n = \langle u_n | \psi \rangle \quad (5.3.12)$$

$$\iff \psi(x) = \sum c_n u_n(x) : c_n = \int_{-\infty}^{\infty} u_n^*(x) \psi(x) dx \quad (5.3.13)$$

Proposition 5.3.7 (ISW: Time Evolution of an Arbitrary State)

An arbitrary state $|\psi\rangle$ with wavefunction $\psi(x)$ evolves in time according to

$$\Psi(x, t) = \sum_n c_n \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \exp\left(-\frac{i\hbar n^2 \pi^2 t}{2ma^2}\right) \quad (5.3.14)$$

for

$$c_n = \langle u_n | \psi \rangle = \int_{-\infty}^{\infty} u_n^*(x) \psi(x) dx \quad (5.3.15)$$

Proof

We know that the individual energy eigenstate evolves in time according to

$$u_n(x, t) = u_n(x) \exp(-iE_n t/\hbar) \quad (5.3.16)$$

Thus,

$$\Psi(x, t) = \sum c_n u_n(x, t) = \sum c_n u_n(x) \exp(-iE_n t/\hbar) \quad (5.3.17)$$

For the infinite square well,

$$\Psi(x, t) = \sum_n c_n \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \exp\left(-\frac{i\hbar n^2 \pi^2 t}{2ma^2}\right) \quad (5.3.18)$$

for

$$c_n = \langle u_n | \psi \rangle = \int_{-\infty}^{\infty} u_n^*(x) \psi(x) dx \quad (5.3.19)$$

as desired. $\square$

5.4 Probability Current

In cases in which the probability distribution $\psi(x)$ cannot be normalized, such as plane waves, it is more useful to consider the *probability current*.

Definition 5.4.1 (Probability Current)

Let a quantum state $|\psi\rangle$ be described by the (potentially non-normalizable) wavefunction $\psi(x)$. The *probability current* $J(x)$ is given by

$$J(x) = \frac{\hbar}{m} \text{Im} \{ \psi^*(x) \cdot \partial_x \psi(x) \} = \frac{\hbar}{2im} (\psi^*(x) \partial_x \psi(x) - (\partial_x \psi^*(x)) \psi(x)) \quad (5.4.1)$$

Let us consider the basic plane wave, an eigenket of the momentum operator:

$$\psi(x) = u_p(x) = A \exp\left(\pm \frac{ipx}{\hbar}\right) \quad (5.4.2)$$

Proposition 5.4.2 (Probability Current Conservation Equation)

For a quantum state ψ and probability current J , we have

$$\partial_x J + \partial_t |\psi|^2 = 0 \quad (5.4.3)$$

Property 5.4.3 (Probability Current of a Plane Wave)

The probability current associated with a plane wave $\psi(x) = A \exp(\pm ipx/\hbar)$ of momentum p and complex amplitude A is $\pm |A|^2 p/m$.

Proof

The current associated is

$$J(x) = \frac{\hbar}{m} \operatorname{Im} \{ \psi^*(x) \cdot \partial_x \psi(x) \} = \frac{\hbar}{m} \operatorname{Im} \left\{ A^* \exp\left(\mp \frac{ipx}{\hbar}\right) \cdot \partial_x \left(A \exp\left(\pm \frac{ipx}{\hbar}\right) \right) \right\} \quad (5.4.4)$$

$$= \frac{|A|^2 \hbar}{m} \operatorname{Im} \left\{ \exp\left(\mp \frac{ipx}{\hbar}\right) \cdot \exp\left(\pm \frac{ipx}{\hbar}\right) \cdot \frac{\pm ip}{\hbar} \right\} = \pm |A|^2 \frac{p}{m} \quad (5.4.5)$$

We study a potential step.

Definition 5.4.4 (Potential Step)

The potential $V(x)$ given by

$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & x > 0 \end{cases} \quad (5.4.6)$$

is known as a *potential step* of size V_0 .

The right side of the state is either bound or not depending on $V_0 > E$. To solve for the wavefunction, we need to solve the Schrödinger Equation on either side of the step, and apply the following boundary conditions:

$$\psi(0^-) = \psi(0^+) \quad \partial_x \psi(0^-) = \partial_x \psi(0^+) \quad (5.4.7)$$

The Time-Independent Schrödinger Equation states

$$-\frac{2m}{\hbar^2} (E - V(x)) \psi(x) = \psi''(x) \quad (5.4.8)$$

On the left, this is

$$\psi'' = -\frac{2m}{\hbar^2} E \psi \quad (5.4.9)$$

On the right, this is

$$\psi'' = -\frac{2m}{\hbar^2} (E - V_0) \psi \quad (5.4.10)$$

In either case, we see that a superposition of plane waves with a particular momentum (k on the left, κ on the right) is a valid solution, where

$$k \equiv \sqrt{\frac{2mE}{\hbar^2}} \quad \kappa \equiv \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \quad (5.4.11)$$

We observe that κ is real if $E > V_0$, in the free state; otherwise, we define κ_{bound} according to

$$\kappa_b \equiv \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \in \mathbb{R} \quad (5.4.12)$$

and have a real exponential on the right hand side.

Free State

We propose the ansatz of an incident and reflected plane wave superposition:

$$\psi_L(x) = A \exp(ikx) + B \exp(-ikx) \quad (5.4.13)$$

We propose the ansatz of a transmitted plane wave on the right side:

$$\psi_R(x) = C \exp(i\kappa x) \quad (5.4.14)$$

We must match the boundary conditions, which state

$$\psi_L(0) = \psi_R(0) \quad \partial_x \psi_L(0) = \partial_x \psi_R(0) \quad (5.4.15)$$

The first implies $A + B = C$, the second gives

$$ikA - ikB = i\kappa C \Rightarrow A - B = \frac{\kappa}{k}C \quad (5.4.16)$$

We solve the system, leaving the amplitudes in terms of A and finding $r \equiv B/A$ and $\tau \equiv C/A$:

$$\begin{cases} 1 + r = \tau \\ 1 - r = (\kappa/k)\tau \end{cases} \quad (5.4.17)$$

Adding these gives

$$\tau = \frac{2}{\kappa/k + 1} = \frac{2k}{k + \kappa} \quad (5.4.18)$$

and

$$r = \frac{2k}{k + \kappa} - 1 = \frac{k - \kappa}{k + \kappa} \quad (5.4.19)$$

We note that these are not the probabilities; the reflection probability is

$$R = |r|^2 = \frac{(k - \kappa)^2}{(k + \kappa)^2} \quad (5.4.20)$$

and the transmission probability is

$$T = 1 - R = \frac{4k\kappa}{(k + \kappa)^2} = \frac{\kappa}{k} |\tau|^2 \quad (5.4.21)$$

where we note the prefactor accounts for the change in momentum.

The probability current of the constituents is

$$\text{Incident : } |A|^2 \frac{\hbar k}{m} \quad \text{Reflected : } -|A|^2 \frac{(k - \kappa)^2}{(k + \kappa)^2} \frac{\hbar k}{m} \quad \text{Transmitted : } |A|^2 \frac{4k^2}{(k + \kappa)^2} \frac{\hbar \kappa}{m} \quad (5.4.22)$$

Classically Forbidden Transmission

The ansatz is the same on the left:

$$\psi_L(x) = A \exp(ikx) + B \exp(-ikx) \quad (5.4.23)$$

The ansatz on the right is now a real exponential with the second definition for κ :

$$\psi_R(x) = C \exp(-\kappa x) + D \exp(\kappa x) \quad (5.4.24)$$

We can eliminate the D -coefficient term according to a no-divergence condition at infinity, namely

$$\exists B \in \mathbb{R}^+ : |\psi(\pm\infty)| \leq B \quad (5.4.25)$$

The boundary conditions at $x = 0$ have not changed; we still require

$$\psi_L(0) = \psi_R(0) \quad \partial_x \psi_L(0) = \partial_x \psi_R(0) \quad (5.4.26)$$

Thus,

$$A + B = C \quad ; \quad ikA - ikB = -\kappa C \quad (5.4.27)$$

We again work with $r \equiv B/A$ and $\tau \equiv C/A$:

$$1 + r = \tau \quad 1 - r = i\kappa\tau/k \quad (5.4.28)$$

Thus,

$$\tau = \frac{2k}{k + i\kappa} \quad ; \quad r = \frac{k - i\kappa}{k + i\kappa} \quad \Rightarrow \quad R = |r|^2 \equiv \frac{|k - i\kappa|^2}{|k + i\kappa|^2} = 1 \quad ; \quad \Rightarrow \quad T = 0 \quad (5.4.29)$$

While it is impossible to transmit through a potential barrier of height $V_0 > E$, the wavefunction is not zero there; thus, there is hope for tunneling.

We calculate the probability current in the classically forbidden region:

$$J_R(x) \equiv \frac{\hbar}{m} \text{Im} \{ \psi_R^*(x) \cdot \partial_x \psi_R(x) \} = \frac{\hbar}{m} \text{Im} \{ C^* \exp(-\kappa x) \cdot \partial_x (C \exp(-\kappa x)) \} = \frac{\hbar}{m} |C|^2 \text{Im} \{ -\kappa \exp(-2\kappa x) \} \quad (5.4.30)$$

$$= 0 \quad (5.4.31)$$

since clearly ψ_R has no imaginary component other than potentially the factor C which removes itself by virtue of having its complex conjugate in the expression whose imaginary value we take.

Proposition 5.4.5 (Step Potential Scattering)

Consider a potential step of size V_0 , where $V(x) = 0$ for $x < 0$ and $V(x) = V_0 > 0$ for $x > 0$. Consider the wavefunction $\psi(x)$ corresponding to a scattering state, in which plane waves arrive with energy E and are reflected/transmitted by the step in potential.

In the case of $E > V_0$, we have

$$k \equiv \sqrt{\frac{2mE}{\hbar^2}}; \quad \kappa \equiv \sqrt{\frac{2m(E - V_0)}{\hbar^2}} : \quad \psi(x) = A \begin{cases} \exp(ikx) + \frac{k - \kappa}{k + \kappa} \exp(-ikx) & x < 0 \\ \frac{2k}{k + \kappa} \exp(i\kappa x) & x > 0 \end{cases} \quad (5.4.32)$$

where an incident particle is reflected with probability R or transmitted with probability T given by

$$R = \frac{(k - \kappa)^2}{(k + \kappa)^2} = |r|^2 \quad ; \quad T = \frac{4k\kappa}{(k + \kappa)^2} = \frac{\kappa}{k} |\tau|^2 \quad (5.4.33)$$

where r, τ are the reflection and transmission coefficients in $\psi(x)$. The probability current of the constituents is

$$\text{Incident} : |A|^2 \frac{\hbar k}{m} \quad \text{Reflected} : -|A|^2 \frac{(k - \kappa)^2}{(k + \kappa)^2} \frac{\hbar k}{m} \quad \text{Transmitted} : |A|^2 \frac{4k^2}{(k + \kappa)^2} \frac{\hbar \kappa}{m} \quad (5.4.34)$$

In the case of $E < V_0$, we have

$$k \equiv \sqrt{\frac{2mE}{\hbar^2}}; \quad \kappa \equiv \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} : \quad \psi(x) = A \begin{cases} \exp(ikx) + \frac{k - i\kappa}{k + i\kappa} \exp(-ikx) & x < 0 \\ \frac{2k}{k + i\kappa} \exp(-\kappa x) & x > 0 \end{cases} \quad (5.4.35)$$

where an incident particle is reflected with probability $R = 1$ or transmitted with probability $T = 0$. The probability current of the constituents is

$$\text{Incident} : |A|^2 \frac{\hbar k}{m} \quad \text{Reflected} : -|A|^2 \frac{\hbar k}{m} \quad \text{Transmitted} : 0 \quad (5.4.36)$$

The reflected constituent has a phase shift ϕ given by $\tan \phi = \frac{-2k\kappa}{(k^2 - \kappa^2)}$.

We may superimpose step potentials to form a potential barrier:

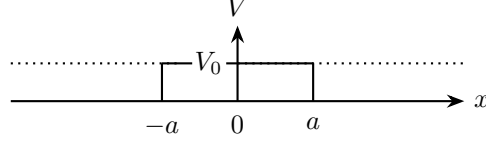


Figure 10. Potential Barrier

For energies $E < V_0$, the wavefunction is, as before, classically forbidden in the potential barrier region, but a superposition of bound states is allowed:

$$u_n(x \mid x \in [-a, a]) = F \exp(-\kappa x) + G \exp(\kappa x) \quad (5.4.37)$$

where F and G are chosen to match the boundary at $-a$ and a . The full ansatz is

$$\text{Left } (x \leq -a): \quad u_n^L = A \exp(ikx) + B \exp(-ikx) \quad (5.4.38)$$

$$\text{Well } (x \in [-a, a]): \quad u_n^W = F \exp(\kappa x) + G \exp(-\kappa x) \quad (5.4.39)$$

$$\text{Right } (x \geq a): \quad u_n^R = C \exp(ikx) \quad (5.4.40)$$

where

$$k = \sqrt{\frac{2m(E)}{\hbar^2}} \quad ; \quad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \quad (5.4.41)$$

We must satisfy the boundary conditions, namely $u(\pm a^-) = u(\pm a^+)$ and $u'(\pm a^-) = u'(\pm a^+)$:

$$x = -a: \quad Ae^{-ika} + Be^{ika} = Ge^{\kappa a} + Fe^{-\kappa a} \quad ; \quad ikA \exp(-ika) - ikB \exp(ika) = \kappa Ge^{\kappa a} + \kappa Fe^{-\kappa a} \quad (5.4.42)$$

$$x = a: \quad Ce^{ika} = Ge^{-\kappa a} + Fe^{\kappa a} \quad ; \quad ikC \exp(ika) = \kappa Ge^{-\kappa a} + \kappa Fe^{\kappa a} \quad (5.4.43)$$

We can eliminate G and F through several steps of algebra. At $x = -a$,

$$\frac{ikAe^{-ika} - ikBe^{ika}}{Ae^{-ika} + Be^{ika}} = \frac{-\kappa Ge^{\kappa a} + \kappa Fe^{-\kappa a}}{Ge^{\kappa a} + Fe^{-\kappa a}} \quad (5.4.44)$$

At $x = a$,

$$ik = \frac{ikCe^{ika}}{Ce^{ika}} = \frac{-\kappa Ge^{-\kappa a} + \kappa Fe^{\kappa a}}{Ge^{-\kappa a} + Fe^{\kappa a}} \quad (5.4.45)$$

From this last equation, we write G as a function of F :

$$ikGe^{-\kappa a} + ikFe^{\kappa a} = -\kappa Ge^{-\kappa a} + \kappa Fe^{\kappa a} \quad (5.4.46)$$

$$\Rightarrow (\kappa + ik)Ge^{-\kappa a} = (\kappa - ik)Fe^{\kappa a} \quad (5.4.47)$$

$$\Rightarrow G = \frac{\kappa - ik}{\kappa + ik} Fe^{2\kappa a} \quad (5.4.48)$$

We recognize the reflection term. We can substitute into the $x = -a$ equation:

$$\frac{ikAe^{-ika} - ikBe^{ika}}{Ae^{-ika} + Be^{ika}} = \frac{-\kappa Ge^{\kappa a} + \kappa Fe^{-\kappa a}}{Ge^{\kappa a} + Fe^{-\kappa a}} \quad (5.4.49)$$

$$= \frac{-\kappa \frac{\kappa - ik}{\kappa + ik} Fe^{3\kappa a} + \kappa Fe^{-\kappa a}}{\frac{\kappa - ik}{\kappa + ik} Fe^{3\kappa a} + Fe^{-\kappa a}} \quad (5.4.50)$$

$$= \frac{-\kappa^2 (e^{2\kappa a} - e^{-2\kappa a}) + ik\kappa (e^{2\kappa a} + e^{-2\kappa a})}{\kappa (e^{2\kappa a} + e^{-2\kappa a}) - ik (e^{2\kappa a} - e^{-2\kappa a})} \quad (5.4.51)$$

$$= -\frac{\kappa^2 \sinh(2\pi a) + ik\kappa \cosh(2\kappa a)}{\kappa \cosh(2\kappa a) - ik \sinh(2\kappa a)} \quad (5.4.52)$$

Thus, we can write

$$Ae^{-ika}((k^2 + \kappa^2) \sinh(2\kappa a)) = Be^{ika}(2ik\kappa \cosh(2\kappa a) + (k^2 - \kappa^2) \sinh(2\kappa a)) \quad (5.4.53)$$

The reflection coefficient will be

$$r \equiv \frac{B}{A} = e^{-2ika} \frac{-i(k^2 + \kappa^2) \sinh(2\kappa a)}{2k\kappa \cosh(2\kappa a) - i(k^2 - \kappa^2) \sinh(2\kappa a)} \quad (5.4.54)$$

The transmission coefficient depends on C :

$$Ce^{ika} = \frac{2\kappa}{\kappa + ik} Fe^{\kappa a} \quad (5.4.55)$$

F is found from continuity at $-a$:

$$Ae^{-ika} + Be^{ika} = Ee^{\kappa a} + Fe^{-\kappa a} \quad (5.4.56)$$

The right hand side can be expressed as

$$Ee^{\kappa a} + Fe^{-\kappa a} = \left(\frac{\kappa - ik}{\kappa + ik} e^{3\kappa a} + e^{-\kappa a} \right) F \quad (5.4.57)$$

$$= Fe^{\kappa a} \left(\frac{\kappa - ik}{\kappa + ik} e^{2\kappa a} + \frac{\kappa + ik}{\kappa + ik} e^{-2\kappa a} \right) \quad (5.4.58)$$

$$= Fe^{\kappa a} \frac{2\kappa \cosh(2\kappa a) - 2ik \sinh(2\kappa a)}{\kappa + ik} \quad (5.4.59)$$

The left hand side can be expressed as

$$Ae^{-ika} + Be^{ika} = Ae^{-ika} \frac{2ik\kappa \cosh(2\kappa a) + 2k^2 \sinh(2\kappa a)}{2ik\kappa \cosh(2\kappa a) + (k^2 - \kappa^2) \sinh(2\kappa a)} \quad (5.4.60)$$

Thus, the transmission coefficient C/A is given by

$$\tau \equiv \frac{C}{A} \quad (5.4.61)$$

$$= \frac{2\kappa}{A} \frac{F}{\kappa + ik} e^{\kappa a - ika} \quad (5.4.62)$$

$$= e^{-2ika} \frac{2k\kappa}{2k\kappa \cosh(2\kappa a) - i(k^2 - \kappa^2) \sinh(2\kappa a)} \quad (5.4.63)$$

Proposition 5.4.6 (Scattering from a Potential Barrier)

Consider a potential barrier of height V_0 and size $2a$, over the interval $[-a, a]$. Let $0 < E < V_0$ be the energy of incoming (scattering) momentum eigenstates. Let

$$k \equiv \sqrt{\frac{2mE}{\hbar^2}} \quad ; \quad \kappa \equiv \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \quad (5.4.64)$$

The reflection coefficient r and transmission coefficient τ are given by

$$r = e^{-2ika} \frac{-i(k^2 + \kappa^2) \sinh(2\kappa a)}{2k\kappa \cosh(2\kappa a) - i(k^2 - \kappa^2) \sinh(2\kappa a)} \quad (5.4.65)$$

$$\tau = e^{-2ika} \frac{2k\kappa}{2k\kappa \cosh(2\kappa a) - i(k^2 - \kappa^2) \sinh(2\kappa a)} \quad (5.4.66)$$

The energy and particle velocity are the same on either side of the barrier, and so $|r|^2 + |\tau|^2 = 1$.

In the limit of $\kappa a \gg 1$, we have

$$T \equiv |\tau|^2 = \frac{(2k\kappa)^2}{(2k\kappa)^2 + (k^2 + \kappa^2)^2 \sinh^2(2\kappa a)} \rightarrow \left(\frac{4k\kappa}{k^2 + \kappa^2} \right)^2 e^{-4\kappa a} \quad (5.4.67)$$

Proposition 5.4.7 (Localization of a Particle in the Forbidden Region)

An energy measurement with sufficient resolution to find a particle in the classically forbidden region transfers sufficient energy to allow the particle to exist in that region.

Proof

$$\langle \Delta E \rangle = \frac{(\Delta p)^2}{2m} \geq \frac{\hbar^2}{2m(\Delta x)^2} \geq \frac{\hbar^2 \kappa^2}{2m} = V_0 - E \quad (5.4.68)$$

as desired. $\square$

5.5 Finite Square Well

We return to the square wells. What if the potential was not infinite outside of the well ?

Definition 5.5.1 (Finite Square Well)

The potential $V(x)$ given by

$$V(x) = \begin{cases} 0 & x \in [-a, a] \\ V_0 & \text{elsewhere} \end{cases} \quad (5.5.1)$$

is known as the *finite square well*.

Proposition 5.5.2 (Energy Eigenstates of the Finite Square Well)

The energy eigenstates of the finite square well are either “scattering” or “bound”; there always exists an even bound state solution, given by solutions to

$$\sqrt{\lambda - y_n^2} = y_n \tan y_n \quad (5.5.2)$$

where $y = a\sqrt{2mE_n/\hbar^2}$ and $\lambda = 2mV_0a^2/\hbar^2$. The odd solutions, should they exist, are given by

$$\sqrt{\lambda - y_n^2} = -y_n \cot y_n \quad (5.5.3)$$

The solutions are approximately given by $y = ka = (n + 1/2)\pi$.

Scattering states have a continuous energy distribution, with reflection and transmission coefficients given by

$$r = e^{-2ika} \frac{2kq}{2kq \cos(2qa) - i(q^2 + k^2) \sin(2qa)} \quad ; \quad \tau = e^{-2ika} \frac{2kq}{2kq \cos(2qa) - i(q^2 + k^2) \sin(2qa)} \quad (5.5.4)$$

No reflection will occur for $E_n^{\text{resonance}} = (\hbar^2 n^2 \pi^2) / 8ma^2 - V_0$.

Proof

We recognize that there exist both bound and free solutions, governed by $E < V_0$ for bound and $E > V_0$ for scattering ones.

Bound States

Outside the potential well, we require

$$\text{Left : } u_n'' = \frac{2m}{\hbar^2}(V_0 - E)u_n \equiv \kappa^2 u_n \Rightarrow u_n = A_L \exp(\kappa x) + B_L \exp(-\kappa x) \quad (5.5.5)$$

$$\text{Right : } u_n'' = \frac{2m}{\hbar^2}(V_0 - E)u_n \equiv \kappa^2 u_n \Rightarrow u_n = A_R \exp(\kappa x) + B_R \exp(-\kappa x) \quad (5.5.6)$$

We note that to prevent divergence of the wavefunction at infinity, we require $B_L = 0$ and $A_R = 0$. Inside the potential well, we require

$$\text{Well : } u_n'' = -\frac{2m}{\hbar^2}Eu_n \equiv -k^2 u_n \Rightarrow u_n = A_W \exp(ikx) + B_W \exp(-ikx) \quad (5.5.7)$$

We observe a symmetry in the potential well⁷, which allows us to conclude that the basis of solutions inside the well must also be symmetric. We thus obtain both even and odd solutions.

Instead of dealing with complex exponentials, we simplify to sines and cosines. The even solutions are of course formed of cosines, so

$$u_n^W = C_W \cos(kx) \quad (5.5.8)$$

We require

$$u_n^L(-a) = u_n^W(-a) \quad ; \quad \partial_x u_n^L(-a) = \partial_x u_n^W(-a) \quad (5.5.9)$$

The first boundary condition gives

$$A_L \exp(-\kappa a) = C_W \cos(-ka) = C_W \cos(ka) \quad (5.5.10)$$

The second boundary condition gives

$$-\kappa A_L \exp(-\kappa a) = k C_W \sin(-ka) = -k C_W \sin(ka) \quad (5.5.11)$$

We can obtain an expression for the tangent of ka :

$$\tan(ka) = \frac{\sin(ka)}{\cos(ka)} = \frac{\kappa A_L \exp(-\kappa a)}{k C_W} \cdot \left(\frac{A_L \exp(-\kappa a)}{C_W} \right)^{-1} \quad (5.5.12)$$

$$= \frac{\kappa}{k} = \sqrt{\frac{2m}{\hbar^2}(V_0 - E)} \cdot \left(\sqrt{\frac{2mE}{\hbar^2}} \right)^{-1} = \sqrt{\frac{V_0}{E} - 1} \quad (5.5.13)$$

For odd solutions, we write with sine rather than cosine:

$$u_n^W = C_W \sin(kx) \quad (5.5.14)$$

The boundary conditions give identical results up to a sign difference and reciprocal:

$$\tan(ka) = -\frac{k}{\kappa} = -\sqrt{\frac{E}{V_0 - E}} \quad (5.5.15)$$

It is now convenient to introduce

$$y \equiv ka \quad ; \quad \lambda = \frac{2m}{\hbar^2} V_0 a^2 \quad (5.5.16)$$

We write

$$\kappa a = \sqrt{\frac{2m}{\hbar^2}(V_0 - E)a^2} = \sqrt{\lambda - \frac{2m}{\hbar^2} E a^2} = \sqrt{\lambda - k^2 a^2} = \sqrt{\lambda - y^2} \quad (5.5.17)$$

We can then express the even and odd boundary conditions as

$$\text{Even : } \frac{\sqrt{\lambda - y^2}}{y} = \tan y \quad ; \quad \text{Odd : } \frac{\sqrt{\lambda - y^2}}{y} = -\cot y \quad (5.5.18)$$

The question then becomes if there exist y that satisfy the above equations for some potential-dependent λ . We note that λ does not depend on E , and so the solution(s) obtained y will implicitly determine E according to

$$E = \frac{\hbar^2}{2m} \frac{y}{a} \quad (5.5.19)$$

We can plot to inspect what kind of solutions exist for various potentials.

⁷One may argue that this symmetry also existed for the infinite square well, and while this is true, we were not forced to harness it since the well from $0 \rightarrow a$ allows for a simple solution basis of the form $\sin(n\pi x/a)$ as opposed to $-a \rightarrow a$, which requires both even and odd solutions. Here, should we have had the well from $0 \rightarrow a$ only, then we have introduced an asymmetry, since $\psi \neq 0$ for $x \notin [0, a]$ for a finite square well. Thus, the treatment is simplified for the infinite square well in a way that cannot be replicated for the finite one.

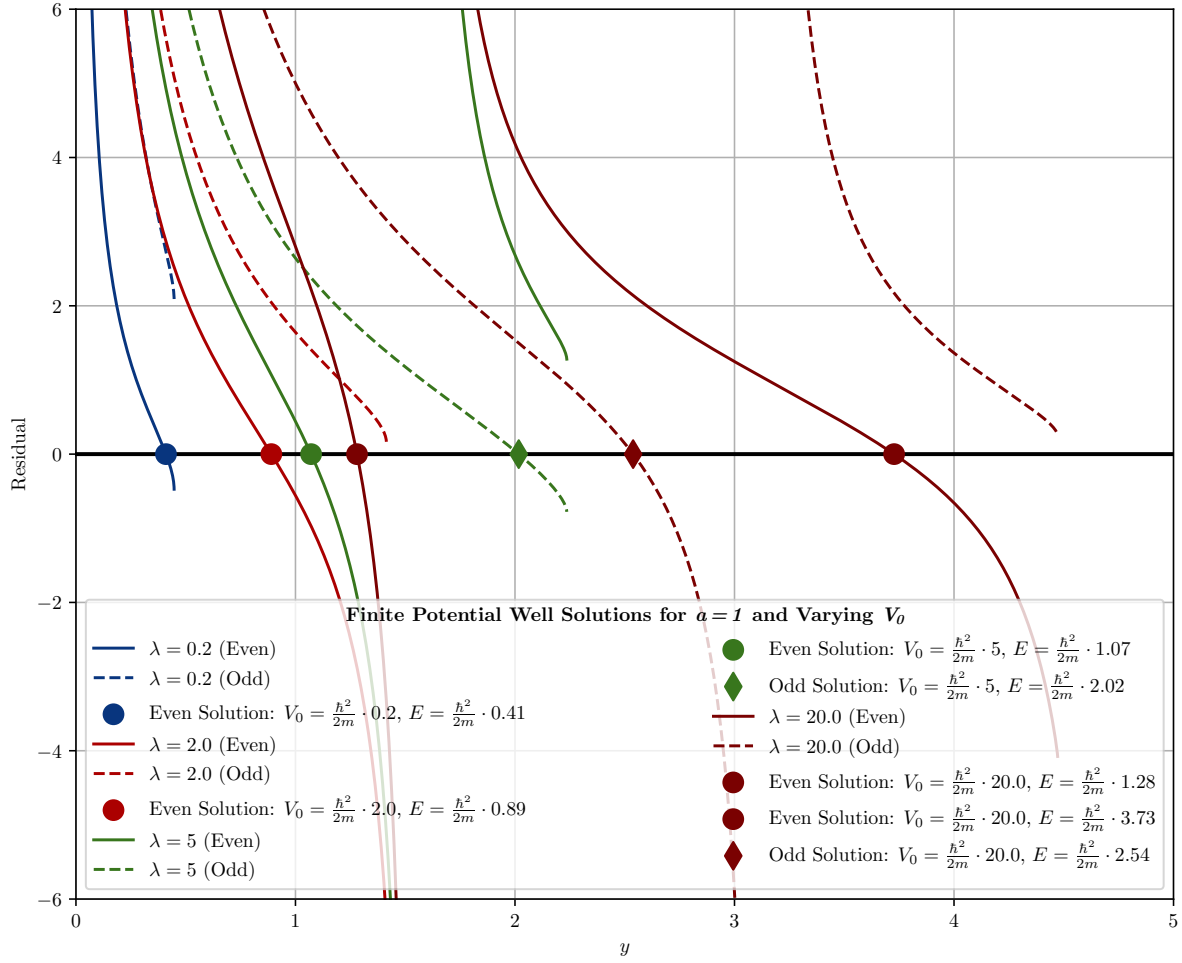


Figure 11. Even and Odd Bound Energy Eigenstates in the Finite Square Well.

We note that there will always exist an even solution. For $\lambda \gg 1$, the solutions are approximately given by

$$y = ka = \left(n + \frac{1}{2}\right) \pi \quad (5.5.20)$$

We see that for $\lambda \sim 10^3$ this holds well for about 7 solutions:

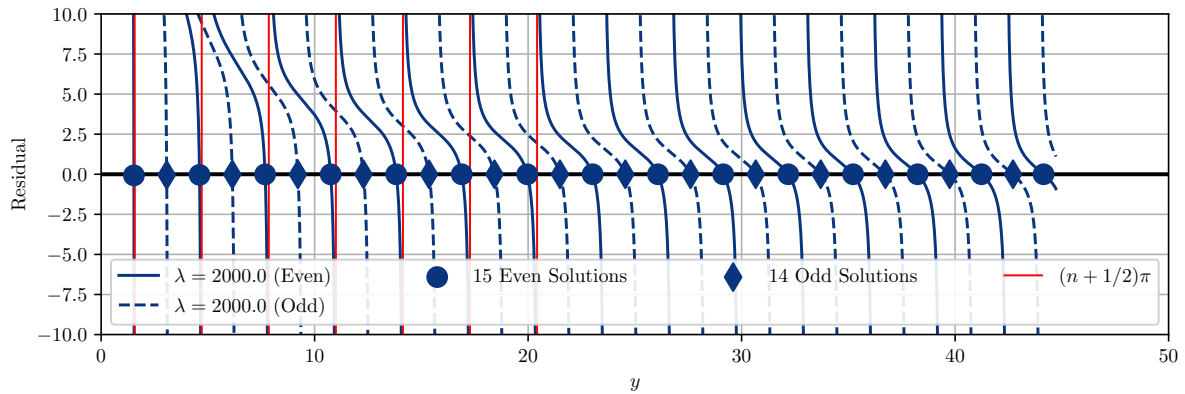


Figure 12. Even and Odd Bound Energy Eigenstates for $\lambda = 2 \times 10^3$; the approximation $y_n \approx (n + 1/2)\pi$ holds for the first half of the states.

Odd solutions will only exist if

$$\sqrt{\lambda - \left(\frac{\pi}{2}\right)^2} \geq 0 \iff \frac{2mV_0a^2}{\hbar^2} \geq \frac{\pi^2}{4} \quad (5.5.21)$$

which clearly does not hold for the red curves in Fig. 11.

Scattering States

Scattering states will have $E > V_0$, and so we can write the ansatz below:

$$\text{Left } (x \leq -a): \quad u_n^L = A \exp(ikx) + B \exp(-ikx) \quad (5.5.22)$$

$$\text{Well } (x \in [-a, a]): \quad u_n^W = G \exp(iqx) + F \exp(-iqx) \quad (5.5.23)$$

$$\text{Right } (x \geq a): \quad u_n^R = C \exp(ikx) \quad (5.5.24)$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}} \quad ; \quad q = \sqrt{\frac{2m(V_0 + E)}{\hbar^2}} \quad (5.5.25)$$

We may replace κ in the potential barrier with $-iq$ to obtain

$$r = e^{-2ika} \frac{-i(k^2 + \kappa^2) \sinh(2\kappa a)}{2k\kappa \cosh(2\kappa a) - i(k^2 - \kappa^2) \sinh(2\kappa a)} \quad (5.5.26)$$

$$\rightarrow e^{-2ika} \frac{-i(k^2 + (-iq)^2) \sinh(2(-iq)a)}{2k(-iq) \cosh(2(-iq)a) - i(k^2 - (-iq)^2) \sinh(2(-iq)a)} \quad (5.5.27)$$

$$= \boxed{ie^{-2ika} \frac{(q^2 - k^2) \sin(2qa)}{2kq \cos(2qa) - i(q^2 + k^2) \sin(2qa)}} \quad (5.5.28)$$

$$\tau = e^{-2ika} \frac{2k\kappa}{2k\kappa \cosh(2\kappa a) - i(k^2 - \kappa^2) \sinh(2\kappa a)} \quad (5.5.29)$$

$$\rightarrow e^{-2ika} \frac{2k(-iq)}{2k(-iq) \cosh(2(-iq)a) - i(k^2 - (-iq)^2) \sinh(2(-iq)a)} \quad (5.5.30)$$

$$= \boxed{e^{-2ika} \frac{2kq}{2kq \cos(2qa) - i(q^2 + k^2) \sin(2qa)}} \quad (5.5.31)$$

where we have used

$$\cosh(-ix) = \cos(x) \quad \text{and} \quad \sinh(-ix) = -i \sin(x) \quad (5.5.32)$$

The “wavelengths” for which reflection goes to zero are

$$\lambda_n = \frac{4a}{n} \Rightarrow q_n = \frac{2\pi}{\lambda_n} = \frac{n\pi}{2a} \quad (5.5.33)$$

which corresponds to an energy

$$E_n = \frac{\hbar^2 q_n^2}{2m} - V_0 = \frac{\hbar^2 n^2 \pi^2}{8ma^2} - V_0 \quad (5.5.34)$$

We have now fully characterized the finite square well. □

5.6 Dirac Potentials

The potential $V(x)$ may contain delta-function terms, for example

$$V(x) = \lambda \delta(x - a) \quad (5.6.1)$$

We derive the condition of continuity at such a potential according to

$$\partial_x u|_{x=0^+} - \partial_x u|_{x=0^-} = \int_{0^-}^{0^+} dx \frac{d}{dx} \frac{du}{dx} = \int_{0^-}^{0^+} dx u'' = -\frac{2m}{\hbar^2} \int_{0^-}^{0^+} dx (E - V(x))u(x) \quad (5.6.2)$$

Proposition 5.6.1 (Continuity Condition at a Delta-Function Potential)

Consider a delta-function potential at $x = a$ of strength λ . The wavefunction ψ must satisfy

$$\psi'(a^+) - \psi'(a^-) = \frac{2m}{\hbar^2} \lambda \psi(a) \quad (5.6.3)$$

Property 5.6.2 (Bound States of a Single Delta-Function Potential)

Consider a delta function potential of the form

$$V(x) = -\lambda \delta(x - a) \quad (5.6.4)$$

There is exactly one bound state; it has an energy $E_1 = -m\lambda^2/2\hbar^2$. The wavefunction is given by

$$u_1(x) = A \exp\left(\frac{-m\lambda \cdot (x - a)}{\hbar^2} \cdot \text{sgn}(x - a)\right) \quad (5.6.5)$$

Proof

We suppose the delta function is at 0, then we shift $x \rightarrow (x - a)$. The ansatz to use is a real exponential on either side; we keep the non-diverging kind on each side:

$$u_L = A \exp(\kappa x) \quad ; \quad u_R = D \exp(-\kappa x) \quad (5.6.6)$$

Boundary conditions require

$$u_L(0) = u_R(0) \quad ; \quad u'_R(0) - u'_L(0) = -\frac{2m\lambda}{\hbar^2} u_L(0) \quad (5.6.7)$$

Thus,

$$A = D \quad \text{and} \quad -D\kappa e^{-\kappa x} - A\kappa e^{\kappa x} = -\frac{2m\lambda}{\hbar^2} A \quad (5.6.8)$$

Thus,

$$\kappa = \frac{m\lambda}{\hbar^2} \quad (5.6.9)$$

which gives

$$E_1 = -\frac{\hbar^2 \kappa^2}{2m} = -\frac{m\lambda^2}{2\hbar^2} \quad (5.6.10)$$

which implies

$$u_1 = A \exp(-\kappa x \cdot \text{sgn}(x)) = A \exp(-\kappa x \cdot \text{sgn}(x)) = A \exp\left(\frac{-m\lambda \cdot x}{\hbar^2} \cdot \text{sgn}(x)\right) \quad (5.6.11)$$

Replacement of $x \rightarrow (x - a)$ gives the desired expressions. $\square$

Property 5.6.3 (Bound States of a Double Delta-Function Potential)

Consider a potential of the form

$$V(x) = -\lambda\delta(x-a) - \lambda\delta(x+a) \quad (5.6.12)$$

for $\lambda > 0$. An even bound state solution always exists, and an odd bound state solution can exist under certain conditions.

Proof

We begin with even solutions. The solution will be of the form

$$u_n(x) = \begin{cases} Ae^{\kappa x} & x < -a \\ 2B \cosh(\kappa x) & x \in [-a, a] \\ Ae^{-\kappa x} & x > a \end{cases} \quad (5.6.13)$$

We enforce continuity at $x = \pm a$:

$$2B \cosh(\kappa a) = Ae^{-\kappa a} \quad (5.6.14)$$

We also enforce continuity of the derivative at $x \pm a$:

$$-\kappa Ae^{-\kappa a} - \kappa \cdot 2B \sinh(\kappa a) = -\frac{2m}{\hbar^2} \lambda Ae^{-\kappa a} \quad (5.6.15)$$

$$\Rightarrow \left(\frac{2m}{\hbar^2} \lambda - \kappa \right) Ae^{-\kappa a} = 2\kappa B \sinh(\kappa a) \quad (5.6.16)$$

$$\Rightarrow \left(\frac{2m}{\hbar^2} \lambda - \kappa \right) 2B \cosh(\kappa a) = 2\kappa B \sinh(\kappa a) \quad (5.6.17)$$

$$\Rightarrow \frac{2ma\lambda}{\hbar^2 \kappa a} - 1 = \tanh(\kappa a) \quad (5.6.18)$$

Thus, we can always find a solution of this form; the energy is given by

$$E = -\frac{\hbar^2 \kappa^2}{2m} < 0 \quad (5.6.19)$$

The dimensionless term κa determines this energy; the hyperbolic tangent starts from 0 and approaches 1 for large κa , while the left side of the equation will start at positive infinity for $\kappa a = 0$ and decrease to -1 , requiring some intersection κa for which both sides are equal. Concretely,

$$\frac{ma\lambda}{\hbar^2} < \kappa a < \frac{2ma\lambda}{\hbar^2} \quad (5.6.20)$$

which is equivalent to

$$\frac{m\lambda}{\hbar^2} < \kappa < \frac{2m\lambda}{\hbar^2} \quad (5.6.21)$$

The particle is more strongly bound here as compared to the single δ -potential, where we had

$$\kappa_1 = \frac{m\lambda}{\hbar^2} < \kappa_2 \quad (5.6.22)$$

5.6.1 Ehrenfest Theorem

We try to relate CM to QM with CM as the limiting case of QM with $\hbar \rightarrow 0$. In CM, of course, we have

$$p = mv = m\dot{x} \quad (5.6.23)$$

Let us relate this to expectations.

Proposition 5.6.4 (Momentum from Product of Mass and “Velocity”)

We claim that for an arbitrary normalizable quantum state Ψ ,

$$\langle p \rangle = m \frac{d}{dt} \langle x \rangle \quad (5.6.24)$$

Proof

$$m \langle \dot{x} \rangle = m \frac{d}{dt} \langle x \rangle \quad (5.6.25)$$

$$= m \frac{d}{dt} \int_{-\infty}^{\infty} dx \Psi^* x \Psi \quad (5.6.26)$$

$$= m \int_{-\infty}^{\infty} dx (\partial_t \Psi^* x \Psi + \Psi^* x \partial_t \Psi) \quad (5.6.27)$$

$$= m \int_{-\infty}^{\infty} dx \left(-\frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \partial_x^2 \Psi^* + V(x) \Psi^* \right) x \Psi + \Psi^* x \frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \partial_x^2 \Psi + V(x) \Psi \right) \right) \quad (5.6.28)$$

$$= \frac{\hbar}{2i} \int_{-\infty}^{\infty} dx (\partial_x^2 \Psi^* x \Psi - \Psi^* x \partial_x^2 \Psi) - \frac{1}{i\hbar} \int_{-\infty}^{\infty} dx (\Psi^* V(x) x \Psi - \Psi^* x V(x) \Psi) \quad (5.6.29)$$

We prepare integrate the first integral by parts:

$$\partial_x^2 \Psi^* x \Psi = \partial_x (\partial_x \Psi^* x \Psi) - \partial_x \Psi^* \partial_x (x \Psi) \quad (5.6.30)$$

$$= \partial_x (\partial_x \Psi^* x \Psi) - \partial_x \Psi^* \Psi - \partial_x \Psi^* x \partial_x \Psi \quad (5.6.31)$$

The second term in the first integral is the complex conjugate of the first since x is real, and so

$$(\partial_x^2 \Psi^* x \Psi - \Psi^* x \partial_x^2 \Psi) = \partial_x (\partial_x \Psi^* \Psi - \Psi^* x \partial_x \Psi) - \partial_x \Psi^* \Psi + \Psi^* \partial_x \Psi \quad (5.6.32)$$

$$= \partial_x (\partial_x \Psi^* x \Psi - \Psi^* x \partial_x \Psi) - \partial_x (\Psi^* \Psi) + 2\Psi^* \partial_x \Psi \quad (5.6.33)$$

A normalizable wavefunction Ψ must vanish at $x = \pm\infty$ faster than $1/\sqrt{x}$; thus, the first two terms will easily be zero and so we only integrate

$$m \frac{d}{dt} \langle x \rangle = \frac{\hbar}{2i} \int_{-\infty}^{\infty} dx d\Psi^* \partial_x \Psi \quad (5.6.34)$$

$$= \int_{-\infty}^{\infty} dx \Psi^* \left(\frac{\hbar}{i} \partial_x \right) \Psi \quad (5.6.35)$$

$$= \int_{-\infty}^{\infty} dx \Psi^* \hat{p} \Psi \quad (5.6.36)$$

$$= \langle p \rangle \quad (5.6.37)$$

as desired. $\square$

Theorem 5.6.5 (Ehrenfest)

For an arbitrary quantum state Ψ ,

$$\frac{d}{dt} \langle p \rangle = - \langle \partial_x V \rangle \quad (5.6.38)$$

Proof

We calculate the expectation of dp/dt :

$$\left\langle \frac{dp}{dt} \right\rangle = \frac{d}{dt} \langle p \rangle \quad (5.6.39)$$

$$= \partial_t \int_{-\infty}^{\infty} dx \Psi^* \left(\frac{\hbar}{i} \partial_x \right) \Psi \quad (5.6.40)$$

$$= \frac{\hbar}{i} \int_{-\infty}^{\infty} dx (\partial_t \Psi^* \partial_x \Psi + \Psi^* \partial_x \partial_t \Psi) \quad (5.6.41)$$

$$= \int_{-\infty}^{\infty} dx \left(-\frac{\hbar^2}{2m} \partial_x^2 \Psi^* \partial_x \Psi + V \Psi^* \partial_x \Psi - \Psi^* \partial_x \left(-\frac{\hbar^2}{2m} \partial_x^2 \Psi + V \Psi \right) \right) \quad (5.6.42)$$

The integrand simplifies to

$$A = -\frac{\hbar^2}{2m} \partial_x^2 \Psi^* \partial_x \Psi + V \Psi^* \partial_x \Psi + \frac{\hbar^2}{2m} \partial_x (\Psi^* \partial_x^2 \Psi) - \frac{\hbar^2}{2m} \partial_x \Psi^* \partial_x^2 \Psi - \Psi^* \partial_x V \Psi - \Psi^* V \partial_x \Psi \quad (5.6.43)$$

$$= \frac{\hbar^2}{2m} \partial_x (\Psi^* \partial_x^2 \Psi - \partial_x \Psi^* \partial_x \Psi) - \Psi^* \partial_x V \Psi \quad (5.6.44)$$

The first term gives zero since $\Psi \rightarrow 0$ for $x \rightarrow \pm\infty$. Thus,

$$\left\langle \frac{dp}{dt} \right\rangle = \int_{-\infty}^{\infty} dx \Psi^* (-\partial_x V) \Psi = \langle -\partial_x V \rangle \quad (5.6.45)$$

which is equivalent to

$$\frac{d}{dt} \langle p \rangle = - \langle \partial_x V \rangle \quad (5.6.46)$$

as desired. $\square$

Property 5.6.6 (“Force” on a Quantum State)

If we define $F(x) \equiv -\partial_x V$ in the classical sense, then we can obtain

$$\frac{d}{dt} \langle p \rangle = \int_{-\infty}^{\infty} dx F(x) |\Psi|^2 = \langle \Psi | F | \Psi \rangle \quad (5.6.47)$$

Consider a two-peaked wavefunction – the potential in between the peaks thus cannot be of much consequence for the particle’s motion.

5.7 Free Particle Motion

The free particle is another fundamental behavior.

Property 5.7.1 (Momentum Eigenstates)

The eigenstates of the momentum operator are

$$u_p(x) = \frac{1}{\sqrt{2\pi\hbar}} \exp(ipx/\hbar) \quad (5.7.1)$$

that satisfy

$$\hat{p}|u_p\rangle = \frac{\hbar}{i} \partial_x u_p(x) = p u_p(x) = p|u_p\rangle \quad (5.7.2)$$

We can characterize these eigenstates further.

Property 5.7.2 (Probability Current of a Momentum Eigenstate)

The probability current of a momentum eigenstate ψ with probability density

$$\psi(x) = C e^{ipx} \quad (5.7.3)$$

is given by

$$J(x) = \text{Re} \left(\psi^* \frac{\hat{p}}{m} \psi \right) = |C|^2 \frac{p}{m} \quad (5.7.4)$$

which corresponds to a speed p/m of particles with density $|C|^2$.

We consider superpositions of momentum eigenstates.

Definition 5.7.3 (Wavepacket)

A *wavepacket* is a superposition of an infinite number of momentum eigenstates, continuously distributed. It can be normalized if the wavefunction falls off faster than $1/\sqrt{x}$ in position space (or equivalently $1/\sqrt{k}$ in momentum space).

In a finite box, we can build the momentum eigenstate basis:

Proposition 5.7.4 (Momentum Basis under Periodic Boundary Conditions)

Consider a finite box of size L where we require $\psi(0) = \psi(L)$. Thus, the allowed momenta are

$$p_n = n\hbar k_0 : \quad k_0 \equiv \frac{2\pi}{L} \quad (5.7.5)$$

and the eigenstates are

$$u_{p_n}(x) = \frac{1}{\sqrt{L}} \exp(ip_n x/\hbar) \quad (5.7.6)$$

Proof

We simply recognize that the boundary conditions require

$$\exp(ipL/\hbar) = 1 \Rightarrow p_n = n\hbar k_0 : \quad k_0 \equiv \frac{2\pi}{L} \quad (5.7.7)$$

and the normalization constant must satisfy $L|C|^2 = 1 \Rightarrow C = 1/\sqrt{L}$. $\square$

In free space, we work with Gaussian wavepackets, as in either (position or momentum) representation they are still Gaussians.

Property 5.7.5 (Gaussian Wavepacket Properties)

Consider a wavefunction of the kind

$$\Psi(x, t = 0) = \frac{1}{\sqrt{\sqrt{2\pi}w_0}} \exp\left(-\frac{x^2}{4w_0^2}\right) \quad (5.7.8)$$

Firstly, this wavefunction is normalized. The probability density is given by

$$|\Psi|^2 = \frac{1}{\sqrt{2\pi}w_0} \exp\left(-\frac{x^2}{2w_0^2}\right) \quad (5.7.9)$$

The expectations are

$$\langle x \rangle = 0 \quad ; \quad \langle x^2 \rangle = w_0^2 \quad ; \quad \Delta x \equiv \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = w_0 \quad (5.7.10)$$

The momentum representation of such a wavepacket is

$$\phi(k) = \frac{1}{\sqrt{\sqrt{2\pi}k_0}} \exp\left(-\frac{k^2}{4k_0^2}\right) \quad (5.7.11)$$

with

$$k_0 w_0 = \frac{1}{2} \quad ; \quad \Delta k = \sqrt{\langle k^2 \rangle - \langle k \rangle^2} = k_0 \quad (5.7.12)$$

The uncertainty associated with such a wavepacket is the minimum allowed by Heisenberg's Uncertainty Principle. Experimentally, thermal distributions tend to be best modeled as Gaussians.

Wavepackets that move at some velocity have a nonzero average momentum $\langle p \rangle = \hbar \langle k \rangle = mv_1$, modeled as

$$\phi(k) = \frac{1}{\sqrt{\sqrt{2\pi}k_0}} \exp\left(-\frac{(k - k_1)^2}{4k_0^2}\right) \quad (5.7.13)$$

where $k_1 = mv_1/\hbar$. The space representation is thus

$$\Psi(x, t = 0) = \frac{1}{\sqrt{\sqrt{2\pi}w_0}} \exp\left(-\frac{x^2}{4w_0^2}\right) \exp(ik_1 x) \quad (5.7.14)$$

Property 5.7.6 (Time Evolution of a Gaussian Wave Packet)

Consider a gaussian wave packet in free space ($V = 0$ everywhere). Thus, since the Hamiltonian and Momentum operators commute (see Prop. 5.1.13), each momentum/energy eigenstate has its own time dependence phase factor, given by

$$u_p(t) = u_p(0) \exp\left(-i \frac{p^2}{2m\hbar} t\right) \quad (5.7.15)$$

Thus,

$$\Psi(x, t) = \int_{-\infty}^{\infty} dp \phi(p) U_p(x, t) : \quad U_p(x, t) \equiv \frac{1}{\sqrt{2\pi\hbar}} \exp(ipx/\hbar) \exp\left(-i \frac{p^2}{2m} \frac{t}{\hbar}\right) \quad (5.7.16)$$

Notably,

- the momentum density $|\phi(k)|^2$ **does not change in time**, while
- the root-mean-square width $\Delta x(t) = w(t)$ **grows in time** as $w(t) = w_0 \sqrt{1 + \frac{\hbar^2 t^2}{m^2 w_0^4}}$.

Property 5.7.7 (Speed of Wavepacket Spread)

For a Gaussian wavepacket originally of width w_0 of particles of mass m , the position-space wavefunction width spread can be approximated as

$$w(t) = w_0 \sqrt{1 + \frac{\hbar^2 t^2}{m^2 w_0^4}} \approx \frac{\hbar}{mw_0} t \longrightarrow v_0 = \frac{\hbar}{mw_0} \quad (5.7.17)$$

That is, the position-space wavefunction spreads at a velocity v_0 inversely proportional to the original width w_0 .

The dispersion associated is

$$\omega(k) = \frac{\hbar k^2}{2m} \quad (5.7.18)$$

which is quadratic (and not linear). We note that massless particles such as photons with $E = pc$ and $\omega = ck$ would not spread, but the SE is nonrelativistic and so does not apply.

Definition 5.7.8 (Phase Velocity)

We define the phase velocity as the ratio of angular frequency to wavevector. For a wavepacket with average momentum $\langle p \rangle = p_1$,

$$v_{\text{ph}} = \frac{\omega}{k} = \frac{p_1}{2m} \quad (5.7.19)$$

Note that the particle does not move at this velocity (on average).

At what speed does the particle move ? The energy is

$$E_1 = \frac{\hbar^2 k_1^2}{2m} \quad (5.7.20)$$

The wavefunction is

$$\Psi(x, t) = \int_{-\infty}^{\infty} dk \exp\left(-\frac{(k - k_1)^2}{4k_0^2}\right) \exp(i\varphi(k, x, t)) \quad (5.7.21)$$

with

$$\varphi(k, x, t) = kx - \omega(k)t = kx - \frac{\hbar k^2}{2m}t \quad (5.7.22)$$

Fermat's Principle allows us to conclude that the path will be that one where the phase $\varphi(k, x, t)$ does not vary between momentum components to first order, concretely

$$\partial_k \varphi = 0 \iff x(t) = \partial_k \omega|_{k=k_1} t \quad (5.7.23)$$

Definition 5.7.9 (Group Velocity)

The particle described by the Gaussian envelope moves at a *group velocity* of

$$v_{\text{gr}} = \partial_k \omega|_{k=k_1} = \frac{p_1}{m} \quad (5.7.24)$$

as expected.

6 Quantum Harmonic Oscillator

6.1 Series, Operator Methods

The *Quantum Harmonic Oscillator* is the QM version of the classic approximation of conserved-energy bounded-displacement systems.

Definition 6.1.1 (QHO)

The *Quantum Harmonic Oscillator* (QHO) is that system with a Hamiltonian given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 \hat{x}^2}{2} \quad (6.1.1)$$

We observe a symmetry in momentum and position.

6.1.1 Series Method

Let us define a nondimensional energy coefficient ε according to

$$\varepsilon \equiv \frac{2E}{\hbar\omega} \quad (6.1.2)$$

We can also define a displacement coordinate y , also dimensionless:

$$y \equiv \frac{x}{\sqrt{\hbar/m\omega}} = \sqrt{\frac{m\omega}{\hbar}} x \quad (6.1.3)$$

The TISE gives

$$-\frac{\hbar^2}{2m} \partial_x^2 u(x) + \frac{m\omega^2 x^2}{2} u(x) = E u(x) \quad (6.1.4)$$

We rearrange to write $u(y)$:

$$\partial_x^2 u = \left(-\frac{2mE}{\hbar^2} + \frac{m^2\omega^2 x^2}{\hbar^2} \right) u(x) \quad (6.1.5)$$

$$\Rightarrow \partial_y^2 u = \frac{\hbar}{m\omega} \partial_x^2 u \quad (6.1.6)$$

$$= \left(-2\frac{E}{\hbar\omega} + \frac{\hbar}{m\omega} \frac{m\omega^2}{\hbar^2} \frac{\hbar}{m\omega} y^2 \right) u \quad (6.1.7)$$

$$\Rightarrow \partial_y^2 u = (-\varepsilon + y^2) u \quad (6.1.8)$$

Thus, the differential equation to solve is simplified to the last line above. We search for solutions with $E > 0$ and thus $\varepsilon > 0$, since we require at least one point in space where the potential energy is less than the total, i.e. at least one point where the particle is classically allowed. All states we find are bound, since $V(x)$ diverges as $x \rightarrow \infty$ and so any finite energy will result in $E < V$ for all x satisfying $|x| > B$ for some positive B .

We will solve for the solutions to the above differential equations using a series expansion.

Proposition 6.1.2 (Series Method to find the QHO Eigenstates)

The differential equation

$$\partial_y^2 u = (-\varepsilon + y^2)u \quad (6.1.9)$$

has solutions

$$u_n(y) = \begin{cases} \exp\left(-\frac{y}{2}\right) \sum_{k=0}^{n/2} c_{2k} y^{2k} & \text{for even } n \\ \exp\left(-\frac{y}{2}\right) \sum_{k=0}^{(n-1)/2} c_{2k+1} y^{2k+1} & \text{for odd } n \end{cases} : \begin{cases} c_{2k} = (-2)^k \frac{n(n-2)\cdots(n-2k+2)}{(2k)!} c_0 \\ c_{2k+1} = (-2)^k \frac{(n-1)(n-3)\cdots(n-2k+1)}{(2k+1)!} c_1 \end{cases} \quad (6.1.10)$$

where $c_0 = 1$, $c_1 = 2$. The associated energy levels are

$$E_n = \hbar\omega \left(\frac{1}{2} + n \right) \quad (6.1.11)$$

Proof

In short, the procedure is to factor u into u_0 and h , where u_0 converges quickly, for example $e^{-y/2}$, and h is a Taylor series, which allows for a recursion relation

$$h'' - 2yh' + (\varepsilon - 1)h = 0 \Rightarrow (m+1)(m+2)c_{m+2} = (2m - \varepsilon + 1)c_m \quad (6.1.12)$$

The series must terminate since otherwise the Taylor series term is unbounded. If the series terminates, then $\varepsilon = 2m + 1$ for some $m = n \in \mathbb{N}$, which implies

$$E_n = \frac{\hbar\omega}{2} \varepsilon_n = \hbar\omega \left(n + \frac{1}{2} \right) \quad (6.1.13)$$

as desired. The recursion relation can be solved explicitly into the presented expressions, and we normalize appropriately. We note that in fact the functions $h_n(y)$ are the Hermite Polynomials. $\square$

Definition 6.1.3 (Hermite Polynomials)

Hermite Polynomials are a set of indexed functions with two equivalent definitions:

$$H_n''(y) - 2yH_n'(y) + 2nH_n(y) = 0 \iff H_n(y) = (-1)^n \exp(y^2) \left(\frac{d}{dy} \right)^n \exp(-y^2) \quad (6.1.14)$$

One can show that

$$\int_{-\infty}^{\infty} \exp(-y^2) H_n^2(y) dy = 2^n n! \sqrt{\pi} \quad (6.1.15)$$

The first six polynomials are

$$H_0(y) = 1 \quad (6.1.16)$$

$$H_1(y) = 2y \quad (6.1.17)$$

$$H_2(y) = 4y^2 - 2 \quad (6.1.18)$$

$$H_3(y) = 8y^3 - 12y \quad (6.1.19)$$

$$H_4(y) = 16y^4 - 48y^2 + 12 \quad (6.1.20)$$

$$H_5(y) = 32y^5 - 160y^3 + 120y \quad (6.1.21)$$

Property 6.1.4 (Energy Levels of the QHO)

The allowed energies of the QHO are

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right) \quad (6.1.22)$$

Property 6.1.5 (Uncertainties of the QHO)

In the QHO we have

$$\Delta x \Delta p = \frac{\hbar}{2} \text{ for the ground state} \quad ; \quad \Delta x \Delta p > \frac{\hbar}{2} \text{ for any excited state} \quad (6.1.23)$$

6.1.2 Operator Method

We now do not try to find a functional solution, but rather an operator-based one. Writing the Hamiltonian in a dimensionless form gives

$$H = \hbar\omega \left(\frac{p^2}{2m\hbar\omega} + \frac{m\omega}{2\hbar} x^2 \right) \quad (6.1.24)$$

$$= \hbar\omega \left(\left(\frac{p}{\sqrt{2m\hbar\omega}} \right)^2 + \left(\frac{x}{\sqrt{2\hbar/m\omega}} \right)^2 \right) \quad (6.1.25)$$

$$= \hbar\omega \left(\left(\frac{p}{p_0} \right)^2 + \left(\frac{x}{x_0} \right)^2 \right) \quad (6.1.26)$$

for

$$p_0^2 \equiv 2m\hbar\omega \quad \text{and} \quad x_0^2 \equiv \frac{2\hbar}{m\omega} \quad (6.1.27)$$

Thus,

$$\frac{\hat{H}}{\hbar\omega} = \left(\frac{\hat{p}}{p_0} \right)^2 + \left(\frac{\hat{x}}{x_0} \right)^2 \quad (6.1.28)$$

$$= \left(\frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \right) \left(\frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} \right) - \frac{i}{x_0 p_0} [\hat{x}, \hat{p}] \quad (6.1.29)$$

$$= \left(\frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \right) \left(\frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} \right) - \frac{i}{x_0 p_0} \cdot i\hbar \quad (6.1.30)$$

$$= \left(\frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \right) \left(\frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} \right) + \frac{\hbar}{x_0 p_0} \quad (6.1.31)$$

$$= \left(\frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \right) \left(\frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} \right) + \frac{1}{2} \quad (6.1.32)$$

where

$$\frac{1}{x_0 p_0} = \sqrt{\frac{m\omega}{2\hbar}} \frac{1}{\sqrt{2m\hbar\omega}} = \frac{1}{2\hbar} \quad (6.1.33)$$

We can define the *ladder operator* as

$$\hat{a} \equiv \frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} \quad (6.1.34)$$

with its conjugate transpose

$$\hat{a}^\dagger \equiv \frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \quad (6.1.35)$$

so that a third operator, the *number operator*, defined as

$$\hat{N} \equiv \hat{a}^\dagger \hat{a} = \left(\frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \right) \left(\frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} \right) \quad (6.1.36)$$

gives

$$\hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2} \right) \quad (6.1.37)$$

which now seems quite familiar. We summarize:

Definition 6.1.6 (Ladder Operators)

In the QHO, the *ladder operators* are

$$\hat{a}^\dagger \equiv \frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \quad (6.1.38)$$

$$\hat{a} \equiv \frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} \quad (6.1.39)$$

where

$$\hat{a}^\dagger |n\rangle \propto |n+1\rangle \quad (6.1.40)$$

and

$$\hat{a} |n\rangle \propto |n-1\rangle \quad (6.1.41)$$

for $n > 0$ and $|n\rangle$ the n th energy eigenstate.

Definition 6.1.7 (Number Operator)

We define the *number operator* $\hat{N}$ in the context of the QHO according to

$$\hat{N} \equiv \hat{a}^\dagger \hat{a} \quad (6.1.42)$$

so that eigenstates of the number operator are also eigenstates of the Hamiltonian, labeled according to

$$\hat{N}|n\rangle = n|n\rangle \iff \hat{H}|n\rangle = \hbar\omega \left(n + \frac{1}{2} \right) \quad (6.1.43)$$

We have implicitly confirmed that the energies are again spaced by $\hbar\omega$ with the ground state at $\hbar\omega/2$.

6.2 QHO Operators

Let us explore the full properties of the operators we have just defined.

Property 6.2.1 (QHO: Commutator of the Ladder Operators with Each Other)

$$[\hat{a}, \hat{a}^\dagger] = 1 \quad (6.2.1)$$

Proof

$$[\hat{a}, \hat{a}^\dagger] = \left[\frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0}, \frac{\hat{x}}{x_0} - i \frac{\hat{p}}{p_0} \right] = i \left[\frac{\hat{p}}{p_0}, \frac{\hat{x}}{x_0} \right] - i \left[\frac{\hat{x}}{x_0}, \frac{\hat{p}}{p_0} \right] = \frac{i}{p_0 x_0} ([\hat{p}, \hat{x}] - [\hat{x}, \hat{p}]) = \frac{i}{2\hbar} 2 [\hat{p}, \hat{x}] = \frac{i\hbar}{\hbar i} = \boxed{1} \quad (6.2.2)$$

Property 6.2.2 (QHO: Commutator of the Ladder Operators with the Hamiltonian)

$$[\hat{H}, \hat{a}] = -\hbar\omega \hat{a} \quad ; \quad [\hat{H}, \hat{a}^\dagger] = \hbar\omega \hat{a}^\dagger \quad (6.2.3)$$

Proof

We start with the first one:

$$[\hat{H}, \hat{a}] = \hbar\omega [\hat{a}^\dagger \hat{a}, \hat{a}] = \hbar\omega (\hat{a}^\dagger \hat{a} \hat{a} - \hat{a} \hat{a}^\dagger \hat{a}) = \hbar\omega (\hat{a}^\dagger \hat{a} \hat{a} - (1 + \hat{a}^\dagger \hat{a}) \hat{a}) = -\hbar\omega \hat{a} \quad (6.2.4)$$

The second is similarly shown:

$$[\hat{H}, \hat{a}^\dagger] = \hbar\omega [\hat{a}^\dagger \hat{a}, \hat{a}^\dagger] = \hbar\omega (\hat{a}^\dagger \hat{a} \hat{a}^\dagger - \hat{a}^\dagger \hat{a}^\dagger \hat{a}) = \hbar\omega (\hat{a}^\dagger \hat{a} \hat{a}^\dagger - \hat{a}^\dagger (\hat{a} \hat{a}^\dagger - 1)) = \hbar\omega \hat{a}^\dagger \quad (6.2.5)$$

again as desired. $\square$

Definition 6.2.3 (QHO: Ground State)

The *ground state* $|0\rangle$ satisfies

$$\hat{a}|0\rangle = 0 \quad (6.2.6)$$

Property 6.2.4 (QHO: Lowering Operator Effect)

Let $|\psi\rangle$ be a state defined by $\hat{a}|E\rangle$ for $|E\rangle$ some energy eigenstate with energy E . The energy of $|\psi\rangle \equiv \hat{a}|E\rangle$ is

$$\hat{H}|\psi\rangle = (E - \hbar\omega)|\psi\rangle \quad (6.2.7)$$

We generalize:

$$\hat{a}^n|E\rangle = |E - n\hbar\omega\rangle \quad (6.2.8)$$

Proof

We find the energy:

$$\hat{H}|\psi\rangle = \hat{H}\hat{a}|E\rangle = ([\hat{H}, \hat{a}] + \hat{a}\hat{H})|E\rangle = (-\hbar\omega\hat{a} + \hat{a}E)|E\rangle = (E - \hbar\omega)\hat{a}|E\rangle = (E - \hbar\omega)|\psi\rangle \quad (6.2.9)$$

The second statement arises from repeated application of $\hat{a}$. This holds for $n \leq \hat{N}|E\rangle$; for n larger than this, the result is the zero ket (not the ground state). $\square$

Property 6.2.5 (QHO: Raising Operator Effect)

The raising operator increases the energy by $\hbar\omega$:

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \quad (6.2.10)$$

In this language,

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle \quad (6.2.11)$$

with

$$|n\rangle \equiv \frac{1}{\sqrt{n!}} (\hat{a}^\dagger)^n |0\rangle \quad (6.2.12)$$

Property 6.2.6 (QHO: Eigenstate Representations from Operators)

The eigenstate wavefunctions are

$$u_0(x) = \sqrt{\frac{m\omega}{\pi\hbar}} \exp\left(-\frac{m\omega}{2\hbar}x^2\right) \quad (6.2.13)$$

$$u_n(x) = \frac{1}{\sqrt{n!}} \left(\sqrt{\frac{m\omega}{2\hbar}}x - \frac{i}{\sqrt{2\hbar m\omega}}\frac{\hbar}{i}\partial_x \right)^n u_0(x) \quad (6.2.14)$$

Proof

We write in wavefunction (position) representation what the statement $\hat{a}|0\rangle = 0$ implies:

$$\hat{a} \equiv \frac{\hat{x}}{x_0} + i \frac{\hat{p}}{p_0} = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2\hbar m\omega}} \hat{p} \quad (6.2.15)$$

$$\Rightarrow \hat{a}u_0(x) = \left(\sqrt{\frac{m\omega}{2\hbar}} x + \frac{i}{\sqrt{2\hbar m\omega}} \frac{\hbar}{i} \partial_x \right) u_0(x) = 0 \quad (6.2.16)$$

$$\Rightarrow (m\omega x + \hbar \partial_x) u_0(x) = 0 \quad (6.2.17)$$

The solution to this ODE is:

$$u_0 = C \exp\left(-\frac{m\omega}{2\hbar} x^2\right) \quad (6.2.18)$$

for C the normalization constant satisfying

$$1 = |C|^2 \cdot \frac{\pi\hbar}{m\omega} \Rightarrow C = \sqrt{\frac{m\omega}{\pi\hbar}} \quad (6.2.19)$$

as desired. We form $u_n(x)$ by repeated application of $\hat{a}^\dagger$ in accordance with the previous property. $\square$

Definition 6.2.7 (QHO: Coherent State)

A *coherent state* $|\alpha\rangle$ of the QHO satisfies

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle \quad (6.2.20)$$

Proposition 6.2.8 (QHO: Coherent State Expression)

The QHO coherent state parametrized by α is

$$|\alpha\rangle = \exp\left(-\frac{|\alpha|^2}{2}\right) \exp(\alpha\hat{a}^\dagger)|0\rangle \quad (6.2.21)$$

The probability of finding the QHO in the state $|n\rangle$ is given by a Poisson distribution:

$$P(N = n) = \exp(-|\alpha|^2) \frac{(|\alpha|^2)^n}{n!} \quad (6.2.22)$$

where $\lambda \leftrightarrow |\alpha|^2$ is the rate parameter and $N, n \leftrightarrow X, x$ are the random variable and its value respectively.

Proof

We verify that the proposed ansatz satisfies the definition:

$$|\alpha\rangle = C \exp(\alpha\hat{a}^\dagger) |0\rangle \quad (6.2.23)$$

$$\Rightarrow \hat{a}|\alpha\rangle = \hat{a}C \sum \frac{\alpha^n (\hat{a}^\dagger)^n}{n!} |0\rangle \quad (6.2.24)$$

$$= C \sum \frac{\alpha^n \hat{a} (\hat{a}^\dagger)^n}{n!} |0\rangle \quad (6.2.25)$$

$$= C \sum \frac{\alpha^n \cdot n (\hat{a}^\dagger)^{n-1}}{n!} |0\rangle \quad (6.2.26)$$

$$= C\alpha \sum \frac{\alpha^{n-1} (\hat{a}^\dagger)^{n-1}}{(n-1)!} |0\rangle \quad (6.2.27)$$

$$\sim C\alpha \sum \frac{\alpha^n (\hat{a}^\dagger)^n}{n!} |0\rangle \quad (6.2.28)$$

$$= \alpha C \sum \exp(\alpha\hat{a}^\dagger) |0\rangle \quad (6.2.29)$$

$$= \alpha|\alpha\rangle \quad (6.2.30)$$

as desired. We find C : We require

$$\langle \alpha | \alpha \rangle = 1 \quad (6.2.31)$$

We calculate this in terms of C :

$$\langle \alpha | \alpha \rangle = |C|^2 \langle \exp(\alpha \hat{a}^\dagger) | 0 \rangle \left| \exp(\alpha \hat{a}^\dagger) | 0 \rangle \right. \quad (6.2.32)$$

$$= |C|^2 \langle 0 | (\exp(\alpha \hat{a}^\dagger))^\dagger \exp(\alpha \hat{a}^\dagger) | 0 \rangle \quad (6.2.33)$$

We take the Hermitian of the exponential:

$$(\exp(\alpha \hat{a}^\dagger))^\dagger = \left(\sum \frac{\alpha^n (\hat{a}^\dagger)^n}{n!} \right)^\dagger = \sum \frac{\alpha^n \hat{a}^n}{n!} = \exp(\alpha \hat{a}) \quad (6.2.34)$$

as might be expected. We thus return to the magnitude calculation:

$$\langle \alpha | \alpha \rangle = |C|^2 \langle 0 | \exp(\alpha^* \hat{a}) \exp(\alpha \hat{a}^\dagger) | 0 \rangle \quad (6.2.35)$$

We recall that $(\hat{a}^\dagger)^n | 0 \rangle = (n!)^{-1/2} | n \rangle$:

$$| \alpha \rangle = \exp(\alpha \hat{a}^\dagger) | 0 \rangle = \sum \frac{C \alpha^n | n \rangle}{\sqrt{n!}} \quad (6.2.36)$$

Thus,

$$\langle \alpha | \alpha \rangle = |C|^2 \sum_m \sum_n \frac{(\alpha^*)^m \alpha^n \langle m | n \rangle}{\sqrt{m!} \sqrt{n!}} \quad (6.2.37)$$

$$= |C|^2 \sum_m \sum_n \frac{(\alpha^*)^m \alpha^n \delta_{mn}}{\sqrt{m!} \sqrt{n!}} \quad (6.2.38)$$

$$= |C|^2 \sum_n \frac{(\alpha^*)^n \alpha^n}{n!} \quad (6.2.39)$$

$$= |C|^2 \sum_n \frac{(|\alpha|^2)^n}{n!} \quad (6.2.40)$$

$$= |C|^2 \exp(|\alpha|^2) \quad (6.2.41)$$

and so

$$|C|^2 = \exp(-|\alpha|^2) \quad (6.2.42)$$

We take the simplest C possible, namely a real positive C :

$$\boxed{C = \exp\left(-\frac{|\alpha|^2}{2}\right)} \quad (6.2.43)$$

as desired.

We now find the probability distribution of finding the HO in the n th state. We write the full form of $| \alpha \rangle$:

$$| \alpha \rangle = C \cdot \exp(\alpha \hat{a}^\dagger) | 0 \rangle \quad (6.2.44)$$

We expand the exponential:

$$| \alpha \rangle = C \cdot \exp(\alpha \hat{a}^\dagger) | 0 \rangle \quad (6.2.45)$$

$$= C \cdot \sum_n \frac{\alpha^n (\hat{a}^\dagger)^n}{n!} | 0 \rangle \quad (6.2.46)$$

$$= C \cdot \sum_n \frac{\alpha^n}{\sqrt{n!}} | n \rangle \quad (6.2.47)$$

$$(6.2.48)$$

and so the probability of finding the harmonic oscillator in the n th state is given by

$$P(N = n) = |\langle n|\alpha\rangle|^2 = \left| \langle n|C \cdot \sum_m \frac{\alpha^m}{\sqrt{m!}}|m\rangle \right|^2 = C^2 \sum_m \left(\frac{\alpha^m}{\sqrt{m!}} \right)^2 \langle m|n\rangle = C^2 \frac{\alpha^{2n}}{n!} \quad (6.2.49)$$

We substitute C^2 :

$$\boxed{P(N = n) = \exp(-|\alpha|^2) \frac{(|\alpha|^2)^n}{n!}} \quad (6.2.50)$$

The generic Poisson distribution is given by

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad (6.2.51)$$

where we recognize $|\alpha|^2 \leftrightarrow \lambda$ and $N, n \leftrightarrow X, x$. □

7 3D Schrödinger Equation

7.1 The Angular Momentum Operator

We generalize the Schrödinger equation; first we write the vector momentum operator.

Definition 7.1.1 (Vector Momentum Operator (3D))

We write

$$\hat{\mathbf{p}}^2 \equiv \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2 \quad (7.1.1)$$

and

$$\hat{\mathbf{p}} = \begin{bmatrix} \frac{\hbar}{i} \partial_x \\ \frac{\hbar}{i} \partial_y \\ \frac{\hbar}{i} \partial_z \end{bmatrix} = \frac{\hbar}{i} \nabla \quad (7.1.2)$$

in position representation.

Definition 7.1.2 (Time-Independent Schrödinger Equation (3D))

$$\left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) \right) \psi(\mathbf{r}) = E\psi(\mathbf{r}) \quad (7.1.3)$$

The time dependence does not change. We consider spherical potentials, so that $V(\mathbf{r}) = V(r)$. We recall the Laplacian in spherical coordinates:

$$\nabla^2 \equiv \partial_x^2 + \partial_y^2 + \partial_z^2 = \partial_r^2 + \frac{2}{r} \partial_r + \frac{1}{r^2} \left(\partial_\theta^2 + \cot \theta \partial_\theta + \frac{1}{\sin^2 \theta} \partial_\varphi^2 \right) \quad (7.1.4)$$

We thus define the “angular momentum operator” as follows.

Definition 7.1.3 (Angular Momentum Operator)

$$\hat{\mathbf{L}}^2 \equiv -\hbar^2 \left(\partial_\theta^2 + \cot \theta \partial_\theta + \frac{1}{\sin^2 \theta} \partial_\varphi^2 \right) \quad (7.1.5)$$

Property 7.1.4 (Laplacian with Angular Momentum)

We can write

$$\nabla^2 = \partial_r^2 + \frac{2}{r} \partial_r - \frac{\hat{\mathbf{L}}^2}{\hbar^2 r^2} \quad (7.1.6)$$

We wish to solve for the general formulation of energy eigenstates in some spherically symmetric potential. To do so, we perform a separation of angular and radial dependence in the wavefunction, according to

$$\psi(\mathbf{r}) = R(r)Y(\theta, \varphi) \quad (7.1.7)$$

which then gives

$$E\psi(\mathbf{r}) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(r) \right) R(r)Y(\theta, \varphi) \quad (7.1.8)$$

$$= \left(-\frac{\hbar^2}{2m} \left(\partial_r^2 + \frac{2}{r} \partial_r \right) + V(r) \right) R(r)Y(\theta, \varphi) + \frac{L^2}{2mr^2} R(r)Y(\theta, \varphi) \quad (7.1.9)$$

If we divide by $\psi(\mathbf{r}) = R(r)Y(\theta, \varphi)$ assuming it is nonzero (true generally), we have

$$\frac{1}{R(r)} \left(-\frac{\hbar^2}{2m} \left(\partial_r^2 + \frac{2}{r} \partial_r \right) + V(r) \right) R(r) + \frac{1}{Y(\theta, \varphi)} \frac{L^2}{2mr^2} Y(\theta, \varphi) = E \quad (7.1.10)$$

Thus, the first term has pure radial dependence, and the second term has pure angular dependence. The right side can only be constant if we have both terms equal to some constants E_R , E_L such that

$$\frac{1}{Y(\theta, \varphi)} \frac{L^2}{2mr^2} Y(\theta, \varphi) = \frac{\text{const}}{2mr^2} = E_L(r) \quad (7.1.11)$$

Thus,

$$\hat{L}^2 Y(\theta, \varphi) = 2mr^2 E_L(r) Y(\theta, \varphi) \quad (7.1.12)$$

and so Y must be an eigenfunction of the $\hat{L}^2$ operator.

Definition 7.1.5 (Spherical Harmonics)

The *spherical harmonics* are the eigenfunctions of the differential operator

$$\partial_\theta^2 + \cot \theta \partial_\theta + \frac{1}{\sin^2 \theta} \partial_\varphi^2 \quad (7.1.13)$$

We present some properties of the angular momentum.

Property 7.1.6 (Commutator of the Angular Momentum Components)

The angular momentum components satisfy

$$[L_i, L_j] = i\hbar \epsilon_{ijk} L_k \quad (7.1.14)$$

Proof

We write

$$[L_i, L_j] = [r_j p_k - r_k p_j, r_k p_i - r_i p_k] \quad (7.1.15)$$

$$= r_j [p_k, r_k] p_i + r_i [r_k, p_j] p_j \quad (7.1.16)$$

$$= \frac{\hbar}{i} r_j p_i + i\hbar r_i p_j \quad (7.1.17)$$

$$= i\hbar (r_i p_j - r_j p_i) \quad (7.1.18)$$

$$= i\hbar \epsilon_{ijk} L_k \quad (7.1.19)$$

as desired. $\square$

Property 7.1.7 (Commutator of the Angular Momentum Components with the Total)

The angular momentum components satisfy

$$[L_k, L^2] = 0 \quad (7.1.20)$$

Proof

We calculate:

$$[L_k, L^2] = [L_k, L_i^2] + [L_k, L_j^2] \quad (7.1.21)$$

$$= L_i [L_k, L_i] + [L_k, L_i] L_i + L_j [L_k, L_j] + [L_k, L_j] L_j \quad (7.1.22)$$

$$= i\hbar L_i L_j + i\hbar L_j L_i - i\hbar L_j L_i - i\hbar L_i L_j \quad (7.1.23)$$

$$= 0 \quad (7.1.24)$$

Property 7.1.8 (Commutator of the Angular Momentum with other Operators)

We enumerate several properties.

1. $[\hat{L}_j, \hat{r}_k] = i\hbar\epsilon_{jkl}\hat{r}_l$
2. $[\hat{L}_j, \hat{p}_k] = i\hbar\epsilon_{jkl}\hat{p}_l$
3. $[\hat{\mathbf{L}}, \hat{p}^2] = \mathbf{0}$
4. $[\hat{\mathbf{L}}, \hat{r}^2] = \mathbf{0}$

Proof

In this proof, we use $[A, B]$ to represent the commutator $[A, B]$. For the first statement, we begin with the base case. We wish to show that

$$[\hat{L}_x, \hat{y}] = i\hbar\hat{z} \quad (7.1.25)$$

We recall that

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y \quad (7.1.26)$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z \quad (7.1.27)$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \quad (7.1.28)$$

$$(7.1.29)$$

and so

$$[\hat{L}_x, \hat{y}] = \hat{L}_x\hat{y} - \hat{y}\hat{L}_x \quad (7.1.30)$$

$$= (\hat{y}\hat{p}_z - \hat{z}\hat{p}_y)\hat{y} - \hat{y}(\hat{y}\hat{p}_z - \hat{z}\hat{p}_y) \quad (7.1.31)$$

$$= \hat{y}\hat{p}_z\hat{y} - \hat{z}\hat{p}_y\hat{y} - \hat{y}^2\hat{p}_z + \hat{y}\hat{z}\hat{p}_y \quad (7.1.32)$$

$$= \hat{y}[\hat{p}_z, \hat{y}] + \hat{z}[\hat{y}, \hat{p}_y] + [\hat{y}, \hat{z}]\hat{p}_y \quad (7.1.33)$$

where we have added and subtracted the term $\hat{z}\hat{y}\hat{p}_y$ to create the last two commutators. The first commutator is zero, since z and y are orthogonal and so the momentum and coordinate are independent. The second one is given by $i\hbar$, and the third one is zero since again y and z are orthogonal. Thus,

$$[\hat{L}_x, \hat{y}] = \hat{z}(i\hbar) = i\hbar\hat{z} \quad (7.1.34)$$

as desired. We are free to permute indices so that for jkl satisfying $\epsilon_{jkl} = 1$ the above holds. Should j and k be inverted, we would have

$$[\hat{L}_y, \hat{x}] = \hat{L}_y\hat{x} - \hat{x}\hat{L}_y \quad (7.1.35)$$

$$= (\hat{z}\hat{p}_x - \hat{x}\hat{p}_z)\hat{x} - \hat{x}(\hat{z}\hat{p}_x - \hat{x}\hat{p}_z) \quad (7.1.36)$$

$$= \hat{z}\hat{p}_x\hat{x} - \hat{x}\hat{p}_z\hat{x} - \hat{x}\hat{z}\hat{p}_x + \hat{x}^2\hat{p}_z \quad (7.1.37)$$

$$= \hat{x}[\hat{p}_x, \hat{x}] - \hat{z}[\hat{x}, \hat{p}_x] + [\hat{x}, \hat{z}]\hat{p}_x \quad (7.1.38)$$

$$= -i\hbar\hat{z} \quad (7.1.39)$$

Again we are free to permute indices so that for jkl satisfying $\epsilon_{jkl} = -1$ the above holds. Should we have $j = k$, then

$$[\hat{L}_x, \hat{x}] = \hat{L}_x\hat{x} - \hat{x}\hat{L}_x \quad (7.1.40)$$

$$= (\hat{y}\hat{p}_z - \hat{z}\hat{p}_y)\hat{x} - \hat{x}(\hat{y}\hat{p}_z - \hat{z}\hat{p}_y) \quad (7.1.41)$$

$$= \hat{y}\hat{p}_z\hat{x} - \hat{z}\hat{p}_y\hat{x} - \hat{x}\hat{y}\hat{p}_z + \hat{x}\hat{z}\hat{p}_y \quad (7.1.42)$$

$$= \hat{y}\hat{p}_z\hat{x} - \hat{p}_z\hat{y}\hat{x} + \hat{p}_z\hat{y}\hat{x} - \hat{p}_z\hat{x}\hat{y} + \hat{p}_z\hat{x}\hat{y} - \hat{x}\hat{p}_z\hat{y} + \hat{x}\hat{p}_z\hat{y} - \hat{x}\hat{y}\hat{p}_z + \hat{x}\hat{z}\hat{p}_y - \hat{z}\hat{x}\hat{p}_y + \hat{z}\hat{x}\hat{p}_y - \hat{z}\hat{p}_y\hat{x} \quad (7.1.43)$$

$$= [\hat{y}, \hat{p}_z]\hat{x} + \hat{p}_z[\hat{y}, \hat{x}] + [\hat{p}_z, \hat{x}]\hat{y} + \hat{x}[\hat{p}_z, \hat{y}] + [\hat{x}, \hat{z}]\hat{p}_y + \hat{z}[\hat{x}, \hat{p}_y] \quad (7.1.44)$$

$$= 0 \quad (7.1.45)$$

since all commutators above have momenta/coordinates of different indices. This is in accordance with

$$j = k \Rightarrow \epsilon_{jkl} = 0 \quad (7.1.46)$$

Thus,

$$\boxed{[\hat{L}_j, \hat{r}_k] = i\hbar\epsilon_{jkl}\hat{r}_l} \quad (7.1.47)$$

as desired.

For the second statement, we apply a similar reasoning. We begin with the base case. We wish to show that

$$[\hat{L}_x, \hat{p}_y] = i\hbar\hat{p}_z \quad (7.1.48)$$

We recall that

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y \quad (7.1.49)$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z \quad (7.1.50)$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \quad (7.1.51)$$

$$(7.1.52)$$

and so

$$[\hat{L}_x, \hat{p}_y] = \hat{L}_x\hat{p}_y - \hat{p}_y\hat{L}_x \quad (7.1.53)$$

$$= (\hat{y}\hat{p}_z - \hat{z}\hat{p}_y)\hat{p}_y - \hat{p}_y(\hat{y}\hat{p}_z - \hat{z}\hat{p}_y) \quad (7.1.54)$$

$$= \hat{y}\hat{p}_z\hat{p}_y - \hat{z}\hat{p}_y\hat{p}_y - \hat{p}_y\hat{y}\hat{p}_z + \hat{p}_y\hat{z}\hat{p}_y \quad (7.1.55)$$

$$= [\hat{p}_y, \hat{z}]\hat{p}_y + [\hat{y}, \hat{p}_y]\hat{p}_z + \hat{y}[\hat{p}_z, \hat{p}_y] \quad (7.1.56)$$

$$= i\hbar\hat{p}_z \quad (7.1.57)$$

where we have added and subtracted the term $\hat{y}\hat{p}_y\hat{p}_z$ to create the last two commutators. All but the middle commutator evaluate to zero since orthogonal dimensions are independent. We are free to permute the indices above according to $\epsilon_{jkl} = 1$. Should we invert the first two indices, we obtain

$$[\hat{L}_y, \hat{p}_x] = \hat{L}_y\hat{p}_x - \hat{p}_x\hat{L}_y \quad (7.1.58)$$

$$= (\hat{z}\hat{p}_x - \hat{x}\hat{p}_z)\hat{p}_x - \hat{p}_x(\hat{z}\hat{p}_x - \hat{x}\hat{p}_z) \quad (7.1.59)$$

$$= \hat{z}\hat{p}_x\hat{p}_x - \hat{x}\hat{p}_z\hat{p}_x - \hat{p}_x\hat{z}\hat{p}_x + \hat{p}_x\hat{x}\hat{p}_z \quad (7.1.60)$$

$$= [\hat{z}, \hat{p}_x]\hat{p}_x + \hat{x}[\hat{p}_x, \hat{p}_z] - [\hat{x}, \hat{p}_x]\hat{p}_z \quad (7.1.61)$$

$$= -i\hbar\hat{p}_z \quad (7.1.62)$$

where we have added and subtracted $\hat{x}\hat{p}_x\hat{p}_z$ to create the last two commutators. All but the last commutator evaluate to zero by the same reasoning, giving the expected result. We are again free to permute the indices above according to $\epsilon_{jkl} = -1$. Should the first two indices be equal, we have

$$[\hat{L}_x, \hat{p}_x] = \hat{L}_x\hat{p}_x - \hat{p}_x\hat{L}_x \quad (7.1.63)$$

$$= (\hat{y}\hat{p}_z - \hat{z}\hat{p}_y)\hat{p}_x - \hat{p}_x(\hat{y}\hat{p}_z - \hat{z}\hat{p}_y) \quad (7.1.64)$$

$$= \hat{y}\hat{p}_z\hat{p}_x - \hat{z}\hat{p}_y\hat{p}_x - \hat{p}_x\hat{y}\hat{p}_z + \hat{p}_x\hat{z}\hat{p}_y \quad (7.1.65)$$

$$= \hat{y}\hat{p}_z\hat{p}_x - \hat{y}\hat{p}_x\hat{p}_z + \hat{y}\hat{p}_z\hat{p}_z - \hat{p}_z\hat{y}\hat{p}_z + \hat{p}_x\hat{z}\hat{p}_y - \hat{z}\hat{p}_x\hat{p}_y + \hat{z}\hat{p}_x\hat{p}_y - \hat{z}\hat{p}_y\hat{p}_x \quad (7.1.66)$$

$$= \hat{y}[\hat{p}_z, \hat{p}_x] + [\hat{y}, \hat{p}_z]\hat{p}_x + [\hat{p}_x, \hat{z}]\hat{p}_y + \hat{z}[\hat{p}_x, \hat{p}_y] \quad (7.1.67)$$

$$= 0 \quad (7.1.68)$$

again since each commutator involves orthogonal dimensions, in accordance with

$$j = k \Rightarrow \epsilon_{jkl} = 0 \quad (7.1.69)$$

Thus,

$$\boxed{[\hat{L}_j, \hat{p}_k] = i\hbar\epsilon_{jkl}\hat{p}_l} \quad (7.1.70)$$

as desired.

For the third statement, using Einstein index notation, we start from

$$[\hat{L}_j, \hat{p}_k] = i\hbar \epsilon_{jkl} \hat{p}_l \quad (7.1.71)$$

We express $\hat{\mathbf{L}}$ as

$$\hat{\mathbf{L}} = \begin{bmatrix} \hat{L}_x \\ \hat{L}_y \\ \hat{L}_z \end{bmatrix} \quad (7.1.72)$$

and so we really wish to show that

$$[\hat{L}_j, \hat{p}^2] = 0 \quad (7.1.73)$$

for all j . We recall that

$$\hat{L}_j = \hat{r}_k \hat{p}_l - \hat{r}_l \hat{p}_k \quad (7.1.74)$$

and that

$$\hat{p}^2 = \hat{p}_j^2 \quad (7.1.75)$$

and so

$$[\hat{L}_j, \hat{p}^2] = [\hat{L}_j, \hat{p}_j^2] + [\hat{L}_j, \hat{p}_k^2] + [\hat{L}_j, \hat{p}_l^2] \quad (7.1.76)$$

$$= [\hat{L}_j, \hat{p}_j] \hat{p}_j + \hat{p}_j [\hat{L}_j, \hat{p}_j] + [\hat{L}_j, \hat{p}_k] \hat{p}_k + \hat{p}_k [\hat{L}_j, \hat{p}_k] + [\hat{L}_j, \hat{p}_l] \hat{p}_l + \hat{p}_l [\hat{L}_j, \hat{p}_l] \quad (7.1.77)$$

$$= i\hbar(0 + 0 + \cancel{\hat{p}_l \hat{p}_k} + \cancel{\hat{p}_k \hat{p}_l} - \cancel{\hat{p}_k \hat{p}_l} - \cancel{\hat{p}_l \hat{p}_k}) \quad (7.1.78)$$

$$= 0 \quad (7.1.79)$$

and so

$$\boxed{[\hat{\mathbf{L}}, \hat{p}^2] = \mathbf{0}} \quad (7.1.80)$$

as desired.

For the fourth statement, we apply a similar procedure to the third. We have previously shown that

$$[\hat{L}_j, \hat{r}_k] = i\hbar \epsilon_{jkl} \hat{r}_l \quad (7.1.81)$$

We wish to show that

$$[\hat{L}_j, \hat{r}^2] = 0 \quad (7.1.82)$$

We expand:

$$[\hat{L}_j, \hat{r}^2] = [\hat{L}_j, \hat{r}_j^2] + [\hat{L}_j, \hat{r}_k^2] + [\hat{L}_j, \hat{r}_l^2] \quad (7.1.83)$$

$$= [\hat{L}_j, \hat{r}_j] \hat{r}_j + \hat{r}_j [\hat{L}_j, \hat{r}_j] + [\hat{L}_j, \hat{r}_k] \hat{r}_k + \hat{r}_k [\hat{L}_j, \hat{r}_k] + [\hat{L}_j, \hat{r}_l] \hat{r}_l + \hat{r}_l [\hat{L}_j, \hat{r}_l] \quad (7.1.84)$$

$$= i\hbar(0 + 0 + \cancel{\hat{r}_l \hat{r}_k} + \cancel{\hat{r}_k \hat{r}_l} - \cancel{\hat{r}_k \hat{r}_l} - \cancel{\hat{r}_l \hat{r}_k}) \quad (7.1.85)$$

$$= 0 \quad (7.1.86)$$

and so

$$[\hat{\mathbf{L}}, \hat{r}^2] = \mathbf{0} \quad (7.1.87)$$

as desired. □

Property 7.1.9 (Conditions for Simultaneous Eigenstates of Diff. Ang. Moment. Components) One can construct simultaneous eigenstates of L_i and L_j only if the angular momentum is zero.

Proof

Suppose that $|\psi\rangle$ is a simultaneous eigenstate of L_i and L_j :

$$L_i|\psi\rangle = l_1|\psi\rangle \quad \text{and} \quad L_j|\psi\rangle = l_2|\psi\rangle \quad (7.1.88)$$

Thus,

$$[L_i, L_j]|\psi\rangle = (l_1l_2 - l_2l_1)|\psi\rangle = 0 = \epsilon_{ijk}L_k|\psi\rangle \quad (7.1.89)$$

and

$$l_2|\psi\rangle = L_j|\psi\rangle = \frac{1}{i\hbar} [L_k, L_i]|\psi\rangle = (L_kL_i - L_iL_k)|\psi\rangle = 0 \Rightarrow l_2 = 0 \quad (7.1.90)$$

Similarly, $l_1 = 0$. Thus, the condition for simultaneous eigenstates is that $\langle \mathbf{L} \rangle = 0$ as desired. $\square$

We can have simultaneous eigenstates between L_k and $\mathbf{L}^2$; by convention we choose L_z to be that “special” component.

Definition 7.1.10 (Angular Momentum Eigenstate Notation)

A simultaneous eigenstate of $\mathbf{L}^2$ and L_z is denoted by

$$|l, m\rangle \quad (7.1.91)$$

where

$$L_z|l, m\rangle = m\hbar|l, m\rangle \quad (7.1.92)$$

$$\mathbf{L}^2|l, m\rangle = \hbar^2 l(l+1)|l, m\rangle \quad (7.1.93)$$

Property 7.1.11 (Orthonormality of the Angular Momentum Eigenstates)

We have that

$$\langle l', m' | l, m \rangle = \delta_{ll'} \delta_{mm'} \quad (7.1.94)$$

7.2 Spherical Harmonics

We can define the raising and lowering operators for the magnetic moment number m .

Definition 7.2.1 (Angular Momentum Raising/Lowering Operators)

We define

$$L_{\pm} = L_x \pm iL_y \quad (7.2.1)$$

These trivially satisfy

$$L_{\pm}^{\dagger} = L_{\mp} \quad (7.2.2)$$

We write their commutators.

Property 7.2.2 (Commutators of the Angular Momentum Raising/Lower Operators)

Several relevant commutators include

$$[\mathbf{L}^2, L_{\pm}] = 0 \quad (7.2.3)$$

$$[L_+, L_-] = 2\hbar L_z \quad (7.2.4)$$

$$[L_{\pm}, L_z] = \mp\hbar L_{\pm} \quad (7.2.5)$$

Though this last statement is not about commutators, we include it nonetheless:

$$L_{\pm}L_{\mp} = \mathbf{L}^2 - L_z^2 \pm \hbar L_z \quad (7.2.6)$$

Proof

The first statement is trivially true since $[\mathbf{L}^2, L_i] = 0$. The second statement is shown as follows:

$$[L_+, L_-] = [L_x + iL_y, L_x - iL_y] = -i[L_x, L_y] + i[L_y, L_x] = -2i[L_x, L_y] = 2\hbar L_z \quad (7.2.7)$$

The third statement is proved as follows:

$$[L_\pm, L_z] = [L_x \pm iL_y, L_z] \quad (7.2.8)$$

$$= [L_x, L_z] \pm i[L_y, L_z] \quad (7.2.9)$$

$$= -i\hbar L_y \pm i(i\hbar L_x) \quad (7.2.10)$$

$$= \mp\hbar L_x - i\hbar L_y \quad (7.2.11)$$

$$= \mp\hbar(L_x \pm L_y) \quad (7.2.12)$$

$$= \mp\hbar L_\pm \quad (7.2.13)$$

as desired. The last statement is proved as follows:

$$L_\pm L_\mp = (L_x \pm iL_y)(L_x \mp iL_y) = L_x^2 + L_y^2 \mp iL_x L_y + L_y L_x = L^2 - L_z^2 \mp i[L_x, L_y] = L^2 - L_z^2 \pm \hbar L_z \quad (7.2.14)$$

as desired. $\square$

We now clarify the range of the values of l and m .

Proposition 7.2.3 (Allowed values of l, m)

For an arbitrary spherical harmonic $|l, m\rangle$, the active constraints are

$$l \geq 0 \quad \text{and} \quad m \in \{-l, -l+1, \dots, l-1, l\} \quad (7.2.15)$$

Proof

Since the angular momentum operators are all observables, we have

$$\langle l, m | L_i^2 | l, m \rangle = \langle L_i^\dagger(l, m) | L_i(l, m) \rangle = \langle L_i(l, m) | L_i(l, m) \rangle \geq 0 \quad (7.2.16)$$

Thus,

$$\langle l, m | \mathbf{L}^2 | l, m \rangle \geq 0 \quad (7.2.17)$$

which gives

$$0 \leq \langle l, m | \mathbf{L}^2 | l, m \rangle = \hbar^2 l(l+1) \langle l, m | l, m \rangle = \hbar^2 l(l+1) \quad (7.2.18)$$

from orthonormality. Thus, we choose $l \geq 0$, since had we chosen $l \leq -1$ we could easily construct an $l' = -(l+1)$ so that $l' \geq 0$.

We explore the effect of the operators $L_\pm$.

$$|\psi_\pm\rangle \equiv L_\pm |l, m\rangle \quad \implies \quad \mathbf{L}^2 |\psi_\pm\rangle = \mathbf{L}^2 L_\pm |l, m\rangle \quad (7.2.19)$$

$$= L_\pm \mathbf{L}^2 |l, m\rangle \quad (7.2.20)$$

$$= \hbar^2 l(l+1) L_\pm |l, m\rangle \quad (7.2.21)$$

$$= \hbar^2 l(l+1) |\psi_\pm\rangle \quad (7.2.22)$$

$$L_z |\psi_\pm\rangle = L_z L_\pm |l, m\rangle \quad (7.2.23)$$

$$= (L_\pm L_z \pm \hbar L_\pm) |l, m\rangle \quad (7.2.24)$$

$$= (m \pm 1) \hbar L_\pm |l, m\rangle \quad (7.2.25)$$

$$= (m \pm 1) \hbar |\psi_\pm\rangle \quad (7.2.26)$$

Thus, $L_\pm$ raises (lowers) the magnetic quantum number m ; we have

$$\langle l, \widetilde{m \pm 1} | l, \widetilde{m \pm 1} \rangle = \langle l, m | L_\mp L_\pm | l, m \rangle \quad (7.2.27)$$

$$= \langle l, m | \mathbf{L}^2 - L_z^2 \mp \hbar L_z | l, m \rangle \quad (7.2.28)$$

$$= (\hbar^2 l(l+1) - \hbar^2 m^2 \mp \hbar^2 m) \langle l, m | l, m \rangle \quad (7.2.29)$$

$$= \hbar^2 (l \mp m)(l \pm m + 1) \quad (7.2.30)$$

To summarize,

Property 7.2.4 (Effect of the Raising/Lowering Operators)

$$L_{\pm}|l, m\rangle = \hbar\sqrt{(l \mp m)(l \pm m + 1)} |l, m \pm 1\rangle \quad (7.2.31)$$

The range is established by $L_{+}|l, m\rangle = 0$ for some m . This m is thus $m = l$ from the factorization above; the lower bound is similarly set by $L_{-}|l, m\rangle$ for some m , which will then be $-l$. Thus,

$$m \in \{-l, -l + 1, \dots, l - 1, l\} \quad (7.2.32)$$

as desired. $\square$

We note an important consequence – the angular momentum quantum number l must therefore be either integer or half-integer; the half-integer option is called “spin” and will be discussed in the last section of this document.

Property 7.2.5 (Key Angular Momentum Properties)

To summarize:

$$\mathbf{L} \equiv \mathbf{r} \times \mathbf{p} \quad (7.2.33)$$

$$L_i = r_j p_k - r_k p_j \quad (7.2.34)$$

$$[r_i, p_j] = i\hbar\delta_{ij} \quad (7.2.35)$$

$$[L_i, L_j] = i\hbar\epsilon_{ijk}L_k \quad (7.2.36)$$

$$[\mathbf{L}^2, L_x] = [\mathbf{L}^2, L_y] = [\mathbf{L}^2, L_z] = 0 \quad (7.2.37)$$

Without loss of generality, the spherical harmonics are described as angular momentum eigenstates, given by kets $|l, m\rangle$:

$$L_z|l, m\rangle = m\hbar|l, m\rangle \quad (7.2.38)$$

$$\mathbf{L}^2|l, m\rangle = \hbar^2 l(l+1)|l, m\rangle \quad (7.2.39)$$

$$\langle l', m' | l', m' \rangle = \delta_{ll'} \delta_{mm'} \quad (7.2.40)$$

$$(7.2.41)$$

The raising and lowering operators $L_{\pm}$ are defined as

$$L_{\pm} = L_x \pm iL_y \quad (7.2.42)$$

so that

$$[\mathbf{L}^2, L_{\pm}] = 0 \quad (7.2.43)$$

$$L_{\pm}|l, m\rangle = \hbar\sqrt{(l \mp m)(l \pm m + 1)} |l, m \pm 1\rangle \quad (7.2.44)$$

We recall that a state may be decomposed along a basis of continuously or discretely distributed eigenkets:

$$|\psi\rangle = \sum |\xi' d\rangle \langle \xi' d | \psi \rangle + \int d(\xi' c) |\xi' c\rangle \langle \xi' c | \psi \rangle \quad (7.2.45)$$

For angular momentum, we return to the spherical harmonic function $Y_{l,m}(\theta, \varphi)$. We assign meaning to this function according to

$$Y(\theta, \varphi) \equiv \langle \theta, \varphi | \psi \rangle \quad (7.2.46)$$

for some state ψ , where $|\theta, \varphi\rangle$ is the state corresponding to the location of a particle being specified by the angular coordinates θ and φ . Thus,

$$Y_{lm}(\theta, \varphi) \equiv \langle \theta, \varphi | l, m \rangle \quad (7.2.47)$$

Property 7.2.6 (Polar Representation of Angular Momentum Operators)

We can write

$$L_z = \frac{\hbar}{i} \partial_\varphi \quad \text{and} \quad L_\pm = \hbar e^{\pm i\varphi} (\pm \partial_\theta + i(\cot \theta) \partial_\varphi) \quad (7.2.48)$$

Thus, the eigenequation is

$$\langle \theta, \varphi | L_z | l, m \rangle = \begin{cases} \hbar m \langle \theta, \varphi | l, m \rangle = \hbar m Y_{lm}(\theta, \varphi) \\ \frac{\hbar}{i} \partial_\varphi \langle \theta, \varphi | l, m \rangle = \frac{\hbar}{i} \partial_\varphi Y_{lm}(\theta, \varphi) \end{cases} \Rightarrow \boxed{\partial_\varphi Y_{lm}(\theta, \varphi) = im Y_{lm}(\theta, \varphi)} \quad (7.2.49)$$

This is a separable PDE that gives

$$Y_{lm}(\theta, \varphi) = P_{lm}(\theta) e^{im\varphi} \quad (7.2.50)$$

Thus, the constraint $L_+ |l, m = l\rangle = 0$ gives

$$\hbar e^{i\varphi} (\partial_\theta + i(\cot \theta) \partial_\varphi) Y_{ll}(\theta, \varphi) = 0 \quad (7.2.51)$$

$$\Rightarrow e^{i\varphi} (\partial_\theta + i(\cot \theta) \partial_\varphi) P_{ll}(\theta) e^{il\varphi} = 0 \quad (7.2.52)$$

$$\Rightarrow (\partial_\theta - l \cot \theta) P_{ll}(\theta) e^{(l+1)\varphi} = 0 \quad (7.2.53)$$

$$(\partial_\theta - l \cot \theta) P_{ll}(\theta) = 0 \quad (7.2.54)$$

This ODE has a solution of

$$P_{ll} = (\sin \theta)^l \quad \Longrightarrow \quad Y_{ll}(\theta, \varphi) = C_{ll} (\sin \theta)^l e^{il\varphi} \quad (7.2.55)$$

Thus, the general solutions are

$$Y_{lm}(\theta, \varphi) = c(L_-)^{l-m} (\sin \theta)^l e^{il\varphi} \quad (7.2.56)$$

We can then formalize the above results.

Proposition 7.2.7 (Form of the Spherical Harmonics)

Spherical Harmonics are characterized by two indices, the angular momentum number l and the magnetic quantum number m . The spherical harmonics are given by

$$Y_{lm}(\theta, \varphi) = \begin{cases} (-1)^m \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) \exp(im\varphi) & m \geq 0 \\ (-1)^{-m} Y_{l,-m}^* & m \leq 0 \end{cases} \quad (7.2.57)$$

where P_l^m are the Legendre Polynomials given by

$$P_l^m(u) = (-1)^{l+m} \frac{(l+m)!}{(l-m)!} \frac{(1-u^2)^{-m/2}}{2^l l!} \left(\frac{d}{du} \right)^{l-m} (l-u^2)^l \quad m \geq 0 \quad (7.2.58)$$

$$P_l^{-m}(u) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(u) \quad m \geq 0 \quad (7.2.59)$$

Property 7.2.8 (First Three Sets of Spherical Harmonics)

In order of increasing l :

$$Y_{0,0} = \frac{1}{\sqrt{4\pi}} \left. \vphantom{\frac{1}{\sqrt{4\pi}}} \right\} l = 0 \quad (7.2.60)$$

$$\left. \begin{aligned} Y_{1,0} &= \sqrt{\frac{3}{4\pi}} \cos \theta \\ Y_{1,\pm 1} &= \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\varphi} \end{aligned} \right\} l = 1 \quad (7.2.61)$$

$$\left. \begin{aligned} Y_{2,0} &= \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1) \\ Y_{2,\pm 1} &= \mp \sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{\pm i\varphi} \\ Y_{2,\pm 2} &= \sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i\varphi} \end{aligned} \right\} l = 2 \quad (7.2.62)$$

Property 7.2.9 (Mapping to Electron Orbital Shells)

The spherical harmonics describe exactly the shells used to model atomic spectra and intramolecular bonds: the s shell is $l = 0$, p is $l = 1$, d is $l = 2$, and f is $l = 3$.

We recall that the components of angular momentum do not commute unless the total angular momentum is nonzero. Consider the state $|l, m = l\rangle$:

$$\langle l, m = l | \mathbf{L}^2 | l, m = l \rangle = \hbar^2 l(l+1) > \hbar^2 l^2 = \langle l, m = l | L_z^2 | l, m = l \rangle \quad (7.2.63)$$

Thus, there must be some angular momentum pointing in a different direction, such that

$$L_x^2 + L_y^2 = \mathbf{L}^2 - L_z^2 = \hbar^2 l \neq 0 \quad (7.2.64)$$

Property 7.2.10 (Angular Momentum Expectations in an $\mathbf{L}^2$ Eigenstate)

Consider the ket $|l, m\rangle$. The expectations of the angular momentum operators are given by

$$\langle L_x \rangle = 0 \quad ; \quad \langle L_y \rangle = 0 \quad ; \quad \langle L_z \rangle = \hbar m \quad ; \quad \langle \mathbf{L}^2 \rangle = \hbar^2 l(l+1) \quad (7.2.65)$$

$$\langle L_x^2 \rangle = \langle L_y^2 \rangle = \frac{\hbar^2(l^2 - m^2 + l)}{2} \quad ; \quad \langle L_z^2 \rangle = \hbar^2 m^2 \quad (7.2.66)$$

Proof

We have previously shown the third, fourth, and last statements; we show the first two:

$$\langle L_x \rangle = \langle l, m | L_x | l, m \rangle = \langle l, m | (L_+ + L_-)/2 | l, m \rangle = \frac{1}{2} \left(\langle l, m | \widetilde{m+1} \rangle + \langle l, m | \widetilde{m-1} \rangle \right) = 0 \quad (7.2.67)$$

and similarly for L_y . The second-to-last statement is shown as follows:

$$\langle L_x^2 \rangle = \frac{1}{4} \langle l, m | (L_+ + L_-)^2 | l, m \rangle \quad (7.2.68)$$

$$= \frac{1}{4} \langle l, m | L_+^2 + L_+ L_+ L_- L_+ + L_-^2 | l, m \rangle \quad (7.2.69)$$

$$= \frac{1}{4} \langle l, m | L_+ L_+ + L_- L_+ | l, m \rangle \quad (7.2.70)$$

$$= \frac{1}{4} \langle l, m | (\mathbf{L}^2 - L_z^2 + \hbar L_z) + (\mathbf{L}^2 - L_z^2 - \hbar L_z) | l, m \rangle \quad (7.2.71)$$

$$= \frac{1}{2} \langle l, m | (\mathbf{L}^2 - L_z^2) | l, m \rangle \quad (7.2.72)$$

$$= \frac{1}{2} (\hbar^2 l(l+1) - \hbar^2 m^2) \quad (7.2.73)$$

$$= \frac{\hbar^2 (l^2 + l - m^2)}{2} \rightarrow \frac{\hbar^2 l}{2} \text{ for } l = m \quad (7.2.74)$$

and similarly for L_y , where

$$L_y^2 = \left(\frac{L_+ - L_-}{i} \right)^2 = L_+ L_- + L_- L_+ - L_+^2 - L_-^2 \quad (7.2.75)$$

which becomes equivalent after the second line, all as desired. $\square$

7.3 Hydrogen Atom

We still need to characterize the radial component of a spherical-coordinate wavefunction. It was specified by Eq. (7.1.10):

$$\frac{1}{R(r)} \left(-\frac{\hbar^2}{2m} \left(\partial_r^2 + \frac{2}{r} \partial_r \right) + V(r) \right) R(r) + \frac{1}{Y(\theta, \varphi)} \frac{L^2}{2mr^2} Y(\theta, \varphi) = E \quad (7.3.1)$$

If we recombine once having solved for $Y(\theta, \varphi)$, we have

$$-\frac{\hbar^2}{2m} \left(\partial_r^2 + \frac{2}{r} \partial_r \right) \psi(\mathbf{r}) + \frac{\mathbf{L}^2}{2mr^2} \psi(\mathbf{r}) + V(\mathbf{r}) \psi(\mathbf{r}) = E \psi(\mathbf{r}) \quad (7.3.2)$$

$$\Rightarrow \left(-\frac{\hbar^2}{2m} \left(\partial_r^2 + \frac{2}{r} \partial_r \right) + \frac{\hbar^2 l(l+1)}{2mr^2} + V(r) \right) R_{nl}(r) = E_{nl} R_{nl}(r) \quad (7.3.3)$$

We note that the magnetic number m does not enter the radial equation. There is another term of pure radial dependence in the above equation that allows for the definition of the “effective potential.”

Definition 7.3.1 (Spherical $V(r)$): Effective Potential)

In the context of a 3D spherical potential $V(r)$, angular momentum can be incorporated to write an *effective potential*

$$V_{\text{eff},l}(r) = V(r) + \frac{\hbar^2 l(l+1)}{2mr^2} \quad (7.3.4)$$

so that radial functions $R_{nl}(r)$ are eigenfunctions of the below differential equation:

$$\left(-\frac{\hbar^2}{2m} \left(\partial_r^2 + \frac{2}{r} \partial_r \right) + V_{\text{eff},l}(r) \right) R_{nl}(r) = E_{nl} R_{nl}(r) \quad (7.3.5)$$

It is often convenient to solve for $u(r) \equiv r R_{nl}(r)$ since this simplifies the differential equation:

$$\left(-\frac{\hbar^2}{2m} \partial_r^2 + \frac{\hbar^2 l(l+1)}{2mr^2} + V(r) \right) u_{nl}(r) = E_{nl} u_{nl}(r) \quad (7.3.6)$$

since

$$R \equiv \frac{u}{r} \Rightarrow R' = \frac{u'}{r} - \frac{u}{r^2}; \quad R'' = \frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3} \quad (7.3.7)$$

and thus

$$\left(\partial_r^2 + \frac{2}{r}\partial_r\right)R(r) = R'' + \frac{2}{r}R' = \frac{u''}{r} \quad (7.3.8)$$

We note that such a substitution is common even in classical mechanics for angular momentum problems. Solutions to the equation in u are in general antisymmetric.

Axiom 7.3.2 (Coulomb Potential)

The potential governing electron motion in a Hydrogen atom is the Coulomb one:

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r} \quad (7.3.9)$$

Thus,

$$\left(-\frac{\hbar^2}{2m}\partial_r^2 - \frac{Ze^2}{4\pi\epsilon_0 r} + \frac{\hbar^2 l(l+1)}{2mr^2} - E\right)u(r) = 0 \quad (7.3.10)$$

Let

$$\rho \equiv \frac{2\sqrt{2m|E|}}{\hbar}r \quad (7.3.11)$$

We also define a quantity λ so that

$$\lambda \equiv \frac{Ze^2}{4\pi\epsilon_0 \hbar} \sqrt{\frac{m}{2|E|}} = Z\alpha \sqrt{\frac{mc^2}{2|E|}} \quad (7.3.12)$$

for

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137} \quad (7.3.13)$$

the fine structure constant. Thus, our governing equation can be simplified to

$$\partial_\rho^2 u(\rho) + \left(\frac{\lambda}{\rho} - \frac{1}{4} - \frac{l(l+1)}{\rho^2}\right)u(\rho) = 0 \quad (7.3.14)$$

Large ρ give

$$\partial_\rho^2 u = \frac{u}{4} \Rightarrow u(\rho) \propto \exp(-\rho/2) \quad (7.3.15)$$

Small ρ give

$$\partial_\rho^2 u = \frac{l(l+1)}{\rho^2}u \Rightarrow u(\rho) \propto \rho^{l+1} \quad (7.3.16)$$

We thus propose a solution of the form

$$u(\rho) = s(\rho)\rho^{l+1}\exp(-\rho/2) \quad (7.3.17)$$

Taking two derivatives gives

$$u'(\rho) = \left(s'(\rho)\rho^{l+1} + s(\rho)(l+1)\rho^l - \frac{1}{2}s\rho^{l+1}\right)\exp(-\rho/2) \quad (7.3.18)$$

$$u''(\rho) = \rho^{l+1}\exp(-\rho/2)\left(s'' + 2(l+1)\frac{s'}{\rho} + \frac{(l+1)l}{\rho^2}s - s' - \frac{l+1}{\rho}s + \frac{1}{4}s\right) \quad (7.3.19)$$

In the differential equation, we obtain a new one for s :

$$s'' + \left(\frac{2l+2}{\rho} - 1\right)s' + \frac{\lambda - l - 1}{\rho}s = 0 \quad (7.3.20)$$

This differential equation is expanded into a Taylor Series, giving

$$s(\rho) = \sum_{k=0}^{\infty} a_k \rho^k \implies \begin{cases} s'' = \sum_{k=0}^{\infty} a_{k+2} (k+2)(k+1) \rho^k \\ s' = \sum_{k=0}^{\infty} a_k k \rho^{k-1} \\ \frac{s}{\rho} = \sum_{k=0}^{\infty} a_{k+1} \rho^k \end{cases} \quad (7.3.21)$$

Thus, we obtain the series

$$\sum_k [(k+2)(k+1)a_{k+2} + 2(l+1)(k+2)a_{k+2} + (\lambda - l - 1 - l - 1)a_{k+1}] \rho^k = 0 \quad (7.3.22)$$

Since each term must be zero, we have the recursion relation

$$(k+2)(k+2l+3)a_{k+2} = (k+l+2-\lambda)a_{k+1} \quad (7.3.23)$$

The series must again terminate (as for the angular momentum) since otherwise the function $s(\rho)$ will be either unbounded or non-normalizable; thus,

$$\frac{a_{k+1}}{a_k} = \frac{k+l+1-\lambda}{(k+1)(k+2(l+1))} \Rightarrow \lambda_{\text{termination}} = k+l+1 \quad (7.3.24)$$

We introduce the following nomenclature.

Definition 7.3.3 (Atomic Quantum Numbers)

In line with the energy eigenstates we have so far characterized, we formally name the four important quantum numbers:

- n : Principal Quantum Number
- n_r : Radial Quantum Number
- l : Angular Momentum Quantum Number
- m : Magnetic Quantum Number

These satisfy $n = n_r + l + 1$.

In the expression above, $\lambda \rightarrow n$ and $k \rightarrow n_r$. Thus,

$$\lambda_n = Z\alpha \sqrt{\frac{mc^2}{2|E_n|}} = n \quad (7.3.25)$$

Proposition 7.3.4 (Bohr Quantized Energies)

$$E_n = -\frac{mc^2}{2} \frac{(Z\alpha)^2}{n^2} \quad (7.3.26)$$

We can thus qualify the eigenstate as $|n, l, m\rangle$ with $n \in \mathbb{N}$, $l \in \mathbb{N} \cap [0, n-1]$, and $m \in \mathbb{Z} \cap [-l, l]$.

Property 7.3.5 (Bohr Atom Energy Eigenstates)

The ket $|n, l, m\rangle$ is described by

$$\psi_{n,l,m}(\mathbf{r}) = \sqrt{\left(\frac{2}{na_0^*}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}} \exp\left(-\frac{r}{na_0^*}\right) \left(\frac{2r}{na_0^*}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2r}{na_0^*}\right) Y_{lm}(\theta, \varphi) \quad (7.3.27)$$

for

$$L_0^\alpha(x) = 1 \quad ; \quad L_1^\alpha(x) = 1 + \alpha - x \quad ; \quad L_{k+1}^\alpha = \frac{(2k+1+\alpha-x)L_k^\alpha(x) - (k+\alpha)L_{k-1}^\alpha(x)}{k+1} \quad (7.3.28)$$

alternatively defined as

$$L_n^\alpha(x) = \sum_{m=0}^n \binom{n+\alpha}{n-m} \frac{(-x)^m}{m!} \quad (7.3.29)$$

and a_0^* is the reduced Bohr radius, μ is the reduced mass of the Hydrogen atom:

$$a_0^* = \frac{4\pi\epsilon_0\hbar^2}{\mu e^2} : \quad \mu \equiv \frac{m_e m_p}{m_e + m_p} \quad (7.3.30)$$

The functions Y_{lm} are the previously defined spherical harmonics:

$$Y_{lm}(\theta, \varphi) = \begin{cases} (-1)^m \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) \exp(im\varphi) & m \geq 0 \\ (-1)^{-m} Y_{l,-m}^* & m \leq 0 \end{cases} \quad (7.3.31)$$

where P_l^m are the Legendre Polynomials given by

$$P_l^m(u) = (-1)^{l+m} \frac{(l+m)!}{(l-m)!} \frac{(1-u^2)^{-m/2}}{2^l l!} \left(\frac{d}{du} \right)^{l-m} (l-u^2)^l \quad m \geq 0 \quad (7.3.32)$$

$$P_l^{-m}(u) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(u) \quad m \geq 0 \quad (7.3.33)$$

Property 7.3.6 (First Few Radial Functions)

$$|1, 0, 0\rangle : \quad R_{1,0}(r) = C \exp\left(-\frac{Zr}{a_0}\right) \quad (7.3.34)$$

$$|2, 0, 0\rangle : \quad R_{2,0}(r) = C' \left(1 - \frac{Zr}{2a_0}\right) \exp\left(-\frac{Zr}{2a_0}\right) \quad (7.3.35)$$

$$|2, 1, 0\rangle : \quad R_{2,1}(r) = C'' \left(\frac{Zr}{a_0}\right) \exp\left(-\frac{Zr}{2a_0}\right) \quad (7.3.36)$$

We recall that the energy is determined uniquely by the principal number n . The constants are found according to

$$1 = \int_0^\infty r^2 dr |R_{nl}(r)|^2 \quad (7.3.37)$$

Property 7.3.7 (First Few Hydrogen Wavefunctions)

$$|1, 0, 0\rangle \rightarrow \psi_{100}(\mathbf{r}) = \frac{2}{a_0^{3/2}} \exp\left(-\frac{r}{a_0}\right) Y_{0,0}(\theta, \varphi) \rightarrow 1s \quad (7.3.38)$$

$$|2, 0, 0\rangle \rightarrow \psi_{200}(\mathbf{r}) = \frac{2}{(2a_0)^{3/2}} \left(1 - \frac{r}{2a_0}\right) \exp\left(-\frac{r}{2a_0}\right) Y_{0,0}(\theta, \varphi) \rightarrow 2s \quad (7.3.39)$$

$$|2, 1, *\rangle \rightarrow \psi_{21*}(\mathbf{r}) = \frac{r}{\sqrt{3}(2a_0)^{3/2}a_0} \exp\left(-\frac{r}{2a_0}\right) Y_{1,*}(\theta, \varphi) \rightarrow 2p \quad (7.3.40)$$

Since energy is determined only by the principal quantum number n , there exists a degeneracy in the energy eigenstates.

Proposition 7.3.8 (Degeneracy of the Hydrogen Spectrum)

There are $2n^2$ degenerate states for an atom in the excited state of principal quantum number n .

Proof

$$\underbrace{1}_{l=0} + \underbrace{3}_{l=1} + \underbrace{5}_{l=2} + \cdots + \underbrace{2(n-1)+1}_{l=n-1} = n^2 \quad (7.3.41)$$

We account for twice as many states (total $2n^2$) since each electron can have spin up or spin down. □

Property 7.3.9 (Expectation Values of Radial Position in a Hydrogen Atom)

$$\langle r^k \rangle = \int_0^\infty dr r^{2+k} |R_{nl}(r)|^2 \quad (7.3.42)$$

$$\langle r \rangle = \frac{a_0}{2Z} (3n^2 - l(l+1)) \quad (7.3.43)$$

$$\langle r^2 \rangle = \frac{a_0^2 n^2}{2Z^2} (5n^2 + 1 - 3l(l+1)) \quad (7.3.44)$$

$$\left\langle \frac{1}{r} \right\rangle = \frac{Z}{a_0 n^2} \quad (7.3.45)$$

$$\left\langle \frac{1}{r^2} \right\rangle = \frac{Z^2}{a_0^2 n^3 (l+1/2)} \quad (7.3.46)$$

Definition 7.3.10 (Bosons, Fermions)

Fermions are particles for which a pair must be describable by a wavefunction that satisfies

$$\psi_f(\mathbf{r}_1, \mathbf{r}_2) = -\psi_f(\mathbf{r}_2, \mathbf{r}_1) \quad (7.3.47)$$

which implies that $\mathbf{r}_1 = \mathbf{r}_2 \Rightarrow \psi_f = 0$: the particles “avoid” each other.

Bosons are the even version (if one would regard fermions as “odd”):

$$\psi_B(\mathbf{r}_1, \mathbf{r}_2) = \psi_B(\mathbf{r}_2, \mathbf{r}_1) \quad (7.3.48)$$

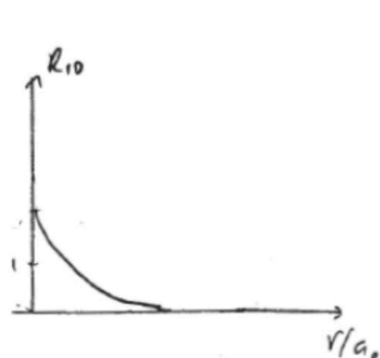
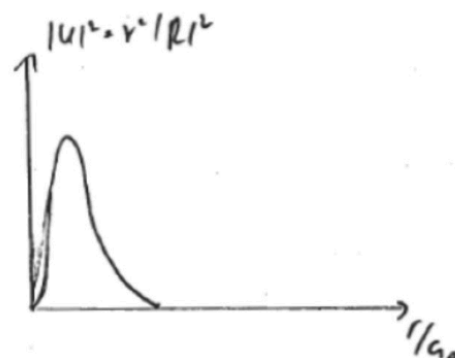
Bosons are more likely to be found at same position, for example in lasers, Bose-Einstein condensation, superconductivity, and the classical notion of fields where amplitudes can be added.

1S

$$n_r = 0$$

$$l = 0$$

$$n = 1$$

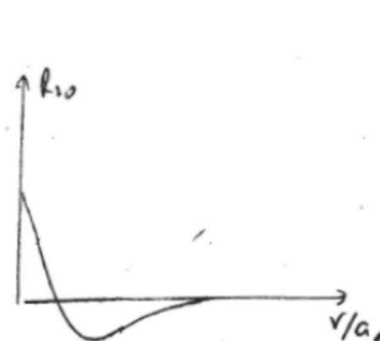
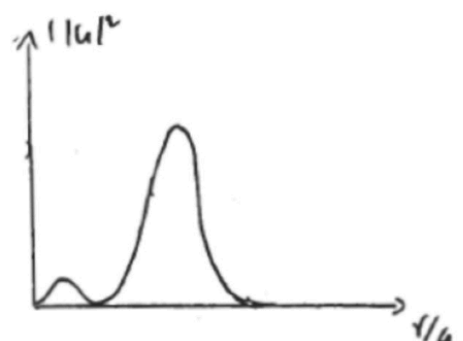
(a) $n_r = 0, l = 0, n = 1$ (b) $n_r = 0, l = 0, n = 1$

(2S)

$$n_r = 1$$

$$l = 0$$

$$n = 2$$

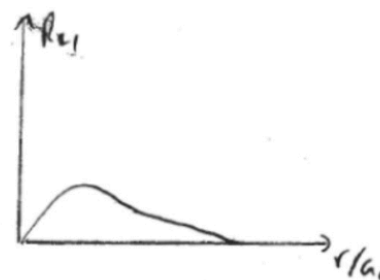
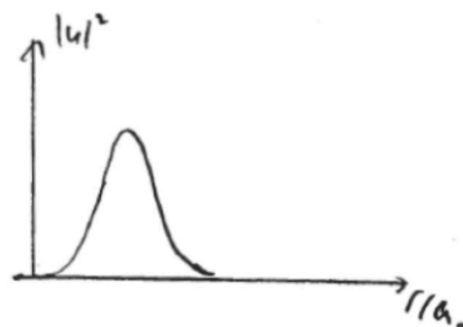
(c) $n_r = 1, l = 0, n = 2$ (d) $n_r = 1, l = 0, n = 2$

(2P)

$$n_r = 0$$

$$l = 1$$

$$n = 2$$

(e) $n_r = 0, l = 1, n = 2$ (f) $n_r = 0, l = 1, n = 2$

(3S)

$$n_r = 2$$

$$l = 0$$

$$n = 3$$

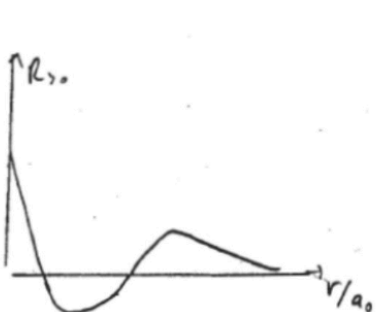
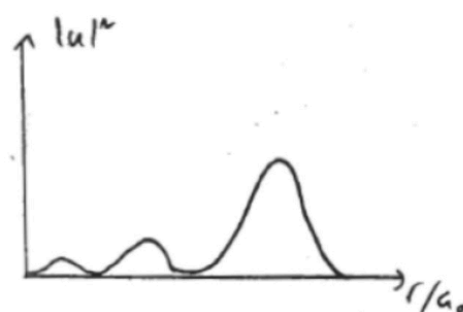
(g) $n_r = 2, l = 0, n = 3$ (h) $n_r = 2, l = 0, n = 3$

Figure 13. Sketches of the electron shells 1s through 3p.

8 Spin-1/2

8.1 Spin States

8.1.1 Magnetic Dipoles

In a circular loop of radius R with particles of mass M with charge Q , the current is defined as

$$I = \frac{Q}{2\pi R} v = \frac{Q}{M} \frac{1}{2\pi R^2} M R v = \frac{Q}{2M} \frac{L}{A} \quad (8.1.1)$$

for L the angular momentum, A the area of the loop. The magnetic dipole moment μ is then

$$\mu \equiv \frac{IA}{c} = \frac{QL}{2Mc} \quad (8.1.2)$$

Thus,

$$\mu \equiv \frac{Q}{2mc} \mathbf{L} \quad (8.1.3)$$

Definition 8.1.1 (Bohr Magneton)

The Bohr Magneton is a unit of magnetic moment, defined by

$$\mu_B = \frac{e}{mc} \frac{\hbar}{2} = 9.274 \cdot 10^{-24} \frac{\text{J}}{\text{T}} = \hbar \cdot 1.4 \frac{\text{MHz}}{\text{G}} = \hbar \cdot 14 \frac{\text{GHz}}{\text{T}} \quad (8.1.4)$$

In vector form, the magnetic moment of an electron is

$$\boldsymbol{\mu}_e = -g_e \mu_B \frac{\mathbf{S}}{\hbar} \quad (8.1.5)$$

and we measure $g_e = 2$, $|\mathbf{S}| = 1/2$. For orbital angular momentum, we measure $g_e = 1$.

Proposition 8.1.2 (Energy of a Magnetic Dipole)

In a magnetic field of strength $\mathbf{B}$, the Hamiltonian is

$$\hat{H} = -\hat{\boldsymbol{\mu}} \cdot \mathbf{B} \quad (8.1.6)$$

For an electron spin,

$$\hat{H} = g_e \frac{\mu_B}{\hbar} \mathbf{S} \cdot \mathbf{B} \quad (8.1.7)$$

for $\mathbf{S}$ the spin vector.

We now clarify the nature of the spin vector $\mathbf{S}$. We consider two eigenstates of the operators $\mathbf{S}^2$ and S_z , analogous to $\mathbf{L}^2$ and L_z :

$$\mathbf{S}^2 |s, m\rangle = \hbar^2 s(s+1) |s, m\rangle = \frac{3}{4} \hbar^2 |s, m\rangle \quad (8.1.8)$$

$$S_z |s, m\rangle = m \hbar |s, m\rangle = \pm \frac{\hbar}{2} |s, m\rangle \quad (8.1.9)$$

We find that it is sufficient to label the kets $|1/2, \pm 1/2\rangle$ as elements of $\mathbb{C}^2$.

Definition 8.1.3 (Pauli Vectors)

We define the *Pauli Vectors* as

$$|s = \frac{1}{2}, m = \frac{1}{2}\rangle \equiv |+\rangle_z = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad ; \quad |s = \frac{1}{2}, m = -\frac{1}{2}\rangle \equiv |-\rangle_z = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (8.1.10)$$

Property 8.1.4 (Decomposition of a Spin State)

We may decompose an arbitrary spin state $|\psi\rangle$ into a linear combination of the up- z and down- z states:

$$|\psi\rangle = c_1|+\rangle_z + c_2|-\rangle_z = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \quad (8.1.11)$$

We now wish to represent the operators S_z and $\mathbf{S}^2$ in this vector language. Thus, they will take the form of matrices. We require

$$S_z \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{\hbar}{2} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad S_z \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -\frac{\hbar}{2} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (8.1.12)$$

The matrix that satisfies this is

$$S_z = \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \equiv \frac{\hbar}{2} \sigma_z \quad (8.1.13)$$

where we call σ_z the z Pauli Matrix.

How do we find the x and y ones? The most general 2×2 matrix has four coefficients, so we expect a basis of four matrices. We have two already (σ_z and the identity); thus the other two are σ_x and σ_y .

Proposition 8.1.5 (Commutators of Spin)

Since spin is a kind of angular momentum, we require

$$[S_i, S_j] = i\hbar\epsilon_{ijk}S_k \iff [\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k \quad (8.1.14)$$

Proposition 8.1.6 (Uncertainty for Spin)

As for angular momentum,

$$\Delta S_x \Delta S_y \geq \frac{\hbar}{2} |\langle S_z \rangle| \quad (8.1.15)$$

If a particle is in $|+\rangle_z$, then $\langle S_z \rangle = \hbar/2$ and $\langle S_x \rangle = \langle S_y \rangle = 0$, giving measurement results of $\pm\hbar/2$ for each of S_x, S_y while maintaining a centered distribution (equally likely). Thus, we finish the basis for $\mathbb{M}_{2 \times 2}^{\mathbb{C}}$.

Definition 8.1.7 (Pauli Matrices)

The cartesian components of the spin operator $\mathbf{S}$ can be defined in terms of the Pauli matrices

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (8.1.16)$$

where $S_k = \sigma_k \cdot \hbar/2$ for $k = x, y, z$.

We can now define eigenstates along x and y :

$$\sigma_x |\pm\rangle_x = \pm |\pm\rangle_x \quad (8.1.17)$$

$$\Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_2 \\ c_1 \end{bmatrix} = \pm \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \quad (8.1.18)$$

$$\Rightarrow |+\rangle_x = \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \frac{|+\rangle_z + |-\rangle_z}{\sqrt{2}} \quad (8.1.19)$$

$$\Rightarrow |-\rangle_x = \frac{|+\rangle_z - |-\rangle_z}{\sqrt{2}} \quad (8.1.20)$$

Property 8.1.8 (Pauli Representation of x and y Spin)

$$|\pm\rangle_x = \frac{|+\rangle_z \pm |-\rangle_z}{\sqrt{2}} \quad (8.1.21)$$

$$|\pm\rangle_y = \frac{|+\rangle_z \pm i|-\rangle_z}{\sqrt{2}} \quad (8.1.22)$$

We can represent measurement along an arbitrary axis $\hat{\mathbf{n}}$ according to

$$\hat{S}_{\hat{\mathbf{n}}} \equiv \hat{\mathbf{n}} \cdot \hat{\mathbf{S}} = n_i \hat{S}_i = \frac{\hbar}{2} \hat{\mathbf{n}} \cdot \boldsymbol{\sigma} \quad (8.1.23)$$

where we represent $\hat{\mathbf{n}} = \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}$ as a unit vector in a Cartesian representation.

Proposition 8.1.9 (Spin Operator in an Arbitrary Polarization)

In Cartesian or Spherical Coordinates, we have

$$\hat{S}_{\hat{\mathbf{n}}} = \frac{\hbar}{2} \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} = \frac{\hbar}{2} \begin{bmatrix} \cos \theta & \sin \theta \exp(-i\varphi) \\ \sin \theta \exp(i\varphi) & -\cos \theta \end{bmatrix} \quad (8.1.24)$$

The eigenvectors of this operator are

$$|+\rangle_{\hat{\mathbf{n}}} = \begin{bmatrix} \cos \frac{\theta}{2} \\ \exp(i\varphi) \sin \frac{\theta}{2} \end{bmatrix} ; \quad |-\rangle_{\hat{\mathbf{n}}} = \begin{bmatrix} \cos \frac{\theta}{2} \\ -\exp(i\varphi) \sin \frac{\theta}{2} \end{bmatrix} \quad (8.1.25)$$

The “sphere” onto which we project these states using spherical coordinates is called the “Bloch Sphere”.

Property 8.1.10 (Spin States)

The spin states in cartesian coordinates are

$$|+\rangle_z = \begin{bmatrix} 1 \\ 0 \end{bmatrix} ; \quad |-\rangle_z = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (8.1.26)$$

$$|+\rangle_x = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} ; \quad |-\rangle_x = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (8.1.27)$$

$$|+\rangle_y = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} ; \quad |-\rangle_y = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} \quad (8.1.28)$$

Definition 8.1.11 (Spinor)

The most general representation of a spin state is

$$|\psi\rangle = e^{i\alpha} \begin{bmatrix} \cos(\theta/2) \\ e^{i\varphi} \sin(\theta/2) \end{bmatrix} \quad (8.1.29)$$

This is called a *spinor*. The overall phase is irrelevant; this state corresponds to a Bloch vector of

$$\hat{\mathbf{n}} = \begin{bmatrix} \sin \theta \cos \varphi \\ \sin \theta \sin \varphi \\ \cos \theta \end{bmatrix} \quad (8.1.30)$$

with

$$\langle S_x \rangle = \frac{\hbar}{2} \sin \theta \cos \varphi \quad ; \quad \langle S_y \rangle = \frac{\hbar}{2} \sin \theta \sin \varphi \quad ; \quad \langle S_z \rangle = \frac{\hbar}{2} \cos \theta \quad (8.1.31)$$

We can write a compact rule for finding probabilities.

Proposition 8.1.12 (Probability to Measure an $\hat{n}$ Spin from $\hat{m}$ Polarization)

Let Θ be the angle between $\hat{n}$ and $\hat{m}$, namely

$$\Theta = \arccos(\hat{n} \cdot \hat{m}) \quad (8.1.32)$$

Thus,

$$|\langle +_{\hat{m}} | +_{\hat{n}} \rangle|^2 = \cos^2 \frac{\Theta}{2} \quad \text{and} \quad |\langle -_{\hat{m}} | +_{\hat{n}} \rangle|^2 = \sin^2 \frac{\Theta}{2} \quad (8.1.33)$$

8.1.2 Time Dependence

The formal solution to the time dependence in the Schrödinger Equation is

$$|\psi(t)\rangle = \exp\left(-\frac{i\hat{H}t}{\hbar}\right) |\psi(0)\rangle \quad (8.1.34)$$

but this is often not useful since components of the Hamiltonian (namely $\hat{p}^2/2m$ and $V(x)$) do not commute. For the electron spin problem with an external magnetic field $\mathbf{B} = \hat{z}B_z$, we have

$$\hat{H} = \omega_L \hat{S}_z = \frac{\hbar}{2} \omega_L \hat{\sigma}_z \quad (8.1.35)$$

Thus,

$$|\psi(t)\rangle = \exp\left(-\frac{i}{2} \omega_L t \hat{\sigma}_z\right) |\psi(0)\rangle \quad (8.1.36)$$

which gives rotation in a period $4\pi/\omega_L$ as described earlier. This result is notably identical to classical precession of a magnetic dipole. We introduce the *gyromagnetic ratio*

$$\gamma \equiv \frac{g\mu_b}{\hbar} \quad (8.1.37)$$

which satisfies

$$\boldsymbol{\mu} = \pm \gamma \mathbf{S} \quad (8.1.38)$$

Thus, the energy is

$$\hat{H} = \omega_L S_z \quad (8.1.39)$$

where $\omega_L = \gamma B_z$ is the Larmor Frequency. The energy is thus

$$H|\pm\rangle_z = \omega_L S_z |\pm\rangle_z = \pm \frac{\hbar}{2} \omega_L |\pm\rangle_z \quad (8.1.40)$$

Property 8.1.13 (Time Evolution of a Spin State)

The z spin states are stationary, with energies $\pm \hbar \omega_L$. The x and y states evolve according to

$$|\pm\rangle_x(t) = \frac{\exp(-i\omega_L t/2)}{\sqrt{2}} (|+\rangle_z + |-\rangle_z \exp(i\omega_L t)) \quad (8.1.41)$$

$$|\pm\rangle_y(t) = \frac{\exp(-i\omega_L t/2)}{\sqrt{2}} (|+\rangle_z + i|-\rangle_z \exp(i\omega_L t)) \quad (8.1.42)$$

These are 4π periodic, akin to a Möbius band.

8.2 Generation of Rotations

Let us define

$$|\phi\rangle \equiv \exp(i\alpha\hat{\sigma}_z)|\psi\rangle \quad (8.2.1)$$

The expectation of σ_z is thus

$$\langle\phi|\sigma_z|\phi\rangle = \langle\psi|\exp(-i\alpha\sigma_z)\sigma_z\exp(-i\alpha\sigma_z)|\psi\rangle = \langle\psi|\sigma_z|\psi\rangle \quad (8.2.2)$$

which is unchanged between ϕ and ψ . For σ_x , however,

$$\langle\phi|\sigma_x|\phi\rangle = \langle\psi|\exp(-i\alpha\sigma_z)\sigma_x\exp(i\alpha\sigma_z)|\psi\rangle \quad (8.2.3)$$

We apply the matrix exponential according to the Taylor Series definition. We notice that

$$\sigma_z^2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbb{I} \quad (8.2.4)$$

Thus, we split the Taylor series into odd and even powers:

$$\exp(\pm i\alpha\sigma_z) = \sum_{n=0}^{\infty} \frac{1}{n!} (\pm i\alpha)^n \sigma_z^n \quad (8.2.5)$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{(2n)!} (\pm i\alpha)^{2n} \mathbb{I} \right) + \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} (\pm i\alpha)^{2n+1} \sigma_z \quad (8.2.6)$$

$$= (\cos \alpha) \mathbb{I} \pm i(\sin \alpha) \sigma_z \quad (8.2.7)$$

Thus,

$$\langle\phi|\sigma_x|\phi\rangle = \langle\psi|\exp(-i\alpha\sigma_z)\sigma_x\exp(i\alpha\sigma_z)|\psi\rangle \quad (8.2.8)$$

$$= \cos^2 \alpha \langle\psi|\sigma_x|\psi\rangle + \sin^2 \alpha \langle\psi|\sigma_z\sigma_x\sigma_z|\psi\rangle + i \sin \alpha \cos \alpha \langle\psi|\sigma_x\sigma_z - \sigma_z\sigma_x|\psi\rangle \quad (8.2.9)$$

We observe that

$$\sigma_x\sigma_z - \sigma_z\sigma_x = -2i\sigma_y \quad (8.2.10)$$

and that

$$\sigma_z\sigma_x\sigma_z = [\sigma_z, \sigma_x] \sigma_z + \sigma_x\sigma_z^2 = 2i\sigma_y\sigma_z + \sigma_x \quad (8.2.11)$$

We evaluate the first term:

$$\sigma_y\sigma_z = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} = i\sigma_x \quad (8.2.12)$$

Thus,

$$\sigma_z\sigma_x\sigma_z = -\sigma_x \quad (8.2.13)$$

Thus,

$$\langle\phi|\sigma_x|\phi\rangle = (\cos^2 \alpha - \sin^2 \alpha) \langle\psi|\sigma_x|\psi\rangle + 2 \sin \alpha \cos \alpha \langle\psi|\sigma_y|\psi\rangle = \cos(2\alpha) \langle\psi|\sigma_x|\psi\rangle + \sin(2\alpha) \langle\psi|\sigma_y|\psi\rangle \quad (8.2.14)$$

We note that these are very close to the 2×2 rank-2 tensor change of basis expressions.

Proposition 8.2.1 (Spin as the Generator of Rotation)

The operator $S_{\hat{n}}$ is the generator of rotation about the $\hat{n}$ axis: the ket $|\phi\rangle$, defined as below, is the ket $|\psi\rangle$ rotated by an angle 2α .

$$|\phi\rangle \equiv \exp(i\alpha S_{\hat{n}})|\psi\rangle \quad (8.2.15)$$

In particular, the new ket $|\phi\rangle$ will have an x - and y -spin distribution given by

$$\langle\phi|\sigma_x|\phi\rangle = \cos(2\alpha) \langle\psi|\sigma_x|\psi\rangle + \sin(2\alpha) \langle\psi|\sigma_y|\psi\rangle \quad (8.2.16)$$

$$\langle\phi|\sigma_y|\phi\rangle = -\sin(2\alpha) \langle\psi|\sigma_x|\psi\rangle + \cos(2\alpha) \langle\psi|\sigma_y|\psi\rangle \quad (8.2.17)$$

Property 8.2.2 (Similarity to Stress Transformation)

In elasticity theory, the components of the stress tensor σ transform according to

$$\tilde{\sigma}_{11} = \frac{\sigma_{11} + \sigma_{22}}{2} + \frac{\sigma_{11} - \sigma_{22}}{2} \cos(2\alpha) + \sigma_{12} \sin(2\alpha) \quad (8.2.18)$$

$$\tilde{\sigma}_{22} = \frac{\sigma_{11} + \sigma_{22}}{2} - \frac{\sigma_{11} - \sigma_{22}}{2} \cos(2\alpha) - \sigma_{12} \sin(2\alpha) \quad (8.2.19)$$

$$\tilde{\sigma}_{12} = -\frac{\sigma_{11} - \sigma_{22}}{2} \sin(2\alpha) + \sigma_{12} \cos(2\alpha) \quad (8.2.20)$$

In the case of spin, we map

$$\frac{\sigma_{11} - \sigma_{22}}{2} \longleftrightarrow \langle \psi | \sigma_x | \psi \rangle, \quad (8.2.21)$$

$$\sigma_{12} \longleftrightarrow \langle \psi | \sigma_y | \psi \rangle. \quad (8.2.22)$$

The hydrostatic part,

$$\frac{\sigma_{11} + \sigma_{22}}{2}, \quad (8.2.23)$$

has no direct analogue in the spin expectation value, since the Pauli matrices are traceless. Equivalently, it would correspond to an identity component, which does not change under rotations. Thus, if we define

$$r_x = \langle \psi | \sigma_x | \psi \rangle, \quad (8.2.24)$$

$$r_y = \langle \psi | \sigma_y | \psi \rangle, \quad (8.2.25)$$

then under rotation we obtain

$$\tilde{r}_x = r_x \cos(2\alpha) + r_y \sin(2\alpha), \quad (8.2.26)$$

$$\tilde{r}_y = -r_x \sin(2\alpha) + r_y \cos(2\alpha). \quad (8.2.27)$$

This is exactly the same transformation law obeyed by the pair

$$\left(\frac{\sigma_{11} - \sigma_{22}}{2}, \sigma_{12} \right) \quad (8.2.28)$$

in the stress tensor transformation equations. The analogy is between the traceless part of the stress tensor and the transverse components of the spin expectation value. The appearance of 2α highlights the fact that both stress components and spin expectation values transform quadratically in the underlying basis amplitudes.

8.3 Stern-Gerlach Experiment

Aside from theoretical discussions, one can measure the spin using a *Stern-Gerlach Device*, which consists of a magnet with a very strong magnetic field gradient. Any beam passed through this device would split into two clear lines, clearly demonstrating the underlying quantization.

Proposition 8.3.1 (Probability of Measurement)

The probability of measuring the $|\pm\rangle_{\hat{n}}$ state from a beam of $|\psi\rangle$ -state particles passing through an $\hat{n}$ -Stern Gerlach device is

$$P_{\pm\hat{n}} = |\pm\hat{n}\rangle\langle\pm\hat{n}| \quad (8.3.1)$$

We can place such devices in a sequence, one after another.

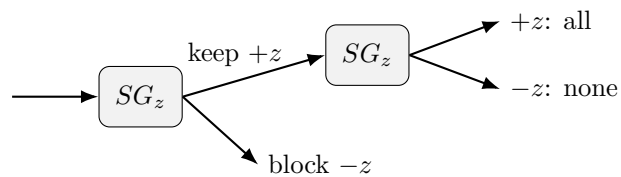


Figure 14. Two Stern-Gerlach Devices, z -oriented, in series.

Once a definite result is obtained, it is repeatable. The most famous sequence is

$$SG_z \rightarrow SG_x \rightarrow SG_z$$

in which we keep the $+z$ from the first device, keep the $+x$ from the second, and measure z again. The result is equal probability, since the measurement on x split the beam into two constituent spins, and so the information was lost.

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